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Chapter 1

Introduction

In recent years, the term *quantum materials* has come to describe a class of materials that, loosely speaking, display quantum mechanical properties and many-body effects over large length and energy scales. These effects manifest in the form of exotic phenomena, examples of which include unconventional superconductivity and the fractional quantum hall effect. Such materials often involve strong inter-electron interactions, low dimensions and topological properties, making them highly sensitive to small changes in parameters or temperature. This typically means that such materials cannot be described using single-particle or weak-coupling approaches, making the study of such materials more challenging and interesting. Examples of quantum materials include the copper oxide superconductors, heavy fermion materials, Moire materials and quantum spin liquids. The exotic phases that emerge from the interplay of competing tendencies in these materials are often referred to as *quantum matter*. Such phases are usually marked by non-trivial patterns of topology and entanglement. There are several examples of phases of quantum matter, some were already provided: unconventional superconducting states, fractional quantum hall states, strange metals and topologically ordered phases, among others. In this thesis, we are concerned with the study of a specific aspect of quantum matter – Mott metal-insulator transition. In doing so, we will deal with a number of correlated phases such as Mott insulators, strange metals and the pseudogap, all of which will be explained in due course. The phenomenon of Mott insulation is also intimately tied to other exotic effects such as high-T_c superconductivity and spin liquid behaviour; these will not be explored in this thesis.

1.1 Mott metal-insulator transition: An Introduction

The rich physics of Mott metal-insulator transitions in strongly correlated systems has been an active subject of study for quite some time [1–4]. Their appearance in various materials such as the nickelates, cuprates, iridates and manganites, among others, is marked by complex phase diagrams that display dramatic changes in properties across changes in filling and/or bandwidth, even at low temperatures. The physics of Mott transitions is driven by the competition between two energy scales in the system: (i) the scale of itineracy, quantified by the bandwidth, and (ii) the scale of electron-electron repulsion. A Mott insulator arises when the electronic correlations are strong enough that low-energy electronic excitations become gapped, and the metallic valence band gets split into lower and upper bands; for an initially half-filled valence band, the lower band is filled at $T = 0$ while the upper band is vacant. Because of strong electronic repulsion, the electrons in the lower band form local moments which can show a variety of behaviour. This includes various symmetry-broken ground states such as an antiferromagnet [5], or symmetry-preserved states such as quantum spin liquids [6]. The physics of Mottness has been proposed as being responsible for several very interesting phenomena, such as high-temperature superconductivity [5], non-Fermi liquids [7] and the various phases in magic-angle twisted bilayer graphene [8, 9].

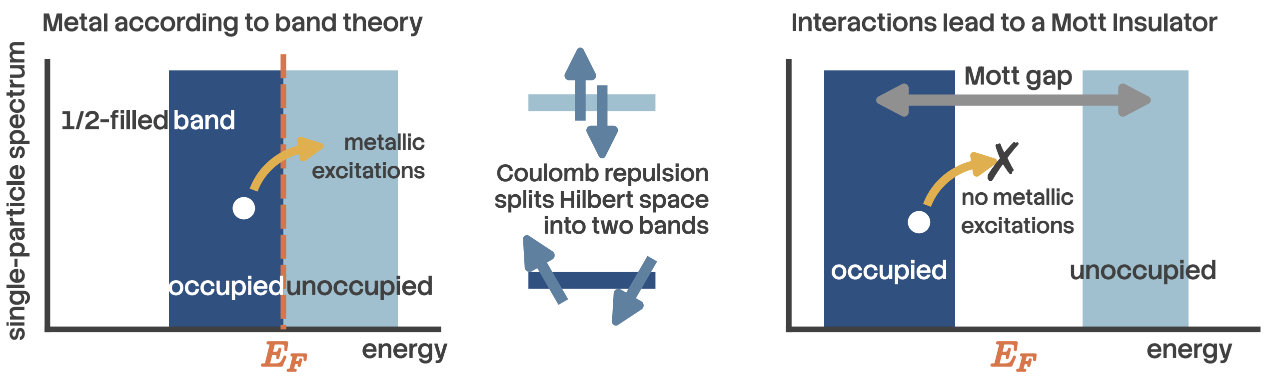

Figure 1.1: *A Mott insulator is an example of strongly correlated phenomena, where inter-electron repulsion leads to insulating behaviour. In the absence of interactions, a half-filled band (left) would result in metallic behaviour according to band theory. Upon including sufficiently strong interactions, the set of states within the band get split into two sub-bands, separated by a spectral gap of the order of the interaction strength. This makes the system metallic, because the lower sub-band remains completely filled while the upper sub-band is empty.*

Initial attempts at explaining metal-insulator transitions were based on a non-interacting picture of electrons (through the work of Bethe, Sommerfeld, Bloch and Wilson among others); in such a picture, electronic bands can arise only from the scattering of electrons off the periodic lattice potential. One then distinguishes between metals and insulators at zero temperature by the position of the Fermi level E_F with respect to the bands: if E_F lies within a band gap, an integer number of bands will be completely filled and the system will be an insulator, whereas if E_F lies within one of the bands, the system will be a metal because there are vacant gapless excitations within the band.

It was first pointed out by Boer and Verwey [10] that certain insulators and semiconductors (such as nickel oxides) have a particle filling that, purely from band theory, would lead to partially filled bands. This poses a challenge because a partially filled band necessarily leads to a metallic phase (see Fig. 1.1 (left)). The insulating behaviour of these materials therefore signals a fundamental limitation of single-particle band theory and points to a qualitatively different mechanism for electron localization—one driven by strong electron–electron interactions rather than by scattering from the lattice potential alone. This realization led to the concept of a Mott insulator, in which charge localization emerges as a genuine many-body effect despite the absence of a band gap in the non-interacting electronic structure. A very simple mechanism for this was put forth by Mott [11–13]: in the presence of a large local Coulomb repulsion cost U on a lattice site, the quantum states $|\Psi_1\rangle$

in which the site is singly occupied get separated in energy from the states $|\Psi_2\rangle$ where the site is doubly occupied by a gap of size U; the former class of states $|\Psi_1\rangle$ form the lower band (LB) while the latter $|\Psi_2\rangle$ form the upper band (UB). If we now have a half-filled system (one electron per site), this electron will completely fill LB and leave UB vacant, leading to an insulating state (see Fig. 1.1 (right)).

Given the above discussion, it is clear that a Mott insulator must involve a gap to charge excitations (excitations that add or remove an electron). It does not however determine the fate of the spin, orbital or other degrees of freedom. One scenario is when the spin or orbital degree of freedom undergoes spontaneous symmetry breaking into an ordered state; in the case of spin, this usually results in a superexchange interaction being generated and leads to an long-range ordered antiferromagnetic Neel state coexisting with the Mott state. Examples of such systems include vanadium sesquioxide V_2O_3 and its derivatives and the high-T_c cuprates (such as La_2CuO_4) [4]. The other scenario is one where the spin degree of freedom retains its symmetry, and we end up with a quantum mechanically non-trivial entangled state (in order to quench the entropy). Examples of such states include quantum spin liquids [14]. Mott insulators without long-range order are of particular interest because of the possibility of topological order, fractional excitations, and novel metallic parent states.

A prototypical model for gaining theoretical understanding of Mott metal-insulator transitions on a lattice is the single-band Hubbard model [1, 15–17]

$$H = -t \sum_{\langle i,j\rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) - \frac{U}{2}\sum_i (n_{i,\uparrow} - n_{i,\downarrow})^2 - \mu N \ . \tag{1.1}$$

The first term incorporates nearest-neighbour hopping on the lattice, while the second term introduces inter-electron repulsion when two electrons site on the same lattice site; in this way, the model incorporates the two competing tendencies of delocalisation and localisation that are responsible for the Mott MIT. The third term fixes the chemical potential and determines the filling of the lattice. Many of the materials that are studied for Mott MIT involve $d-$electrons that are five-fold degenerate, and these orbitals can hybridise with ligand atoms. The Hubbard model avoids these complexities by focusing on only a single band that lies closest to the Fermi energy, and it is the states of this band that are described by the Hubbard model.

This leaves us with two channels for observing Mott MIT [4]: (i) tuning the bandwidth, and (ii) tuning the filling. The former can be done, in experiments, by applying pressure. Filling is tuned by doping specific elements of a compound with elements of a different valence. The former is referred to as bandwidth-control metal-insulator transition, while the latter is called filling-control metal-insulator transition.

1.2 Features of Mott metal-insulator transition

1.2.1 Two complementary pictures of the Mott transition: Brinkman-Rice vs Mott-Hubbard.

Initial insights into the Mott MIT were obtained by Hubbard [1], thinking along lines similar to Mott. Within this picture, one starts from the insulating large U limit of the Hubbard model (eq. 1.1) ($U \gg t$. We stay at half-filling for simplicity ($\mu = 0$). The large on-site repulsion leads to the formation of two bands at each site, the lower Hubbard band (LHB) associated with singly-occupied states and the upper Hubbard band (UHB) associated with doubles and holes (states that incur a cost U). In the large U limit, these bands are well-separated, and the dynamics within the bands is a result of the motion of tightly bound holons and doubles (more on this later) that are generated because of the small but non-zero hopping probablity t. As U/t is lowered, the bands come closer and a metal-insulator transition occurs at the critical value of U at which the gap vanishes. For lower U/t, the bands overlap and we get a metal. This picture fails to provide an accurate description of the metal resulting from the melting of such a Mott insulator [18].

An alternative approach to the Mott MIT is due to Brinkman and Rice [19], where the authors applied Gutzwiller's variational method [20] to the half-filled Hubbard model. Gutzwiller's approach involves modifying, through variational factors, the hopping matrix elements that start from doubly occupied states, in order to reflect the electronic correlation at each site. By requiring that the number of doubly-occupied states vanish in the Mott insulating ground state, they obtained a critical value of the interaction strength. In the Brinkman-Rice scenario, therefore, the transition from the metal to the insulating phase occurs when the hopping matrix elements from the doubly-occupied to the singly-occupied states strictly vanishes. Note that in contrast to the Mott-Hubbard picture, this approach starts from a correlated Fermi liquid (renormalised due to the Gutzwiller factors). While it gives a better low-energy description of the correlated metal compared to the Mott-Hubbard picture, it fails to account for the high-energy physics, and treats the Mott insulator as a direct product state of local moments at each state (on account of the strictly zero hopping). This of course means that this method cannot explain antiferromagnetic superexchange processes in the Mott insulator [18].

1.2.2 Holon-doublon unbinding picture of the Mott MIT.

In 1949, Mott proposed a zero temperature explanation for a correlation-driven insulator and a mechanism for its transition into a metal in terms of the motion of holes and doubles in the background of a half-filled lattice (see Fig. 1.2). For $U \ll t$ (strong correlation), the ground state predominantly involves lattice sites with a single electron. Hopping processes between two nearest-neighbour sites transfer an electron between them and create a pair of hole and double, but the pair remains tightly bound to each other. This is because moving the hole independently involves scattering processes that cross the Mott gap, and these processes are not allowed at $T = 0$. But without the independent motion of the hole (or double), there is no current, and the system is an insulator (left, Fig. 1.2).

As U/t is lowered, the gap closes and processes that transport the hole without transporting the double become gapless. This essentially means that the pair becomes deconfined, and the hole can be moved far away from the double (right, Fig. 1.2). Current can then be passed by having the hole

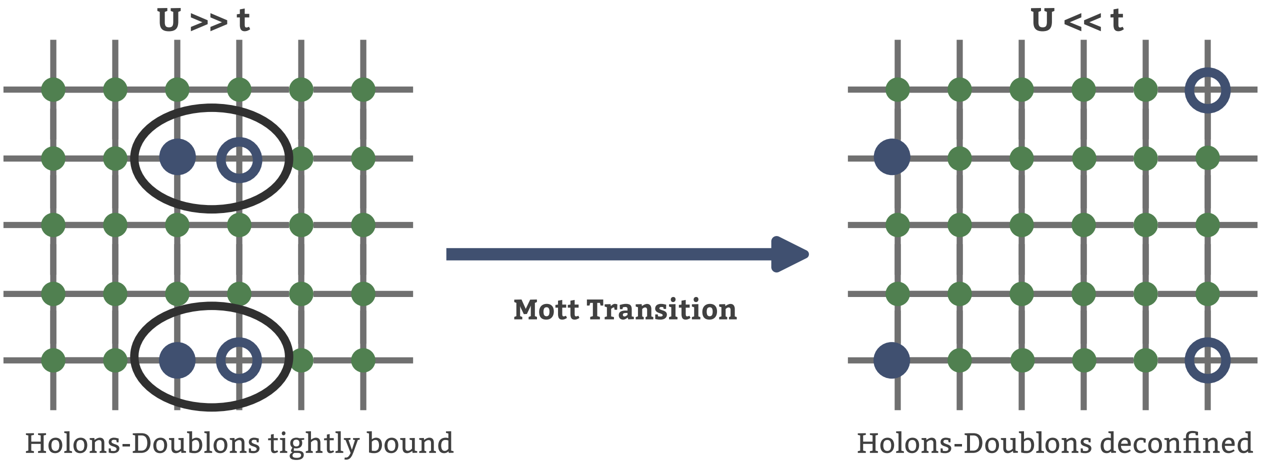

Figure 1.2: *A schematic depiction of Mott's picture of correlation-driven metal-insulator transition. In the insulating state (left), each site is mostly comprised of single spins because of the strong repulsion, but virtual hopping processes can create locally bound pairs of holes and doubles. Because of the gap to charge excitations, these pairs cannot unbind and delocalise. In contrast, in the metallic phase, the presence of gapless excitations means that the hole can move independently of the double, leading to charge transport and metallicity. This frames the Mott MIT as a confinement-deconfinement transition of the holes and doubles.*

move through the lattice, and we get a metallic state. This leads to a picture of the Mott transition into a metal in terms of an unbinding of the hole-double bound states (excitons) that were present in the insulator.

There have been more quantitative studies of this mechanism. I list some of the relatively recent ones here. In [21], Prelovšek et al. studied the motion of holon-doublon pairs in a spin polarised background by rewriting the repulsive Hubbard model in terms of electrons as an attractive Hubbard model of holes and doubles. The model was treated using a BCS-like treatment, and the Mott transition was obtained when the binding energy for the hole-double condensate vanishes. To mark the deconfinement, they show that the approach to the transition from the insulating side is concomitant with an increase in the size of the pairs. This approach however fails to capture the transition for a system away from half-filling. Following a slave-boson treatment of holon-doublon unbinding, Zhou et al. [22] show that the Mott transition from a paramagnetic metal to a Mott insulator on the half-field honeycomb and 2D square lattices occur through an intervening Slater insulator with coherent quasiparticles. In particular, their approach sheds light on the nature of the incoherent excitations in the Hubbard sidebands of the local spectral function. Finally, an older work by Watanabe et al. [23] studies the half-filled Hubbard model on an anisotropic triangular lattice using a Monte carlo method with a variational binding parameter between holons and doublons. Their approach reveals a first order Mott transition, along with robust $d-$wave superconductivity and short-range magnetic order.

1.2.3 Spectral features of the Mott transition.

One of the important diagnostics of the Mott metal-insulator transition is the spectrum for one-particle electron-like excitations, as expressed through the one-particle retarded Greens function $G(\boldsymbol{k},\sigma;t) = -i\theta(t)\langle\{c_{\boldsymbol{k},\sigma}(t), c^\dagger_{\boldsymbol{k},\sigma}\}\rangle$. In general, the presence of poles in the Green function at the Fermi energy and in its neighbourhood imply the existence of gapless excitations and hence a metallic phase. It turns out that this statement is equivalent to Luttinger's theorem [24, 25],

$$\frac{N}{V} = 2\int_{G(0,\boldsymbol{k})>0} \frac{d\boldsymbol{k}}{(2\pi)^3} , \tag{2.1}$$

which states that the particle density in a metal is equal to the number of states in the Fermi volume. As clarified in the equation, the Fermi volume is defined as the region of $k-$space where the Greens function is positive at $\omega = 0$.

In a Mott insulator, strong correlations lead to the destruction of electronic excitations around $\omega = 0$. This results in the Greens function poles transmuting into zeros. This is also tracked by analogous changes in the electronic self-energy $\Sigma(\boldsymbol{k};\omega)$; Greens function zeros leads to a Mott pole at $\omega = 0$ in $\Sigma(\boldsymbol{k};\omega)$. The points in $k-$space where $\Sigma(\boldsymbol{k};0)$ vanishes is called a Luttinger surface [26]. The Mott metal-insulator transition is therefore about the topological change of a Fermi surface (surface of Greens function poles) into a Luttinger surface (surface of self-energy poles).

As discussed earlier, the appearance of Greens function zeros at $\omega = 0$ leads to the formation of a charge gap that freezes one-particle low-energy excitations. The spectrum gets redistributed into two bands on either side of the gap, the upper and lower Hubbard bands. In the ground state, the lower band is obviously filled while the upper band is empty. Bands of a Mott insulator formed from electronic correlation differ from those formed due to one-particle scattering in a particular way: dynamical spectral weight transfer. Unlike a band insulator, the bands of a Mott insulator are not rigid; tuning the lattice filling results in a continuous transfer of states from one band to another.

The non-rigidity of the bands can be seen most easily from the atomic limit ($t = 0$) of the Hubbard model (eq. 1.1). The local retarded Greens function at an arbitrary site i can be obtained using the equation of motion approach. The final result is [27]

$$G_R(i;\omega) = \frac{1+x}{\omega+\mu+U/2} + \frac{1-x}{\omega+\mu-U/2} , \tag{2.2}$$

where $x = 1 - \sum_\sigma \langle n_{i\sigma}\rangle$ is the average number of holes at any site. The spectral weight for the two poles makes it clear that the lower band has $1+x$ number of states while the upper band has $1-x$ states. Changing the number of states in the lower band (by tuning μ and hence x) also changes those in the upper band. For example, by adding 0.1 holes per site to the lower band, the spectral weight in the upper band decreases by the same amount. This is the non-rigidity of the bands mentioned previously.

1.2.4 Preliminary Insights into Mottness From a Toy Model

The Baskaran-Hatsugai-Kohmoto model [28, 29] introduced independently in 1991 and 1992 is an exactly solvable model of electronic correlation that displays a transition from a metallic phase to a

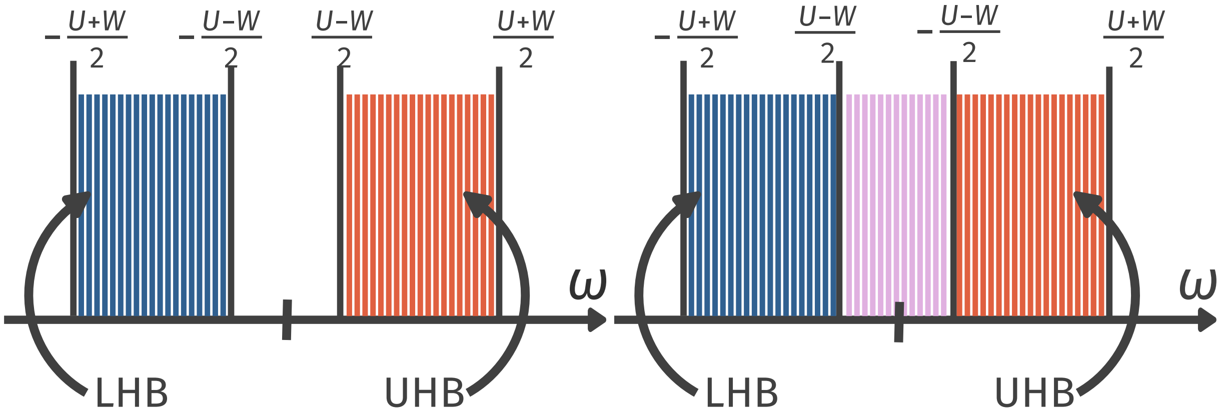

Figure 1.3: *The Mott insulator (left) and non-Fermi liquid (right) phases of the Hatsugai-Kohmoto model. In the Mott insulating phase, each momentum state is singly occupied and the lower (blue) is completely filled, while the upper band consisting of the doubly occupied and empty states is completely empty. There is, thus, a spectral gap to the excitations. In the metallic phase (right), the bands overlap and one can have correlated excitations from singly occupied states to hole and double states. These excitations are non-electronic, and the metal is a non-Fermi liquid.*

Mott insulator. While it's solvability offers an instructive method for studying Mott transition, the infinitely long-ranged nature of its interaction makes it tricky to apply these conclusions to realistic materials.

At half-filling, the model is defined by the following Hamiltonian:

$$H_{\text{BHK}} = \sum_{\mathbf{k},\sigma} \epsilon(\mathbf{k}) n_{\mathbf{k},\sigma} - \frac{U}{2} \sum_{\mathbf{k}} \left(n_{\mathbf{k},\uparrow} - n_{\mathbf{k},\downarrow}\right)^2 . \tag{2.3}$$

Clearly the model is diagonal in momentum-space; the Hamiltonian splits up into commuting 4×4 blocks for every point $\mathbf{k}$:

$$H_{\text{BHK}} = \sum_{\mathbf{k}} H(\mathbf{k}), \quad H(\mathbf{k}) = \epsilon(\mathbf{k}) \sum_{\sigma} n_{\mathbf{k},\sigma} - \frac{U}{2} \left(n_{\mathbf{k},\uparrow} - n_{\mathbf{k},\downarrow}\right)^2 . \tag{2.4}$$

The vacant eigenstate $|0\rangle$ of $H(\mathbf{k})$ has zero energy ($E_0 = 0$), the two singly-occupied states $|\uparrow\rangle$ and $|\downarrow\rangle$ have energy $E_1 = \epsilon(\mathbf{k}) - U/2$ and the doubly-occupied state $|\uparrow, \downarrow\rangle$ has energy $E_2 = 2\epsilon(\mathbf{k})$. This leads to two possible single-particle excitations – one for adding an electron on the vacant state: $(\Delta_1 = E_1 - E_0 = \epsilon(\mathbf{k}) - U/2$, the other for adding an additional electron on a singly-occupied state: $\Delta_2 = E_2 - E_1 = \epsilon(\mathbf{k}) + U/2$.

Upon combining states from other $k-$modes, these isolated excitations broaden into bands around $\pm U/2$ because of the continuum of values available to $\epsilon(\mathbf{k})$. For large enough U (compared to the non-interacting bandwidth W), these bands remain well-separated in energy (Fig. 1.3 left), forming lower and upper Hubbard bands. The half-filled system results in the lower band being completely filled and the upper band empty, leading to a Mott insulator. The states within this band are all singly-occupied (per $k-$state). Upon lowering the correlation strength U, the bands touch each other at the critical value of $U_c = W$ (obtained by equating the top of the lower band to the bottom of the upper band). This mechanism is qualitatively similar to the Mott-Hubbard picture of Mott transition. For lower values of U, the system is a metal due to the presence of gapless excitations within the combined band (Fig. 1.3 right). This therefore provides a simple demonstration of a correlation-driven Mott metal-insulator transition.

1.3 Models and materials for strong local correlation

1.3.1 Introduction to the auxiliary model approach

One of the main results obtained within this thesis is the development of a formalism to study the more realistic case of finite-dimensional interacting models by using appropriately chosen impurity models as auxiliary models. The key to an *auxiliary model method* is to construct an appropriate auxiliary model that while being tractable captures some essential property of the full lattice model [30]. This involves separating the complete degrees of freedom into a system part (H_S) which one treats exactly and the rest of the system (H_R) that one might simplify in order to be able to solve the problem. The non-trivial nature of the problem arises of course from the hybridisation H_{SR} between the system and the bath:

$$H = \begin{bmatrix} H_R & H_{RS} \\ H_{RS}^* & H_S \end{bmatrix} . \tag{3.1}$$

This separation can always be done exactly, if one does not care about tractability. For example, one can take the Hubbard model that describes electrons hopping on a lattice and interacting when two electrons reside on the same lattice site:

$$H_{\text{hub}} = -t \sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sum_i n_{i\uparrow} n_{i\downarrow} , \tag{3.2}$$

and separate this into parts, the system being the local orbitals of a specific site l and the rest of the system being all the other sites:

$$H_S = U n_{l\uparrow} n_{l\downarrow}, \quad H_R = -t \sideset{}{'}\sum_{\langle i,j \rangle,\sigma} (c^\dagger_{i,\sigma} c_{j,\sigma} + \text{h.c.}) + U \sideset{}{'}\sum_i n_{i\uparrow} n_{i\downarrow}, \quad H_{SR} + H_{RS} = -t \sum_{i,\sigma} (c^\dagger_{l,\sigma} c_{i,\sigma} + \text{h.c.}) , \tag{3.3}$$

where the primed summation is carried out over all nearest-neighbour pairs that do not involve l and the summation in $H_{SR} + H_{RS}$ is over all sites adjacent to l.

The smaller system H_S (often called the *cluster*) is typically chosen such that its eigenstates are known exactly. Progress is then made by choosing a simpler form for H_R and its coupling H_{RS} with the smaller system. This combination of the cluster and the simpler bath is then called the *auxiliary system*. A tractable auxiliary system for the Hubbard model is the single-impurity Anderson model (SIAM); the correlated impurity site represents the set of local orbitals of any given site, while the conduction bath represents the rest of the lattice sites in an approximate fashion. Such a construction is shown in fig. 1.4.

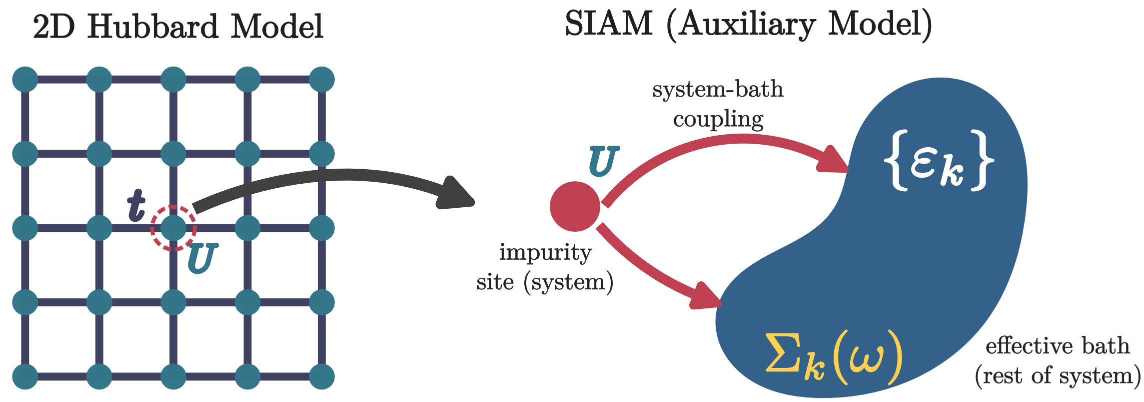

Figure 1.4: *Left: Full Hubbard model lattice with onsite repulsion U^H on all sites and hopping between nearest neighbour sites with strength t^H. Right: Extraction of the auxiliary (cluster+bath) system from the full lattice. The central site on left becomes the impurity site (red) on the right (with an onsite repulsion ϵ_d), while the rest of the $N-1$ sites on the left form a conduction bath (green circles) (with dispersion ϵ_k and correlation modelled by the self-energy $\Sigma_k(\omega)$) that hybridizes with the impurity through the coupling V.*

It should be noted that any reasonable choice of the cluster and bath would break the translational symmetry of the full model: To allow computing quantities, one would need to make the bath (which is a much larger system) simpler than the cluster (which is a single site). This distinction breaks the translational symmetry of the Hubbard model. As a result, calculations that make use of the translation symmetry of the full model (such as $k-$space objects) will require a periodisation operation of an auxiliary model quantity.

1.3.2 Correspondence between impurity and lattice models in $d = \infty$

Dynamical Mean-Field Theory (DMFT) provides a nonperturbative framework for treating strong local electronic correlations in lattice models by mapping them onto a self-consistent quantum impurity problem [18, 31]. The central idea of DMFT is that in the limit of infinite spatial dimension, or equivalently infinite coordination number, the electronic self-energy becomes purely local while retaining its full dynamical (frequency-dependent) structure. As a result, the many-body lattice problem reduces to an effective single-site problem embedded in a self-consistently determined bath, capturing local quantum fluctuations exactly while neglecting nonlocal correlations. DMFT has become a cornerstone of modern correlated-electron theory, successfully describing Mott metal–insulator transitions, quasiparticle formation, and spectral weight transfer in Hubbard-like models [18, 32].

In order to motivate the impurity model-lattice model mapping inherited by DMFT, we first recall the textbook example of classical mean-field theory. Mean-field theory provides one of the earliest and most instructive examples of controlled approximations in many-body physics. In its classical incarnation, exemplified by the Ising model,

$$H_{\text{Ising}} = -\sum_{i\neq j} J_{ij} S_i S_j - h\sum_i S_i = -\sum_i S_i (h + \sum_{j\neq i} J_{ij} S_j), \tag{3.4}$$

mean-field theory proceeds by replacing the interaction between a given spin and its neighbors by an *effective static field* proportional to the average magnetization. Writing $S_j = \langle S_j \rangle + \delta S_j$ and neglecting quadratic fluctuations and constant parts, one obtains an *auxiliary problem* in the form of a mean-field Hamiltonian

$$H^{(\text{MF})}_{\text{Ising}} = -\sum_i (h + h_{\text{MF}}(i)) S_i \ , \tag{3.5}$$

where the mean-field h_{MF} comes out to be

$$h_{\text{MF}}(i) = \sum_{j\neq i} 2\langle S_j \rangle J_{ij} \ . \tag{3.6}$$

The above approach assumes two things: (i) the presence of a non-zero *order parameter* $m_j = \langle S_j \rangle$ (in this case, a non-zero magnetisation), and (ii) the smallness of the fluctuations $S_j - \langle S_j \rangle$. The mean-field Hamiltonian can be used to compute the partition function (at inverse temperature β) and hence the magnetisation at a site i of the lattice:

$$m_i = \tanh(\beta(h_{\text{MF}}(i) + h)) = \tanh(\beta \sum_{j\neq i} 2 m_j J_{ij} + \beta h) \ . \tag{3.7}$$

The problem hasn't yet been solved of course; one needs to determine the appropriate value of m_j. We therefore need an additional equation; this is provided by the so-called *self-consistency condition*. The translation invariance of the lattice ensures that the magnetisation must be uniform across sites i and j. This allows us to replace m_j with m_i in the above equation:

$$m_i = \tanh(\beta m_i \sum_{j\neq i} 2 J_{ij} + \beta h) \ . \tag{3.8}$$

This equation can be solved numerically to obtain the uniform magnetisation $m_i = m$. The mean-field approximation becomes exact in the limit of large coordination number, where fluctuations are suppressed and spatial correlations beyond a single site vanish.

An equivalent way of framing the self-consistency equation is to equate the mean-fields on different lattice sites. Combining eqs. 3.7 and 3.6, we have

$$h_{\text{MF}}(i) = \sum_{j\neq i} 2 m_j J_{ij} = \sum_{j\neq i} 2\tanh(\beta(h_{\text{MF}}(j) + h)) J_{ij} \ . \tag{3.9}$$

The equation in terms of $h_{\text{MF}}(i)$ and $h_{\text{MF}}(j)$ can again be solved numerically by claiming that the mean-field has to be uniform across all the sites owing to translation invariance. This form of the self-consistency requirement (in terms of the field) will be useful to us in the upcoming discussion.

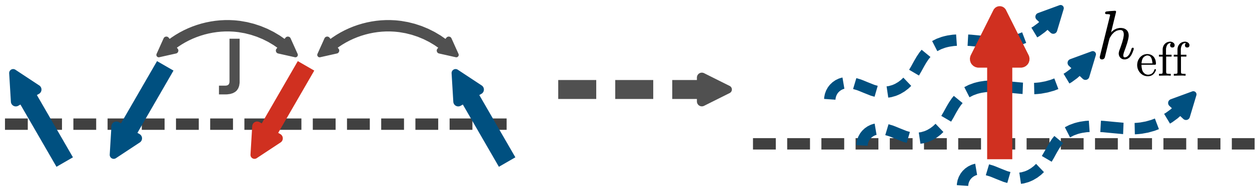

Figure 1.5: *The Ising mode mean-field treatment as an auxiliary model approach. Through the mean-field decoupling followed by the identification of a self-consistent field, the effect of the neighbouring spins is approximated by a magnetic field. This self-consistently determined model of a spin in a magnetic field acts as an auxiliary model for the complete Ising model, representing the local properties of the Ising spins through the properties of the spin subjected to the field.*

The essential feature of classical mean-field theory is therefore the replacement of a many-body problem by an effective *single-site* problem embedded in a self-consistently determined environment. Quantum lattice models admit a closely analogous construction, but with an important conceptual difference: while the classical mean field is static, quantum fluctuations necessarily introduce a tower of excited states that allow for temporal dynamics. This observation underlies the formulation of dynamical mean-field theory (DMFT), which may be viewed as the natural quantum generalization of classical mean-field theory [18, 31].

Similar to the previous case of the Ising model, we will now derive the self-consistency equation for a quantum model. Consider an interacting fermionic lattice Hamiltonian, such as the Hubbard model:

$$H = -\sum_{ij\sigma} t_{ij} c^\dagger_{i\sigma} c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \tag{3.10}$$

defined on a lattice with coordination number z.

The point of this exercise is to calculate the interacting self-energy $\Sigma_k(\omega)$ for the fermions. This therefore acts as an "order parameter" and plays the same role as the local magnetisation in the previous exercise. The self-energy defines the interacting k-space Green's function for a translation-invariant problem:

$$G_{\mathbf{k}}(\omega) = \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}, \tag{3.11}$$

where $\varepsilon_{\mathbf{k}}$ denotes the noninteracting dispersion, and a real-space local Green's function, obtained by summing over momentum:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\mathbf{k}}(\omega)}. \tag{3.12}$$

Similar to the earlier exercise, we need to apply a mean-field decoupling to drop certain fluctuations and obtain a similar effective problem. In the context of interacting electrons, such a decoupling is obtained in the limit of infinite dimensionality or coordination number, where the self-energy becomes $k-$independent [31]:

$$\Sigma_{\mathbf{k}}(\omega) = \Sigma_{\text{loc}}(\omega), \text{ or (equivalently) } \Sigma_{ij}(\omega) = \delta_{ij}\Sigma_{\text{loc}}(\omega)\ . \tag{3.13}$$

The mean-field decoupling here therefore corresponds to ignoring spatial fluctuations of the one-particle self-energy, and leads to a local Greens function that is directly connected to the local self-energy:

$$G_{\text{loc}}(\omega) = \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}. \tag{3.14}$$

It turns out that a local Greens function connected to a local self-energy also defines a quantum impurity problem describing the dynamics of a single correlated site with the other environment sites integrated out:

$$S_{\text{imp}} = -\int_0^\beta d\tau d\tau' \, c^\dagger(\tau) \, G_{\text{MF}}^{-1}(\tau - \tau') \, c(\tau') + U \int_0^\beta d\tau \, n_\uparrow(\tau) n_\downarrow(\tau). \tag{3.15}$$

This acts as an auxiliary problem for the more complete problem of locally interacting electrons on a lattice. The non-interacting Greens function G_{MF} is as of now unknown, and defines the environment of the impurity site and comprises the dynamics of the sites that have been integrated out. Such an impurity problem is solvable, and the impurity Green's function is given by

$$G_{\text{imp}}(\omega) = \frac{1}{G_{\text{MF}}^{-1}(\omega) - \Sigma_{\text{imp}}(\omega)}, \tag{3.16}$$

The first term in eq. 3.15 is therefore what makes this problem dynamical – it represents the probability of electrons hopping into and out of the impurity site, and leads to fluctuations of its occupancy. We can make the identification of the Greens function G_{MF} as the *mean-field* encountered in the previous example. More specifically, G_{MF} acts as a "field" because it is obtained by integrating out the other sites on the lattice, and it acts on the impurity site by affecting its dynamics.

The complete set of equations for the solving such a dynamical mean-field problem is finally obtained by equating the impurity local Greens function G_{imp} to the lattice local Greens function G_{loc}. As discussed earlier, such a solution exists in an exact fashion only in the limit of infinite dimensions, as long as the correct G_{MF} can be found. In order to find the appropriate G_{MF} that allows equating the impurity model Greens function to a lattice mode Greens function, we require an additional equation: the self-consistency condition. Similar to the case of the Ising model, we require that G_{MF} be equal to G_{loc}, owing to translation invariance. The full set of coupled DMFT equations is

$$\begin{aligned}
G_{\text{MF}} &= G_{\text{loc}}, \\
\Sigma_{\text{imp}}(\omega) &= G_{\text{MF}}^{-1}(\omega) - G_{\text{imp}}^{-1}(\omega), \\
\Sigma_{\text{loc}}(\omega) &= \Sigma_{\text{imp}}(\omega), \\
G_{\text{loc}}(\omega) &= \sum_{\mathbf{k}} \frac{1}{\omega - \varepsilon_{\mathbf{k}} - \Sigma_{\text{loc}}(\omega)}.
\end{aligned} \tag{3.17}$$

One can map various features of this self-consistent program with the set of equations obtained above for the Ising model:

- The first equation is equivalent to the feature in the previous exercise where we equated the mean-fields on i and j.

- The second equation is equivalent to computing, in the previous exercise, the magnetisation on the j^{th} site from the partition function.

- The third equation is the essential mean-field approximation, where we equate the self-energy of the simpler decoupled problem to that of the more complete lattice problem and is justified in the limit of infinite dimensions.

- The fourth equation is equivalent, in the previous exercise, to using the computed "magnetisation" of site j in the second step to further compute the mean-field at site i.

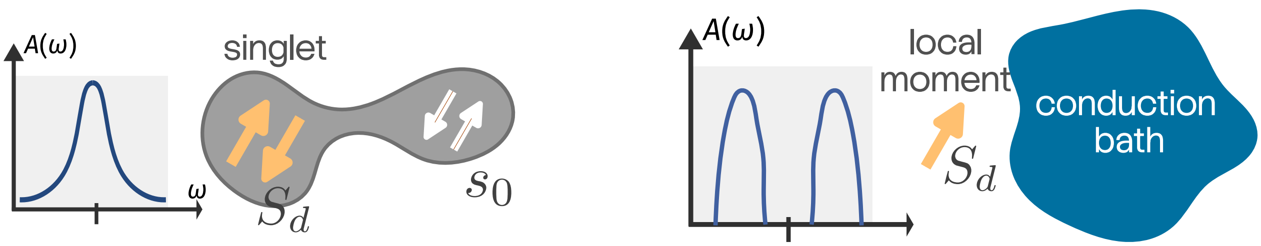

Figure 1.6: *The auxiliary model mapping between interacting lattice models and quantum impurity models involves a dictionary for relating phenomena on either side. For example, Kondo screening and the associated local Fermi liquid excitations on the impurity model (left) map to metallic Fermi liquid behaviour for the lattice model. On the other hand, breakdown in Kondo screening and the presence of a residual unscreened moment at low-energies in the impurity model maps to insulating behaviour in the lattice model.*

In direct analogy with classical mean-field theory, DMFT replaces a lattice model by an effective single-site problem embedded in a self-consistent environment. The crucial distinction is that the mean field is now dynamical rather than static, encoding quantum fluctuations exactly while neglecting spatial correlations. The fact that the correlated self-energy of the impurity site feeds into the impurity-bath hybridisation during the convergence towards self-consistently means that the bath must get correlated in the process. Importantly, this demonstrates the importance of quantum impurity models in investigating correlated lattice models, particularly in the presence of strong local interactions. This is because the effects of local correlations are captured exactly by such an auxiliary model mapping.

1.3.3 Generator of strong local correlations: Quantum impurity models

Within this thesis, we have developed an auxiliary model approach that combines the physics of quantum impurity models with a manybody version of Bloch's theorem to investigate interacting fermionic lattice models. This is motivated by the relative ease in solving impurity models and transparency afforded by such solutions. We now review the phenomenology of some of these impurity models and point out how the various impurity features affect the physics of the lattice models.

The single-impurity Anderson model describes the dynamics of a system of non-interacting itinerant electrons ($c_{k,\sigma}$) hybridising with a localised correlated impurity orbital (d_σ) [15]:

$$H_{\text{AIM}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + \sum_{k,\sigma} V_k (c^\dagger_{k,\sigma} d_\sigma + \text{h.c.}) + \epsilon_d \sum_\sigma n_{d\sigma} + U n_{d\uparrow} n_{d\downarrow} \ . \tag{3.18}$$

V_k controls the hybridisation strength, and ϵ_d and U control the onsite energy and local correlation respectively of the impurity site. In the limit of large U, the impurity dynamics gets projected onto its spin states, and the model then describes the scattering of non-interacting electrons off magnetic impurities, through the Kondo model [33]. Formally, the Kondo model is obtained by carrying out a Schrieffer-Wolff transformation on the Anderson impurity model and then using the condition to $V \ll U$ to drop all contributions beyond second order in the ratio V^2/U [34]:

$$H_{\text{Kondo}} = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1,k_2} \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c^\dagger_{k_1,\alpha} c_{k_2,\beta} \ , \tag{3.19}$$

where $\vec{S}_d$ is the impurity spin formed by projecting the Hilbert space of the impurity to the singly-occupied states.

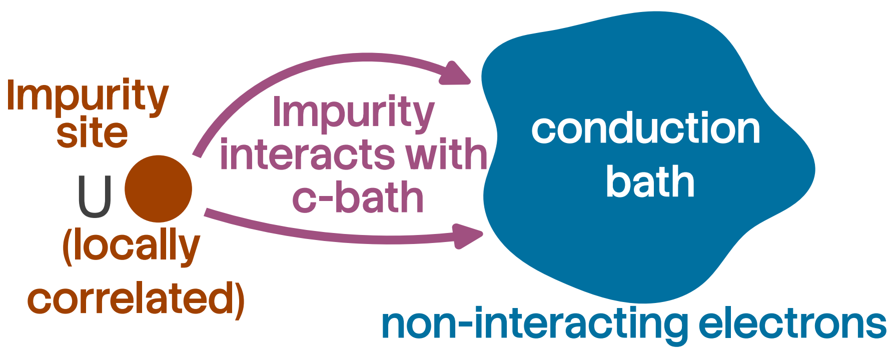

Figure 1.7: *Schematic of a quantum impurity model. The impurity is essentially a locally correlated site that presents a magnetic scattering center for the non-interacting itinerant electrons of the conduction bath. The impurity-bath scattering can appear in multiple forms, such as a one-particle hybridisation through which an electron simply hops between the two regions, or a spin-exchange Kondo interaction that emerges when the impurity site has been reduced to a local moment that can undergo spin-flip scattering with the conduction bath electrons.*

Even though the conduction bath is non-interacting, the presence of local repulsion on the impurity site makes the problem *many-electron* in nature: that is, the "U" term makes it impossible to study the electrons in isolation. This can be seen as follows: imagine two conduction electrons $|k_1 \uparrow\rangle$ and $|k_2 \downarrow\rangle$ trying to hybridise with an empty impurity site. The first electron moves onto the impurity site through a scattering process of the form $c^\dagger_{d\uparrow} c_{k_1\uparrow}$ and makes the impurity site single occupied ($|\uparrow\rangle_d$), but now the second electron k_2 *must pay a cost* of the order of U in order to hybridise with this singly-occupied impurity site through a process of the kind $c^\dagger_{d\downarrow} c_{k_1\downarrow}$. This shows that the conduction electrons *are aware* of the other electrons, and all of them must be treated together.

Another way of seeing this collective behaviour is from the Kondo model. Again imagine two spin-up electrons k_1, k_2 that are about to scatter off the impurity spin (which is presently down) via a *spin-flip* process. The first electron scatters and leaves the impurity spin up while it itself becomes down, but now the second conduction electron cannot spin-flip scatter because the spins cannot be exchanged (both are down!). This is another example of how the conduction electrons can *feel* the presence of the other electrons, and this is what makes the problem many-electron in nature.

Kondo screening Despite the interacting nature of the problem, the problem can be solved in an effectively exact fashion, via Wilson's numerical renormalisation group approach [35, 36], density-matrix renormalisation group approach [37] or the method of *Bethe ansatz* [38, 39]. The former solution is more transparent: for any non-zero value of the hybridisation V (or Kondo coupling J), the low energy physics of the impurity is described by a strong-coupling fixed point ($V = \infty$, or $J = \infty$), where the impurity forms a maximally-entangled singlet state with the local spin density of the conduction bath. The low-energy excitations about such a fixed point can be calculated from renormalised perturbation theory, and describe a local Fermi liquid – one particle excitations in the neighbourhood of the singlet renormalised by a local repulsion term [40, 41]. The crossover from the free local moment at high temperatures to the screened non-magnetic state at low temperatures occurs at an emergent energy scale T_K, the Kondo temperature, which depends exponentially on the strength of the Kondo coupling, and is the only energy scale remaining at low temperatures, so that all thermodynamic quantities must go as a function of T/T_K [35]. Unitary renormalisation group studies show that this crossover is marked by the crystallisation of entanglement between the impurity and the Kondo screening cloud, as well as among the electrons forming the cloud [42]. This entanglement also leads to a non-trivial low-energy theory for the excitations of the Kondo cloud, and it is these excitations that are responsible for the screening mechanism [42].

At half-filling (attained for $\epsilon_d = -U/2$), the only other fixed points are the free orbital fixed point at $V = 0, U = \epsilon_d = 0$ where the impurity site is freely occupied by two independent spins, and the local moment fixed point at $V = 0, U = \infty$, where the impurity site is occupied by a local moment that is decoupled from the bath [36, 43]. Away from half-filling, two other fixed points become available [36, 43]: a valence fluctuation fixed point at $V = 0, \epsilon_d = 0, U = \infty$ where the doubly-occupied state has been projected out from the symmetric free orbital fixed point, and a frozen impurity fixed point at $V = U = 0, \epsilon_d = \infty$ where the impurity site is left completely empty.

The crossover from local moment physics to Kondo screening as the system is tuned from high to low temperatures can also be seen from the impurity spectral function [44–47]. At low frequencies, the spectral function exhibits a sharp resonance around $\omega = 0$ that indicates the presence of gapless single-particle local Fermi liquid excitations (see Fig. 1.6). At higher frequencies, one sees satellite peaks that involve impurity excitations that cost energy of the order of U. The height of the central Kondo resonance is a topological invariant; it is fixed by the impurity occupancy owing to an application of the Friedel sum rule to impurity models [48], and remains unchanged as the interaction U is increased adiabatically while remaining at particle-hole symmetry. The topological nature can also be ascertained by identifying such a sum rule as the impurity model analogue of Luttinger's theorem [24, 25, 49] that relates the total number density of electrons to the number of Greens function poles within the Fermi volume, and which is itself topological [50, 51].

Breakdown of Kondo screening While the Anderson model (or the Kondo model derived from it) only display a strongly-coupled screened phase, it is possible to enhance it in order to introduce Kondo frustration which can eventually lead to a breakdown in the Kondo screening process. Such models are interesting because they provide a means of studying zero temperature *continuous* phase transitions and address experimental signatures of quantum criticality in correlated systems. The canonical example of this is the multichannel Kondo model, involving a single impurity hybridising with multiple independent conduction channels [52]:

$$H_{\text{MCK}} = \sum_{k,\sigma,\alpha} \epsilon_{k,\lambda} n_{k,\sigma,\lambda} + \sum_{k_1,k_2,\lambda} J_\lambda \sum_{\alpha,\beta} \vec{S}_d \cdot \vec{\sigma}_{\alpha,\beta} c^\dagger_{k_1,\alpha,\lambda} c_{k_2,\beta,\lambda} , \tag{3.20}$$

here λ is the channel index. In the case of an isotropic Kondo coupling across the channels ($J_\lambda = J$), the strong-coupling $J = \infty$ fixed-point of the single-channel Kondo model is replaced by an intermediate-coupling fixed-point at a finite value J^* [52, 53]. Powerful methods like boundary conformal field theory [54–59], bosonization [60–64], numerical renormalization group [56,65,66] and Bethe ansatz [67–71] have shown that the low energy theory of such models exhibit non-Fermi liquid excitations. Further, the system displays anomalous behaviour in thermodynamic properties near $T = 0$ and a fractional zero temperature entropy in the thermodynamic limit (upon ensuring that temperature is taken to zero after the system size is taken to infinity [63, 72]). This anomalous divergent behaviour is a signature of the fact that the MCK system with a single channel-symmetric exchange coupling is quantum critical: the ground state is susceptible to perturbations that introduce channel anisotropy in the exchange couplings [52, 56, 62, 70, 73]. The criticality arises from a degeneracy of the impurity states, and it can be shown that the anomalous features – non-Fermi liquid physics, diverging impurity contributions to thermodynamic quantities and the presence of fractional excitations – that appear at the critical point can be tied to this degeneracy [74].

Another class of models that shows Kondo breakdown is that of pseudogap Anderson and Kondo models, where the impurity hybridises with a conduction bath whose density of states (DOS) varies as $\rho(\omega) \sim |\omega|^r$ with a paramter $r > 0$ that controls how fast the DOS vanishes as the Fermi level is approached. Such behaviour can arise in semi-metals and in long-range ordered systems where the order parameter vanishes at certain points on the Fermi surface (such as $d-$wave superconductors). The pseudogap impurity models exhibit a boundary quantum phase transition between a Kondo-screened strong-coupling phase and an unscreened local-moment phase as a function of interaction strength or exchange coupling [75–78]. At the critical Kondo coupling, the impurity spectral function and spin susceptibility exhibit nontrivial power-law scaling with anomalous exponents determined by r [76, 79–82]. For $0 < r < 1/2$ (in the particle-hole symmetric case), both the local-moment and strong-coupling phases are stable, separated by a critical fixed point characterized by fractional residual entropy and nontrivial hyperscaling relations [76, 83]. For larger exponents, the strong-coupling phase may disappear entirely, and the system remains in the local-moment phase for all couplings [78].

The multi-impurity Anderson and Kondo models furnish another canonical realization of Kondo breakdown, driven by the competition between Kondo screening and inter-impurity interaction. The paradigmatic example is the two-impurity Kondo model (TIKM), in which two localized spins couple both to conduction electrons and to each other via an antiferromagnetic exchange K [84, 85]. In the limit of large $K > 0$, the impurities form a non-magnetic singlet decoupled from the conduction electrons and suppressing the Kondo resonance. This fixed point is separated from the strong-coupling Kondo screened fixed point through a quantum critical point [84–86] displaying logarithmic or power-law scaling in thermodynamic quantities and fractional residual entropy [87]. A modified version of this model that uses conduction bath-mediated inter-impurity interaction leads to a metallic spin liquid displaying nonlocal impurity-spin entanglement and fractionalized charge excitations [88]. More intricate geometries can be constructed by arranging antiferromagnetically coupled Kondo impurities along a triangle, each hybridising with its own conduction bath; such a model displays a topological phase transition from a symmetry-preserved gapless spin liquid phase with "topological order" to a screened Fermi liquid phase [89]. Indeed, recent experiments have validated such phase diagrams for the case of two impurities by studying pairs of copper atoms in the presence of strong magnetic fields [90].

1.3.4 Examples of materials with strong local correlations

As discussed in the previous section, the route to studying lattice problems via quantum impurity models makes most sense for materials with strong local correlations. Such systems are often referred to as strongly correlated quantum materials, and their phenomenology often lies beyond conventional approaches like density functional theory or perturbative weak-coupling methods. In practise, such materials often involve transition metal elements or rare earth/actinide elements, due to the presence of partially filled $3d-$ or $4f-$orbitals in such elements. This is because these orbitals are more localised around the nucleus than the $s-$ or $p-$orbitals of comparable energy, and hence lead to a reduced itinerant energy scale as well as amplify the effects of local correlations [32].

The localised nature of these orbitals can be understood as follows. For the $3s-$orbital, the $1s$ and $2s$ orbitals have same orbital angular moment l but lower principal quantum number n; since all such wavefunctions must be orthogonal to each other, the $3s-$ orbital must develop nodes in its radial form and extend farther away from the nucleus in order to be orthogonal to these other orbitals. However, for the $3d-$orbital, there is no orbital with same l but lower n; this means that the orthogonality is already enforced by the behaviour of the angular part (the spherical harmonics), and the radial part can remain localised around the nucleus without having to spread out. A similar argument holds for the localised nature of the $4f-$ orbital.

We now discuss the phenomenology of some of these quantum materials in order to identify the challenges in trying to explain them and the kind of models that are likely to be useful in this endeavour.

Transition Metal Oxides (TMOs) Transition Metal Oxides (TMOs) are defined chemically by the presence of partially filled d-shells in transition metals (e.g., Cu^{2+}, Mn^{3+}, V^{4+}) coordinated by oxygen ligands. The physics of TMOs is governed by the competition between the kinetic energy hopping amplitude t and the on-site Coulomb repulsion U; due to the small overlap between the localised $d-$orbitals, the $d-$electrons can delocalise only by hopping onto the ligand atoms. According to the Zaanen-Sawatzky-Allen (ZSA) classification scheme [91], these materials are categorized as either Mott-Hubbard insulators, where the charge gap is determined by the d-d Coulomb interaction ($E_g \approx U$), or Charge-Transfer insulators, where the gap is dictated by the energy difference between the oxygen p-band and the metal d-band. These derive from considerations of several factors, such as crystal field splitting, arrangement of atoms and lattice symmetries.

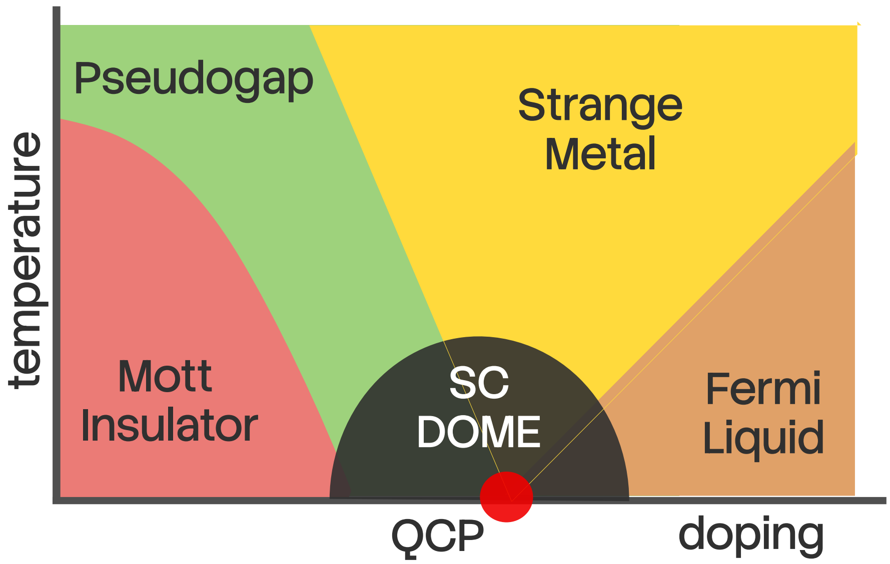

Figure 1.8: *Experimental phase diagram of the high-Tc copper oxide materials. The normal state at half-filling is an antiferromagnetic Mott insulator; doping it leads to a plethora of fascinating phases. Upon (here, hole) doping at low temperatures, the Mott insulator melts and gives rise to a d-wave superconductor with transition temperatures ranging upto 130K at ambient pressures. Further doping the system leads to a correlated Fermi liquid. At higher temperatures, two other regions are observed: the strange metal region with linear-in-temperature resistivity, and the pseudogap with partial gapping of the Fermi surface.*

For example, in the case of octahedral symmetry, repulsion from the surrounding atoms leads to the raising of energy of the $d_{x^2-y^2}$ and $d_{3z^2-r^2}$ orbitals forming the e_g doublet, while the other three orbitals (d_{xy}, d_{yz} and d_{zx}) form the t_{2g} multiplet. For the heavier TMOs (Ti, V, Cr...), the Fermi level lies within e_g. The increased positive nuclear charge (compared to the lighter elements) lowers the energy of the $d-$orbitals and brings them closer to the $p-$orbitals, reducing the charge transfer energy $\Delta = \varepsilon_d - \varepsilon_p$ of transporting an electron from a $d-$state to a $p-$state, compared to the $d-$level repulsion U ($|\Delta| < U$). Geometric reasons (the e_g orbitals point towards the $p-$orbitals) also enhance the hybridisation between $d-$ and $p-$orbitals in such materials. If Δ becomes larger than the bandwidth, we obtain a *charge-transfer insulator*, where the charge gap is decided by Δ. In the lighter TMOs, the Fermi level passes through t_{2g}, and because they point away from the $p-$orbitals, the hybridisation is weaker. Moreover, the reduced positive nuclear charge leads to an increased $\Delta > U$; a U greater than the bandwidth leads to what's called a *Mott-Hubbard insulator*.

Experimental studies of transition metal oxides (TMOs) have established metal–insulator transitions (MITs) as a defining consequence of strong electronic correlations in systems with partially filled (d)-electron shells [92, 93]. Early experiments on vanadates such as V_2O_3 characterised pressure-driven MITs through the pressure- and temperature-dependent conductivity as well as an anomalous enhancement of the optical conductivity [94–96]. Copper oxides constitute the most extensively studied class of correlated TMOs and provide a canonical example of a doping-driven MIT. Transport measurements on parent compounds such as La_2CuO_4 establish a robust insulating state, while carrier doping leads to anomalous metallic behavior characterized by linear-in-temperature resistivity and unconventional superconductivity [97–99]. Such metal-insulator transition is also reported from angle-resolved photoemission spectroscopy (ARPES) measurements in doped La_2CuO_4 and $Bi_2Sr_2CaCu_2O_8$, along with the emergence of pseudogap near the antinodes at high temperatures and a hole-to-electronlike Fermi surface reconstruction [100–102]. Nickelates ($LaNiO_2$, $La_3Ni_2O_7$, $La_4Ni_3O_{10}$, etc) exhibit related but distinct experimental signatures. With Ni^+ sharing the same $3d^9$ valence configuration as Cu^{2+} in the cuprate superconductors, these materials also display superconductivity, although the mechanism in the nicklelates is different due to the strong coupling between the layers [103,104]. The optimal doped regime in thin films shows strange metal behaviour, while infinite-layer nickelates show the absence of long-range antiferromagnetic order [105].

Transition Metal Dichalcogenides Transition Metal Dichalcogenides (TMDs) are layered van der Waals materials with the stoichiometry MX_2, where M is a transition metal (e.g., Mo, W) and X is a chalcogen (e.g. S, Se, Te) [106, 107]. Recent interest has focused on the interplay between strong spin-orbit coupling (SOC), electronic correlations and direct bandgap – properties that make them particularly useful for electronic applications, spintronics and DNA sequencing. Layered TMDs have the structure $X - M - X$, with the transition metal sandwiched between two hexagonal planes of chalcogenides. Different combinations of the metals with the chalcogens lead to widely varying structures and electronic properties in TMDs. Experimental studies show that metal–insulator transitions (MITs) in TMDs are commonly intertwined with charge density wave (CDW) formation rather than arising from purely on-site Coulomb localization [108, 109]. In prototypical compounds such as 1T-TaS_2, ARPES and transport measurements reveal fluctuating magnetic order parameters in the Mott state and a transition to superconductivity driven by pressure [110–112]. Scanning tunnelling microscope (STM) pulses have been successfully used to manipulate the metal–insulator transition of 1T-TaS2, demonstrating the macroscopic controllability of the Mott phase [113].

ARPES measurements in 2H-$NbSe_2$ and 2H-$TaSe_2$ exhibit partial gapping of the Fermi surface restricted to nested momentum regions and a temperature-dependent pseudogap similar to the cuprates [114, 115]. This is further supported by optical conductivity experiments that show spectral weight in the midinfrared region along with a low-frequency Drude contribution [116]. Transport measurements on single crystals of $TaSe_2$ reveal that disorder enhances superconductivity by suppressing other competing orders such as CDW [117]. Collectively, these experimental results establish TMDs as a distinct class of correlated materials in which reduced dimensionality, band narrowing, and lattice reconstruction cooperate to produce metal–insulator behavior beyond conventional band theory.

Iron-Based Superconductors The iron-based superconductors (FeSCs), discovered in 2008, constitute a broad family of layered materials that share a common structure: a layer of iron atoms joined by tetrahedrally coordinated pnictogen (P, As) or chalcogen (S, Se, Te) anions arranged in a stacked sequence; the layers are separated by alkali, alkaline-earth or rare-earth and oxygen/fluorine 'blocking layers' [118–121]. Doping or applying external pressure reveals coupled antiferromagnetic and structural transitions along with superconducting and nematic orders [122–124]. Electrical resistivity measurements in the neighbourhood of the quantum critical point show a crossover from a bad-metal behavior ($T-$linear resistivity) to a Fermi liquid behaviour (T^2-resistivity) [125]. Optical conductivity measurements indicate point to intermediate-to-strong electronic correlations [126].

ARPES measurements reveal a $k-$dependent superconducting gap and de Haas-van Alphen measurements indicate substantial quasiparticle mass enhancements, shedding more light on the correlated nature of the material [127, 128]. Strain-tuning measurements of the nematic susceptibility, and differential elastoresistance measurements have provided compelling evidence for the nematic transition and its surrounding fluctuations being the cause of the structural transition as well as the superconductivity [129, 130]. Neutron scattering and NMR measurements further reveal persistent low-energy spin fluctuations across wide regions of the phase diagram, supporting scenarios in which unconventional superconductivity emerges from the interplay of magnetic, orbital, and nematic degrees of freedom [131, 132].

Heavy-Fermion / Kondo-Lattice Systems Heavy-fermion materials are a class of strongly correlated electron systems typically realized in intermetallic compounds of rare earth metals (such as Ce, Sm and Yb) and actinides (such as U, Np and Pu), where localized f electrons hybridize with itinerant conduction bands [133, 134]. At high temperatures, these systems exhibit local-moment behavior governed by Curie–Weiss susceptibilities and incoherent transport, while at low temperatures the Kondo effect leads to the formation of a coherent many-body state with extremely large quasiparticle effective masses [43]. Experimentally, this crossover is most clearly manifested in transport and thermodynamic measurements: the electrical resistivity shows a characteristic maximum followed by a rapid drop signaling the onset of coherence, while the linear specific heat coefficient $\gamma = C/T$ reaches values orders of magnitude larger than in conventional metals, directly reflecting the heavy quasiparticle mass [135].

A defining feature of heavy-fermion materials is their proximity to magnetic quantum phase transitions, which can be accessed experimentally by pressure, chemical substitution, or magnetic field [136]. Near such quantum critical points (QCPs), transport measurements reveal pronounced non-Fermi-liquid behavior ($T-$linear resistivity) and a vanishing Hall response, along with a divergent effective quasiparticle mass as the transition is approached [137, 138]; these are all indications of the breakdown of electronic excitations near the transition. Neutron scattering experiments provide direct evidence for spin fluctuations down to very low temperatures, interplaying with the itinerant electrons at the QCP and potentially driving the superconducting transition [139, 140]. Quantum oscillation measurements through de Haas–van Alphen experiments further demonstrate Fermi surface reconstruction across the quantum critical point, highlighting the drastic change in Fermi volume [141]. Together, these experimental observations establish heavy-fermion compounds as paradigmatic systems in which strong local correlations, hybridization-driven coherence, and quantum critical fluctuations conspire to produce emergent low-energy quasiparticles and unconventional metallic behavior [142].

1.4 Phenomenology of Mott transition in lattice models

1.4.1 The Infinite-Dimensional Hubbard Model

The Hubbard model (introduced in eq. 1.1) presents one of the most fruitful playgrounds for investigating correlation-driven metal-insulator transitions and its various facets. This model, despite its conceptual simplicity, exhibits a rich variety of phenomena, including antiferromagnetism, superconductivity, and, most notably for our purposes, the correlation-driven Mott metal–insulator transition (MIT) [4, 143–145]. While exact solutions exist in the limits of $d = 1$ [146] and $d = \infty$ [147], the problem remains largely open in more realistic situations such as two and three spatial dimensions.

As discussed earlier, the limit of infinite dimensions presents a drastic simplification for the manybody scattering processes within the problem: the one-particle $k-$space self-energy becomes momentum-independent due to the vanishing of all non-local contributions (they scale as negative powers of the coordination number, which diverges in the limit of $d = \infty$). This simplification lies at the heart of dynamical mean-field theory and its ability to capture the Mott metal–insulator transition at both zero and finite temperatures for this limiting case [18, 147–151]. At half-filling and zero temperature, DMFT predicts a continuous transition as a function of U wherein the quasiparticle weight Z tends to zero at a critical interaction strength U_{c2}, signaling the vanishing of coherent Fermi liquid behavior and the opening of a Mott gap in the single-particle spectral function [18, 149].

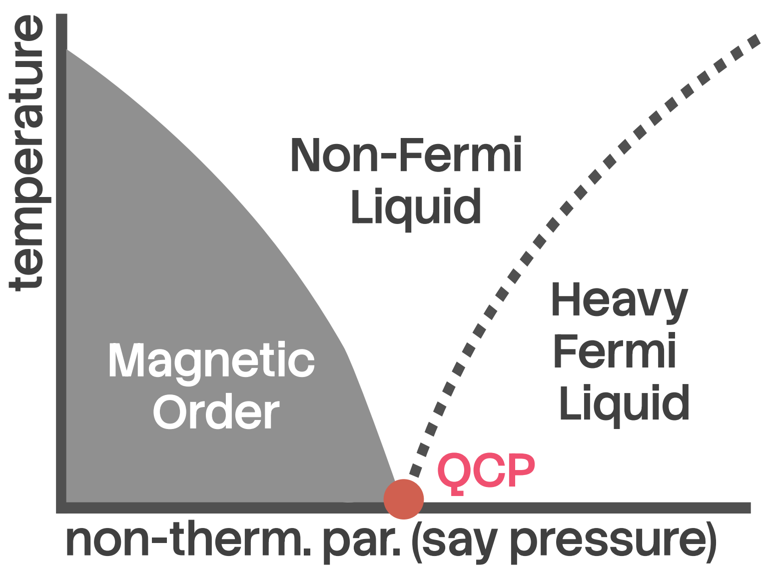

Figure 1.9: *Experimental phase diagram of the Kondo-breakdown class of heavy fermions compounds, as a function of the inter-layer hybridisation. For small hybridisation, the correlated $f-$layer remains decoupled from the itinerant layer in the ground state and undergoes magnetic ordering. For larger values, the $f-$layer undergoes Kondo screening by the itinerant layer, forming inter-layer entangled states. Lying in between these states is a quantum critical point with strange metallic behaviour arising from it at finite temperatures. The transition across the QCP at zero temperature is characterised by a change from a small to a large Fermi volume.*

The three-peak structure of the interacting spectral function, with a central quasiparticle resonance flanked by incoherent lower and upper Hubbard bands, indicates the presence of highly renormalised Landau quasiparticles [149]. A doping-driven continuous metal-insulator transition also exists for this model at zero temperature, marked by the vanishing of electronic compressibility at a critical value of the chemical potential [152]. The quantum critical point itself (at $T = 0$) is likely to be an exotic state, and important questions about it remain unanswered. *Is it possible to obtain a low-energy theory for the local gapless excitations precisely at the MIT, where the metal is on the brink of destruction? How do these excitations compare with those of the local Fermi liquid, e.g., in terms of self-energies and two-particle correlation functions?*

While the DMFT solution captures the zero temperature transition accurately, the self-consistently determined nature of the conduction bath renders the solution somewhat opaque and makes it difficult to write down an effective Hamiltonian for the low-energy theories. Since the simplest version Anderson impurity model does not contain the mechanism to stabilise a localised phase over a finite-sized parameter regime, an important question arising from the DMFT solution is: *what are the minimal changes that one must implement into the Anderson impurity model in order to capture such a metal-insulator transition (*$1/2$*-filled Hubbard model on the Bethe lattice in* $d = \infty$*), in sufficient accuracy and precision?* Finding such an impurity model would also reduce the considerable computational effort that is presently required in self-consistent approaches.

At finite temperatures, the interaction-driven as well as doping-driven Mott transition become first order, and is characterized by a region of phase coexistence and hysteresis [150, 153]. Two spinodal lines, $U_{c1}(T)$ and $U_{c2}(T)$, emerge; within this region metallic and insulating DMFT solutions coexist, and responses such as resistivity show a significant discontinuity [154]. These lines meet at a finite-temperature critical endpoint (U_c, T_c), above which there is no sharp distinction between metallic and insulating solutions and only a crossover exists between the Mott insulator and the Fermi liquid through a bad-metal regime, as the chemical potential is tuned [150,152,155–157]. The coexistence of metallic and insulating phases shows that the insulating solution is present within the many-body spectrum of the metallic phase. This naturally leads to the following question: *since a quantum impurity model lies at the heart of the self-consistency procedure, what does the presence of coexistence region mean for the Hamiltonian dynamics and the spectrum of this impurity model? Can the impurity model Hamiltonian display the emergence of the insulating state prior to the transition, through changes in the eigenstates?* Analytic insight of this kind is particularly important because approaching a coexistence region numerically is often fairly tricky.

The bad metal regime displays anomalous linear-in-temperature resistivity which corresponds to quantum critical ω/T scaling, along the quantum widom line [158, 159]. This is accompanied by the disappearance of the Drude peak in optical conductivity, signalling the loss of quasiparticle transport [158]. Similar quantum critical scaling is also obtained from DMFT solutions at very low temperatures, hinting at a "hidden" quantum criticality that is continuously connected to and responsible for the critical behaviour above the finite temperature endpoint [160]. Luttinger's theorem, which relates the poles of the Greens function within the Fermi volume to the number of electrons in the system at zero temperature, is satisfied within the Fermi liquid phase but is violated in the particle-hole asymmetric Mott insulator by a universal magnitude [161]. The emergence of a quantum critical regime at high temperatures and its connection to similar signatures at low temperatures suggests the presence of quantum fluctuations in the underlying impurity model that modify

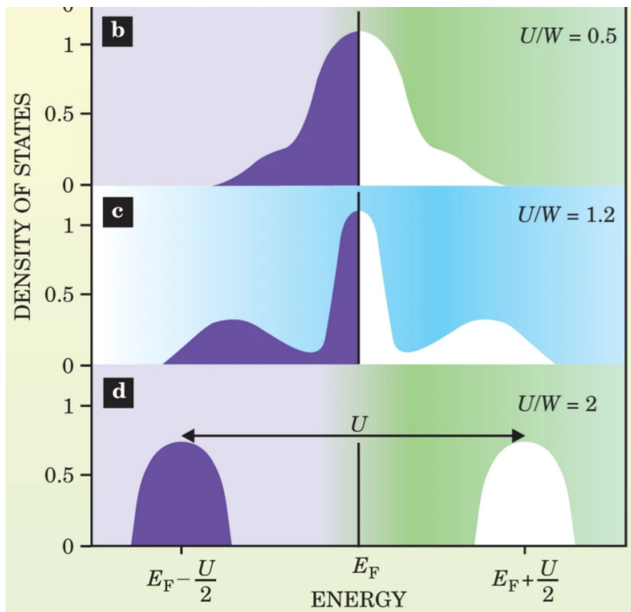

Figure 1.10: *Schematic depiction of the evolution of the local spectral function of the infinite-dimensional Hubbard model as obtained from dynamical mean-field theory (DMFT). At small values of the Hubbard interaction U, the spectral function has a central resonance, indicating good metallic behaviour. For larger values, the central resonance splits and satellite peaks emerge, indicating loss of some spectral weight from zero frequency. Beyond a critical value, the central peak disappears and only the satellite peaks remain, leading to a gapped Mott insulating state.*

the impurity-bath scattering processes and the associated impurity responses. One can therefore ask: *What are the fluctuations that destroy the metal and lead to the insulating phase? Can we obtain a universal theory for these competing tendencies?*

The above studies of single band models describe the Mott transition to be of the Mott-Hubbard kind (described in previous sections), where the size of the gap is determined by the double-occupancy cost U. In order to obtain transition of the charge-transfer kind, one needs to consider at least a two-band model. Indeed, in a non-degenerate two-band Hubbard model with a Charge-Transfer energy Δ for hopping from a $d-$orbital to a $p-$orbital, one sees a Mott-Hubbard transition when $U < \Delta$ and a charge-transfer transition when $U > \Delta$ [162]. The degenerate two-band model with inter-band local correlation shows successive Mott-Hubbard transitions at low temperatures whenever the filling passes through integer values and the interaction strength is intermediate [163].

In the later chapters of this thesis, we will be addressing several of these questions.

1.4.2 The two-dimensional Hubbard model

The 2D Hubbard model is one of the most well-studied models in condensed matter physics, particularly because of its paradigmatic role in Mott metal-insulator transition and its connections to the physics of copper oxide materials and $d-$wave superconductivity. Photoemission experiments reveal the presence of p and d bands close to the Fermi surface in such materials [100]; further work by Zhang and Rice showed that a specific combination of these orbitals governs the low-energy physics, leading to an effective description in terms of a single band Hubbard model [164]. The discovery of high-temperature superconductivity in nickelates and organic charge transfer salts has provided more platforms where the Hubbard model and its extensions can prove to be useful [165, 166].

While simple in appearance, the model has proved to be notoriously hard to solve except in highly specific parameter regimes, despite several decades of computational effort. This is due to several reasons [145]: (i) the presence of closely-lying energy scales arising from competing tendencies, (ii) the inability of several methods to either reach zero temperature or extrapolate to the thermodynamic limit, (iii) the likelihood of approximate/finite-size methods methods to overestimate tendencies towards spontaneous symmetry breaking, and (iv) the presence of differing phases separated by slow crossovers without any sharp boundaries.

At half-filling, the 2D Hubbard model has a spectral gap to charge excitations for any non-zero interaction strength [167–171]. The behaviour below the charge gap is described by a nonlinear sigma model with parameters that depend on the interaction strength U [172]. Upon tuning to finite temperatures, the system goes through a sequence of crossovers from the insulating ordered state to a Fermi liquid state with coherent quasiparticles into a non-Fermi liquid state with partial gapping of the Fermi surface [169, 171, 173]. The excitations of the non-Fermi liquid are described by a marginal Fermi liquid theory, and the Mott insulator displays topological order in the absence of spontaneous symmetry breaking [174]. *Capturing the momentum dependence of static and dynamical quantities in sufficient detail remains an open challenge in such problems, since results from cluster methods suffer from sensitivity to periodisation schemes [175].*

Away from half-filling but at weak coupling, the Neel order is accompanied by incommensurate magnetic orders (also called spiral magnetic state) with wave vectors away from (π, π), as seen from mean-field studies [176, 177]. Functional RG studies indicate that such solutions can exist up

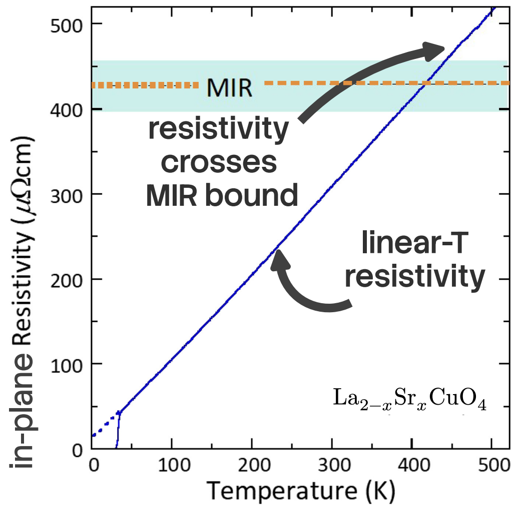

Figure 1.11: *Resistivity measurement in the strange metal phase of the overdoped cuprates. There are two striking features. The first is the linear dependence of the resistivity on temperature, which cannot be explained from Landau's Fermi liquid theory. The second is the unbounded growth of the resistivity beyond the Mott-Ioffe-Regel (MIR) limit as temperature is increased. For a system of electrons on a lattice, the minimum possible mean-free path for propagating electrons is the lattice spacing, and this sets an upper bound for the resistivity, which is the MIR bound. Resistivity crossing the MIR bound implies that electrons are not the propagating degrees of freedom in such a system.*

to fairly high doping if the superconducting instability is suppressed [178, 179]. Superconductivity with $d-$wave pairing has been achieved from renormalised perturbation studies as well as from diagrammatic Monte Carlo (diagMC), functional RG (fRG) calculations and unitary RG calculations [180–183]. More intricate competition between the superconducting and stripe states is found in three-band Hubbard models from variational Monte Carlo, cluster DMFT and DMRG calculations [184–186]. *Further work is necessary to resolve the closely-lying phases in such models.*

The phenomena of pseudogap (PG), where the Fermi surface acquires a momentum-dependent suppression of spectral weight, has also been observed above a certain temperature from methods like diagMC and fRG [171, 187]. The gap is seen to open first near the antinodal point $(\pi, 0)$ and spread across the Fermi surface until it reaches the node $(\pi/2, \pi/2)$. The mechanism of the pseudogap at weak to intermediate coupling is found to be long-range antiferromagnetic (AF) correlations [188]. At underdoping, the pseudog is marked by a dominant charge gapping of the antinodes accompanied by strong spin-nematic fluctuations and sub-dominant superconducting phase fluctuations [183]. The pseudogap disappears at large doping when the Fermi surface undergoes a topological transition from hole-like to electron-like. At strong coupling, the competition between various phases such as $d-$wave superconductivity, antiferromagnetic order and stripe states (spin/charge order modulating in real space) is an active subject of study. Recent studies using various methods like DMRG and iPEPS found that the period 8 charge stripe state is lower in energy than the superconducting state at strong coupling $U/t = 8$ and no next-nearest-neighbour hopping [189]. This was later corroborated by studies from variational Monte Carlo and other DMRG studies [190, 191]. *It is not yet clear whether the stripe period gets lowered as the interaction strength is increased.*

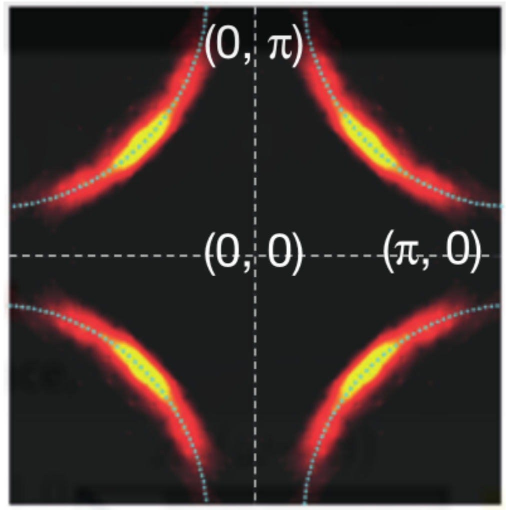

Figure 1.12: *ARPES measurement of the Fermi surface in the pseudogap of underdoped cuprates. The Fermi surface shows partial gapping around the antinodes and gapless behaviour around the nodes. Further doping leads to a reconstruction of the Fermi surface into the strange metal.*

At weaker coupling, the above methods report a reduced stability for the strip state and increased tendency to realise a $d-$wave SC state [181, 192]. *Further investigation is necessary to map the crossovers between stripe and superconducting phases as doping is tuned. Moreover, the precise parameter regime in which the $d-$wave superconducting state becomes more stable than the stripe states is still an open question.*

At strong-coupling, the pseudogap is obtained below optimal doping and has been studied using dynamical methods such as cluster DMFT (DCA, dynamical cluster approximation) and cellular DMFT (CDMFT) [193–196]; in contrast to the weak-coupling feature (where the PG was borne from long-range AF fluctuations), it is believed that the momentum-space differentiation in this regime is brought about by the proximity to the Mott insulator [195]. Conflicting indications are however provided by a fluctuation analysis that finds that long-lived long-ranged spin-fluctuations at the antinode are responsible for the opening of the PG [197]. The PG is found to be accompanied by non-Fermi liquid behaviour in the one-particle self-energy [194]. The doping value at which the non-Fermi liquid first appears was found to be lower for electron-doping than hole-doping [198]. At half-filling, DCA calculations reveal that the extent of the PG decreases as the next-nearest-neighbour hopping strength is increased [199]. Moreover, while DCA+continuous-time auxiliary field methods do not show any effect of van Hove singularities on the generation of the PG [199], DCA+determinantal Monte Carlo approaches identify the proximity to van Hoves as favouring the formation of a PG [200]. *Methods with better momentum-space resolution is necessary in order to identify the role of nesting and van Hoves on the pseudogap.*

The question of whether superconductivity can arise from repulsion was one of the primary motivations for studying the Hubbard model. DiagMC calculations reveal superconductivity in the ground state at weak-coupling and moderate doping, and with multiple symmetries of the gap [181, 201]. For higher couplings (which is relevant to the cuprates) and moderate doping, stripe states take over. Quantum Monte Carlo and DCA calculations suggest a $d_{x^2-y^2}$ pairing symmetry [168, 202]. DCA calculations on the 2D Hubbard mode suggest low-frequency spin fluctuations as the pairing glue for the superconductivity [203]. Unitary RG studies show the presence of preformed Cooper pairs near the antinodes at finite temperature states above the superconducting dome [183]. *The description of how various parent phases such as the pseudogap and strange metal precipitate superconductivity, in terms of the nature of fluctuations and the appropriate degrees of freedom, is still an open question..* Apart from superconductivity, several strongly correlated materials also exhibit strange metallicity, with transport properties that defy Fermi liquid theory, such as a linear-in-temperature resistivity that extends across large regimes of temperature. Within the Hubbard model, evidence for such $T-$linear resistivity and robustness across temperatures is available from CDMFT and determinantal Monte Carlo calculations [204, 205]. Studies using the unitary renormalisation group method reveal a two electron-one hole composite composite as the dominant excitation within the strange metal [183]. *More reliable methods are needed to access and understand the intermediate temperature regime of strange metals [145, 206].*

In the later chapters of this thesis, we will be addressing several of these questions.

1.4.3 Theories of heavy fermion materials

Heavy fermion materials represent the canonical environment for studying quantum phase transitions in correlated systems, owing to the explicit observation of quantum critical points at the boundary of antiferromagnetic order [207]. These systems involve the coexistence of magnetic moments (arising from localised $4f-$ and $5f-$electrons) and itinerant conduction electrons. The ground state of such a system is dictated by the competition between two kinds of interactions: internal spin-spin interaction J_{RKKY} between localised moments in the $f-$layer, and interaction J_K between $f-$electrons and conduction electrons [208].

Analogous to the screening of a single moment by a conduction bath, the lattice of $f-$moments are prone to undergoing Kondo screening by the conduction electrons through the inter-orbital hybridisation J_K. The local Fermi liquid excitations associated with the spins overlap with each other and become delocalised, leading to a paramagnetic heavy Fermi liquid with highly renormalised quasiparticle mass [209]. Because of the delocalisation of the excitations of the $f-$spin through Kondo screening, the number of carriers increases to $1 + x$ compared to x coming purely from the conduction electrons; this is commonly referred to as the large Fermi surface phase [210]. Indeed, a perturbation theoretic calculation in the on-site interaction strength of the $f-$layer allows one to rewrite the two-orbital model as two decoupled quasiparticle bands with dispersions renormalised by phenomenological parameters like the quasiparticle residue [43]. These phenomenological parameters can be estimated from mean-field approaches [211, 212]. While the temperature T^* at which coherent Landau quasiparticles can proliferate was found to be generically lower than the single impurity Kondo temperature T_K [213], T^* was found to be higher than T_K near half-filling of the conduction layer [214]. At half-filling in both layers, the chemical potential passes through the middle of the gap between the two quasiparticle bands, and the system is a Kondo insulator [215]; for bipartite lattices, the ground state can be generally shown to be unique and a singlet ($S = 0$) [216–218].

Tuning the $f-$layer J_{RKKY} to large values compared to the inter-layer hybridisation can bring forth a quantum phase transition into a different ground state [219]. Experimental evidence suggests that this transition usually falls into one of two classes, depending on the behaviour of the Kondo singlet binding energy E^* in the paramagnetic phase. If E^* remains non-zero as the QCP is approached from the heavy Fermi liquid into the ordered state, the transition is said to fall in the SDW type, because it can be described as an instability of the heavy quasiparticles into a spin-density-wave ordered phase [220–222], in the form of a quantum action that involves the spatial and temporal fluctuations of an order parameter. $CeCu_2Si_2$ is a prototypical material that exhibits quantum criticality of the SDW type [134]. Because of the non-zero binding energy of the singlet state at the QCP, heavy quasiparticles survive across the transition and the Fermi surface is expected to evolve continuously, along with the order parameter [223, 224].

On the other hand, if E^* vanishes at the QCP, the transition belongs to the Kondo breakdown type, which is not compatible with the spin-density-wave scenario [225]. This happens when gapless spin fluctuations within the $f-$layer compete with the Kondo spin-flip processes and lead to the destruction of Kondo screening on the Fermi surface [136]. This is tracked by the vanishing of the quasiparticle residue of the heavy Fermi liquid as one approaches the QCP from the paramagnetic phase [82, 224]; this in also implies the critical non-Fermi liquid nature of the large Fermi surface at the transition [226]. *A complete understanding of the Kondo breakdown class of QCP is still missing in the literature.*

Due to the destruction of Kondo screening across the QCP in this class of materials, the localised $f-$spins cease to contribute to the Fermi surface away from the paramagnetic phase, and the Fermi surface shrinks discontinuously from large $(1 + x)$ to small (x) across the QCP [227]. *Such a change in the Fermi volume violates Luttinger's theorem which equates the particles density to the Fermi surface volume, and Oshikawa's topological flux-tuning argument suggests the presence of additional degrees of freedom that respond to the flux while not being part of the Fermi surface. Additional work is needed to identify these degrees of freedom.* Materials like $YbRh_2Si_2$ and $CuCu_{6-x}Au_x$, among others, appear to exhibit behaviour that falls in this class [228, 229]. As mentioned before, several features of this class of experiments are not compatible with the SDW instability transition of the heavy Fermi liquid; these include the jump discontinuous change in Fermi volume mentioned previously (and an associated discontinuous change in the hall conductivity) [230], vanishing of the scale T_{FL} (associated with the onset of Fermi liquid behaviour) and the divergence of the quasiparticle mass upon approaching the QCP from the paramagnet [225], a universal linear-in-temperature strange metal behaviour of the resistivity in an intermediate finite temperature quantum critical regime above the QCP [228, 231] and the possible lack of magnetic ordering immediately away from the QCP [232, 233]. *At present, the connection of the physics of heavy fermion models like the periodic Anderson model to strange metal behaviour is an open question; the mechanism for generating such behaviour from the QCP remains to be completely understood.*

A fairly comprehensive model for describing the phase transition in heavy fermion systems is the periodic Anderson model (PAM):

$$H_{\text{PAM}} = \varepsilon_f \sum_{i,\sigma} n_{i,f,\sigma} + U_f \sum_i n_{i,f,\uparrow} n_{i,f,\downarrow} + \sum_{k,\sigma} \epsilon_k n_{k,c,\sigma} + V_\perp \sum_{i,\sigma} \left(c^\dagger_{i,f,\sigma} c_{i,c,\sigma} + \text{h.c.} \right) , \qquad (4.1)$$

where the subscripts f and c in the operators indicate whether they belong to the $f-$layer or the conduction electron layer. At the Kondo breakdown QCP and in the RKKY-dominated phase, the

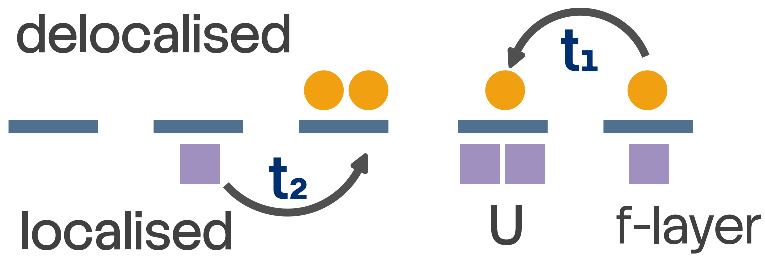

Figure 1.13: *Schematic model of heavy fermions. There are two layers: one uncorrelated delocalised layer (top), and a correlated localised layer (bottom). The delocalised layer has very weak correlation and a large bandwidth, leading to small masses for the carriers. The localised layer has very strong local repulsion leading to extremely narrow bands and highly correlated hopping for the carriers. These two layers hybridise with each other through hopping or spin-exchange processes. This results in a modification of the properties of the weakly correlated layer by the strongly-correlated layer.*

inter-layer hybridisation is expected to be irrelevant and the $f-$ and conduction electrons fail to bind into a Kondo singlet in the ground state [234, 235]. Several numerical and analytical methods have been applied to investigate various parts of the phase diagram and probe varying aspects of the phenomenology. *However, many of these methods suffer from a lack of predictions of various transport measures like strange metallicity, and they often suffer from insufficient resolution in frequency or momentum space.*

One of the major approaches applied to this problem is that of extended DMFT (EMDFT), in which one maps the PAM to a self-consistently determined Bose-Fermi-Kondo model [82, 236]. One of the smoking gun signatures of quantum criticality is the scaling of dynamical properties as ω/k_BT due to temperature being the only remaining energy scale in the system; this is captured from EMDFT calculations [224]. Numerically exact unbiased methods like DMRG and auxiliary field quantum Monte Carlo demonstrate the abrupt change in Fermi surface volume across the transition [237, 238]. Parton methods that implement a slave-particle representation followed by a $1/N$ expansion reveal jump in the conductivity across the transition, with the small Fermi surface arising from a spin-charge separation of the $f-$electrons into a fractionalised Fermi liquid phase [239, 240]. *Currently, a complete description of the Kondo breakdown driven transition that captures the behaviour of both the heavy Fermi liquid and the RKKY-dominated phase is missing.*

In the later chapters of this thesis, we will be addressing several of these questions.

1.5 An alternative perspective on metal-insulator transitions: Entanglement renormalisation and holography

Role of entanglement in fermionic systems The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [241–246]. Being non-local by nature, it is helpful in tracking correlations among particles across both small and large distances, giving valuable insight into the nature of the ground state and the low-energy excitations. Entanglement, in the form of various measures like entanglement entropy and mutual information, is the engine that gives rise to new low-energy degrees of freedom from a renormalisation group (RG) perspective, and is therefore key to understanding and characterising phase transitions and criticality in quantum matter. For instance, entanglement entropy is at the heart of understanding the so-called topological states of matter that can be described in terms of a purely topological effective Lagrangian and have a robust ground state degeneracy and gapless edge states [247–267]. In contrast to systems with local interactions whose subsystem entanglement entropy scales with the area of the subsystem [243, 268, 269], topologically ordered systems have an additional topological term dubbed the topological entanglement entropy [270–288] that quantifies the long-ranged nature of the entanglement in such systems. The non-local nature arises from the fact that the ground state of such a system cannot be made separable (non-entangled) by the application of purely local unitary transformations [245, 289, 290]. Although the topological piece of the entanglement entropy is subdominant compared to the non-topological piece that scales with the subsystem area, it can be isolated by considering the *tripartite information* I_3 of a specific arrangement of subsystems [277,280]. More generally, all multipartite information I_N beyond mutual information ($N \geq 3$) can be expressed as the product of the topological entanglement entropy and the Euler characteristic of the chosen arrangement of subsystems [291].

More pertinent to the work at hand, however, is the entanglement in free fermionic systems. In this thesis, we will consider relativistic Dirac fermions in two dimensions, both massless and massive. They emerge in several condensed matter systems [292], appearing at the surfaces of three-dimensional topological insulators [293–296], quasi-2D organic materials [297–299] and in arficial materials like cold atoms, molecular and microwave crystals [300–306]. For massless critical systems (or conformal field theories (CFTs)) living in one spatial dimension, the entanglement entropy of a single subsystem of length l scales as $\sim \frac{c}{3} \ln l$ [307–316], where c is the conformal central charge of the CFT. The case of multiple subsystems is more involved, but has been solved [314, 317, 318]. For massive fermions, the entanglement entropy (or rather the entropic $c-$function, which is the spatial derivative of the entropy) can be expressed in the form of differential equations which need to be solved numerically [317, 319–321]. The solutions lend themselves to small and large mass expansions, reducing to $\frac{1}{3} \ln \frac{l}{\epsilon} - \frac{1}{6} (ml \ln ml)^2 + O((ml)^2)$ for the case of small ml, and to $\sim (ml)^{-1/2} e^{-2ml}$ for that of large ml, both for a single interval. The $\ln l$ behaviour is a specific case of the more general $l^{d-1} \ln l$ violation of the area law, as observed in $d-$dimensional fermionic critical systems [317, 322–336]. This violation arises from the non-local nature of the real-space quantum fluctuations that exist in quantum critical fermionic systems.

While there is no geometry-independent subdominant term in the entanglement entropy of free fermionic systems [337,338], it *is* possible to generate such a term by placing the system on a manifold with a non-trivial topology, say a cylinder, and inserting a magnetic flux through the cylinder [337, 339–342]. For a massless Dirac fermion, if there are ϕ number of electronic flux quanta piercing the system, the additional term that is generated in the entanglement entropy of a subsystem of sufficiently (as defined in Ref. [337]) large length is of the form $\frac{1}{6} \ln \left|2 \sin \frac{\phi}{2}\right|$ [337, 340]. There is evidence from numerical computations on discrete models that such a term is a signature of the specific field theory and encodes universality [342–344]. In this thesis, we put this idea on a stronger analytical footing by relating the flux-dependent term to a topological invariant of the field theory, its Luttinger volume [24, 25, 50, 51, 345–347]. Given by the number of $k-$states enclosed by the Fermi surface, Luttinger volume V_L has proved to be one of the most important indicators that distinguish metals from insulating states of matter. In metals, V_L is constrained by the number of particles in the system, following Luttinger's theorem [24, 25, 50]. As shown most prominently in Ref. [50], Luttinger's theorem depends crucially on the presence of extended states in the metal that can respond to flux insertion experiments. We demonstrate in Chapter 7 that indeed it is this subdominant term in the entanglement entropy that carries this dependence and hence encodes Luttinger's theorem. It is known that for strongly-interacting states like Mott insulators or quantum hall states, V_L is modified due to changes in the topology of the Fermi surface [51, 345–347]. In this context, we relate the topological invariant on the metallic side (Luttinger volume) to the topological invariants (the Chern numbers) in the integer quantum hall system that the metal would lead to when placed in a perpendicular magnetic field. Such a relation makes explicit the transmutation of the extended states into the localised states of the insulating phase.

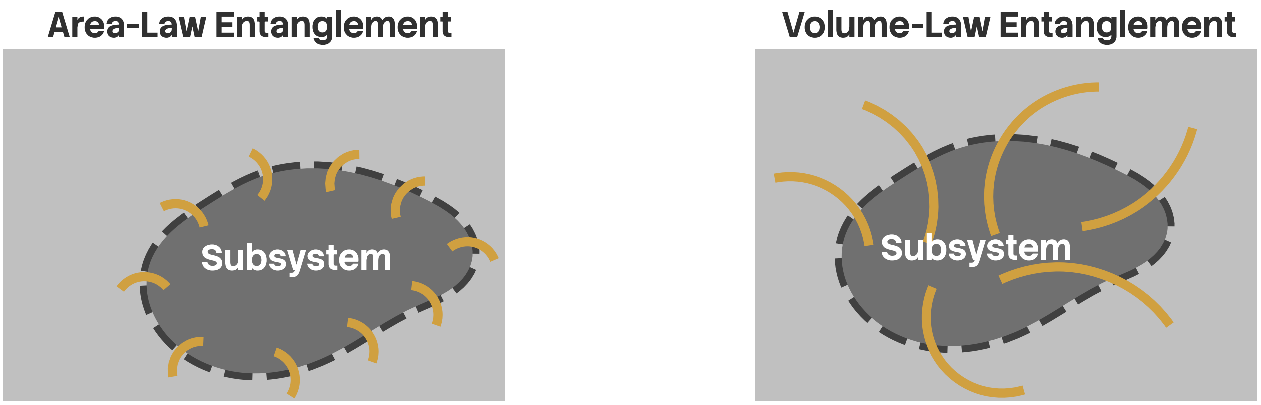

Figure 1.14: *Two classes of entanglement found in quantum systems, grouped by how entanglement measures scale with subsystem size. In systems with a spectral gap (left), correlations decay exponentially with distance, and only the degrees of freedom on the boundary of the subsystem entangle appreciably with the rest of the system. As a result, the entanglement entropy of the subsystem scales with the area L^{d-1} of the subsystem: $S_{EE} \propto L^{d-1}$. In contrast, for gapless systems, correlations are more long-ranged and may even span the entire system (for critical states). In such cases, degrees of freedom beyond the proximity of the boundary can entangle with the rest of the system, and such systems can achieve modified area-law $S_{EE} \propto L^{d-1} \ln L$ or even volume law $S_{EE} \propto L^d$*

.

Entanglement hierarchy and the holographic principle The leading geometry-dependent piece is also found to have intriguing properties of its own. The fact that the free fermion Lagrangian is diagonal in momentum space means that the entanglement arises purely from real space quantum fluctuations, and this allows us to study the entanglement of subsystems of varying sizes in momentum space. We find that the entanglement entropy hence obtained has a Russian doll-like nested structure, such that increasing the size of the subsystem in momentum space gradually reveals layers of entanglement. In fact, we argue that such transformations between the subsystems amount to a renormalisation group (RG) flow in the space of the field theories with varying spectral gaps, and the above-mentioned hierarchy in the entanglement entropy then stretches across the RG transformations. The precise prescription for choosing the subsystems is described in Chapter 7. We also show, simultaneously, that the hierarchy holds across multipartite entanglement measures, leading to an onion-like structure where increasing the number of parties in the entanglement measure (mutual information, tripartite information, 4-partite information and so on) leads to additional entanglement that did not exist before. More importantly, we interpret the hierarchy of entanglement along the RG flow as the emergence of an additional dimension [348–350], allowing us to define a measure of distance using some non-negative measure of entanglement, like the mutual information [351–354]: the more the entanglement, the less will be the distance between those points. Generation of an additional dimension is, of course, the essence of the AdS/CFT conjecture or, more generally, the holographic principle.

The holographic principle [355–363], also known as the gauge-gravity duality, posits that there exists a duality relation between a quantum field theory with a gauge field, and a gravity theory having one additional spatial dimension [364–367]. The additional dimension can be visualised as a renormalisation group flow that starts from the boundary field theory and extends into the bulk [368–377]. The most popular realisation of the holographic principle is the AdS/CFT correspondence that links a super-symmetric gauge theory with a string theory [378–380]. The essence of holography is that all the information about the bulk gravity theory is already contained within the boundary gauge theory; this idea is very similar to the discovery by Bekenstein and Hawking [381–383] that the entropy of a black hole is determined by its surface area. In fact, Ryu and Takayanagi proposed a generalisation of this result where the entanglement entropy of subsystem region a CFT is determined by the area of the minimal surface that bounds the region in the bulk [384, 385]. The AdS/CFT correspondence happens to be a strong-weak duality - it relates a strongly coupled CFT to a theory of weak classical gravity. This allows physicists to use the holographic principle and convert a problem of strong quantum correlation to one of classical gravity which can be solved using well-known methods [358, 359]. This has led to new insights in various topics of high energy and condensed matter physics, such as the viscosity of the quark-gluon plasma [386], the response functions of the Bose-Hubbard model [387, 388], and the phenomenology of non-Fermi liquids [389–392], holographic superconductors [393–398], striped phases [399–401] and holographic Fermi liquids [402–406]. An overview of the metallic states (phases that satisfy Luttinger's theorem) and their connections to the gauge-gravity duality have been provided in Refs. [407–409].

Of course, on the flip side, the holographic principle also implies that studying the boundary conformal theory can provide insights into the nature of the bulk quantum gravity theory. This however requires constructing, *ab initio*, the emergent dimension by starting from the field theory, and has proved to be more difficult than the converse, mostly because the field theories are often strongly coupled. Nevertheless, there exist some examples of constructive attempts towards a bulk gravity theory [346, 347, 353, 354, 410–443]. Some well-known approaches include the momentum shell renormalisation approach of Sung-Sik Lee [423–426] and the multi-scale entanglement renormalisation ansatz (MERA) of Vidal [427–431]; both use certain forms of RG flow to generate the extra spatial dimension. The former integrates out high momentum degrees of freedom in favour of generating dynamics for the boundary gauge field, with the RG trajectory variable giving rise to the additional spatial dimension [357]. On the other hand, the MERA works by disentan-

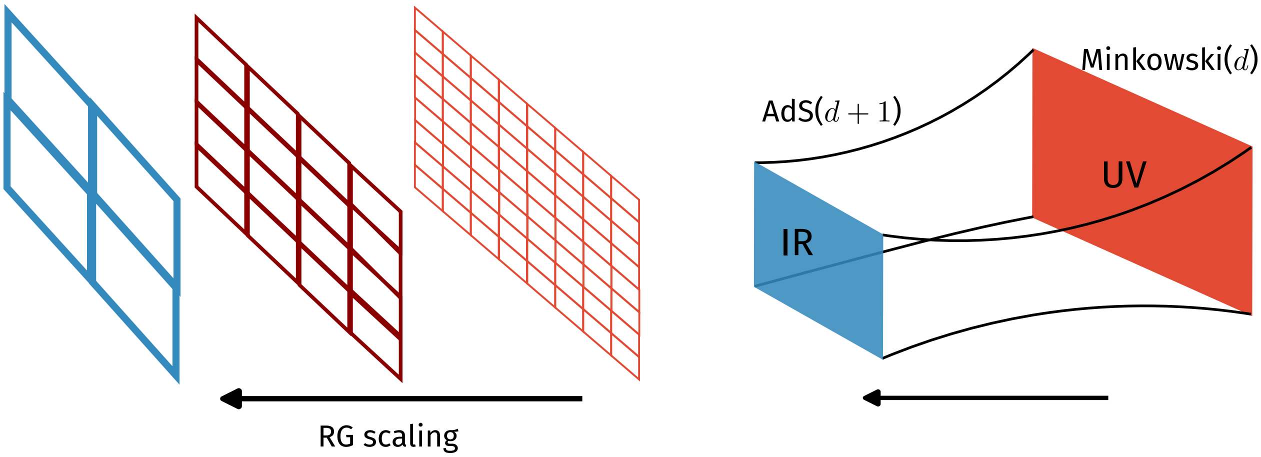

Figure 1.15: *The AdS-CFT correspondence (right), or more generally, the holographic principle maps a quantum field theory in $d-$dimensions to a gravity theory in one extra dimension. The quantum field theory is said to reside at the boundary of this $d+1$-dimensional space. The additional dimension in the gravity theory arises from the hierarchy of quantum fluctuation scales present within the electronic interactions of the quantum field theory, and can be revealed by considering renormalisation group flows (left) of the boundary theory.*

gling degrees of freedom in real space, creating a tensor network in the process [432–434] and leading to a renormalisation in the entanglement. The renormalised entanglement is interpreted as the additional dimension [429]. There exist other approaches that also proceed via entanglement renormalisation, such as the exact holographic mapping [353, 354] (EHM) and the unitary renormalisation group [346, 347, 435] (URG) that operate purely through unitary transformations and have been explicitly shown to lead to a holographic tensor network along the RG flow. Apart from these, there is a body of work due largely to Ki-Seok Kim that provides a geometric description of Wilson's numerical renormalisation group [36,436–438,444], and creates a more general RG framework in terms of an order-parameter field that extends the Ads/CFT correspondence to non-critical theories [439–443].

Questions on entanglement and holography become particularly interesting in systems of critical fermions. As mentioned earlier, such systems differ from bosonic and gapped systems through the presence of longer-ranged entanglement that follows a modified area-law [328, 330] as well as topological features arising from the incompressibility of the Fermi volume [50, 51]. Fermi surface gapping phase transitions in such systems are often driven by changes in topological quantum numbers and the nature of the inter-particle entanglement [136,346,347]. Constructing a framework for fermionic criticality is therefore an active area of research with consequences for important systems such as unconventional superconductors, strange metals and heavy fermions [137,174,183, 445]. Still less is known about the holographic implications of such systems. We extend this body of work by providing an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

1.6 A New Auxiliary Model Approach Towards Fermionic Criticality

To motivate the construction of a new auxiliary model approach in this thesis, we now write down some important outstanding questions that we will address using our approach.

1. *Can we construct an impurity-based auxiliary model approach that describes interaction-driven electronic phase transitions without imposing self-consistency?* As discussed in the preceding sections, one of the most prominent examples of auxiliary model approaches in the study of correlated fermionic models is dynamical mean-field theory (DMFT). However, solutions arising from DMFT and it's extensions such as cellular DMFT and dynamical cluster approximation suffer from a lack of transparency owing to the numerical approach towards self-consistency. To go beyond such approaches, can we explicitly construct an impurity model that maps to the local physics of some interacting lattice model, and recover the low energy physics of the latter by solving the former? Apart from being much more transparent due to the explicit nature of the construction, such an approach would also be more tractable – because of the simplicity of impurity models – than studying the full lattice model.

2. *Can such an approach map the renormalisation group flows of an auxiliary impurity model to those of a correlated lattice model?* Both impurity and lattice models, due to the presence of many-particle interactions, typically involve a hierarchy of energy scales that are coupled to one another. The framework of renormalisation group (RG) is a powerful approach to identify the "important" physics at various energy scales. In particular, it can be used to identify low-energy effective Hamiltonians that describe the essential physics in the ground state and at low temperatures, and the microscopic Hamiltonian is said to "flow" into such low-energy fixed-point Hamiltonians. Can our auxiliary model approach map the renormalisation group flows between the impurity and lattice models, at least in terms of the local physics? This would make the underlying mechanism for the RG flows as well as the various fixed-point theories much more transparent.

3. *Does such an auxiliary model approach preserve symmetries of the microscopic/effective theories of the lattice model, and obey various conservation principles such as Luttinger's theorem?* Importantly, an auxiliary model mapping for a lattice model should preserve the symmetries of the full Hamiltonian. It should also, ideally, show the emergence of conservation principles and spectral sum rules. Luttinger's theorem, as an example of such a sum rule, equates the electron density in a system with the Fermi volume in the presence of interactions, as long as the excitations are electronic. Violations in Luttinger's theorem often signal interaction-driven transitions from Fermi liquids to Mott insulators or non-Fermi liquids. Can the above auxiliary model approach capture changes in Luttinger's theorem? An associated question is whether such an approach can show the emergence of Luttinger surfaces through the appearance of zeros in the one-particle Greens function. Such signatures would

have topological consequences because the Luttinger sum rule is protected by topological invariants.

4. *Is it possible to describe topological transitions using such an approach?* It is becoming increasingly clear that various materials exhibit phase transitions that are not necessarily tied to the spontaneous breaking of a local symmetry and fall outside the Ginzburg-Landau-Wilson paradigm. Examples include (i) the Mott transition driven by a charge gap but without necessarily a spin gap, (ii) the Kondo breakdown QCP scenario in heavy fermion materials where the large-to-small Fermi surface transition is observed to to be separate from magnetic ordering and (iii) transition from a trivial insulator to a topological insulator driven by changes in band topology. Such transitions do not exhibit the appearance of non-zero local order parameters, and are instead often described by changes in topological invariants and entanglement patterns. One can therefore ask: is such an auxiliary model approach capable of tracking such generalised order parameters?

5. *Can such an auxiliary model approach capture the effects of multiple orbitals and momentum-space differentiation?* One of the primary difficulties with many of the numerical methods typically employed in such problems is the high computational complexity (in time as well as memory) and the inability to resolve momentum-space features to sufficient precision. The increased complexity becomes particularly important in several classes of problems such as multi-band models, which are pertinent to investigations into orbital-selective transitions, heavy fermion physics and twisted bilayer graphene, among many others. The lack of momentum-space resolution is relevant to studies of the pseudogap (PG) phase that is observed in the underdoped high-Tc superconductors; the PG is known to display partial gapping of the Fermi surface around the antinodes while remaining gapless around the nodes. The approach proposed here can benefit in this regard in two ways: the explicit construction might be helpful to avoid the computational cost of a self-consistency loop whilst providing access to a large number of momentum states, and (ii) the mapping to an impurity model makes the analysis much more tractable.

Within the thesis, we have developed a new approach towards studying correlated models with strong local interactions. The strategy goes as follows: We first identify a quantum impurity model that represents faithfully the local physics of the lattice model we are targeting. This is done by solving the impurity and studying it's phenomenology. In order to capture specific aspects of the physics, the impurity model may need to be enhanced by, say, allowing for multiple impurities, or embedding it onto a lattice instead of a continuum, etc. Having identifies the appropriate impurity model, we apply manybody translation operators to the impurity model Hamiltonian in order to restore translation invariance and map it to the target lattice Hamiltonian in the process.

For single-particle wavefunctions $\psi_i(r_i)$, the one-particle momentum operator p_i generates a one-particle translation operator $T_i(\mathbf{a})$ that translates the real-space operators associated with the particle i by the vector $\mathbf{a}$. By applying them to the dynamics of local potential wells at each site, these translation operators can be used to construct a translation-invariant Hamiltonians such as the tight-binding model; a similar application on the local wavefunctions generates delocalised translation-invariant Bloch states. Analogous to the single-particle case, one can construct a many-

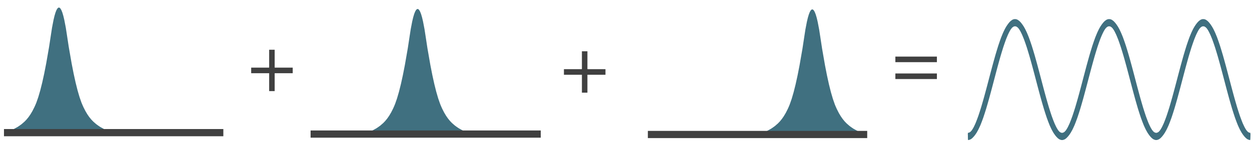

Figure 1.16: For the problem of a single electron, one can construct the full delocalised translation-invariant eigenstate by taking a superposition of the localised wavefunctions at each site. This is equivalent to repeatedly applying translation operations of varying distances on a single localised wavefunction associated with an arbitrarily chosen lattice site. A similar construction also works for the Hamiltonian, where by applying translation operators on the nearest-neighbour hybridisation processes of a single site, one can reconstruct the full tight-binding model.

particle translation operator T [446] that act on an $N-$particle state $\Psi(r_1, r_2, ...r_N)$: $T(\mathbf{a}) = e^{-i\mathbf{a}\cdot\mathbf{P}} = \prod_{i=1}^{N} T_i(\mathbf{a})$, where $\mathbf{P}$ is the total momentum operator. The action of T is to shift all local operators by $\mathbf{a}$ and to multiply all $k-$space operators with a phase shift $e^{i\mathbf{a}\cdot\mathbf{k}}$.

Using such a manybody translation operator, we are able to map both the Hamiltonian as well as eigenstates between the "local" auxiliary impurity model and the translation-invariant lattice model. This leads to specific relations between various observables such as correlations and spectral functions, as well as between entanglement measures. This set of mappings can be used to reconstruct the low-energy physics of the lattice model by starting from that of the auxiliary model.

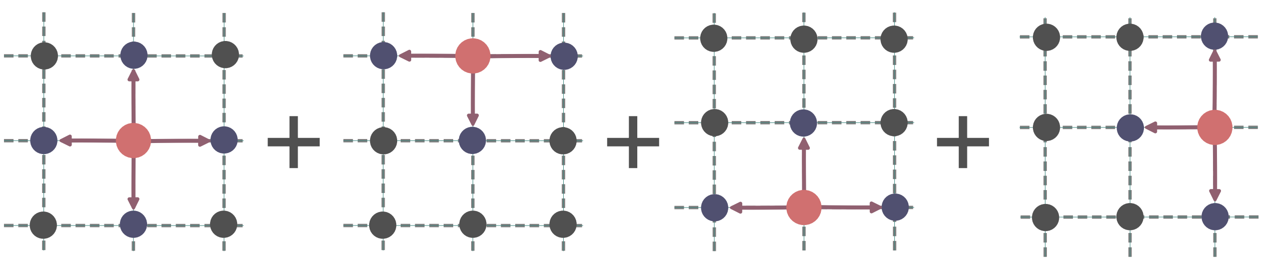

Figure 1.17: Schematic depiction of the "tiling" step in our auxiliary model approach. In order to restore translation invariance and map our impurity model to a lattice model, we apply manybody translation operators on our impurity model Hamiltonian for all lattice sites; this reconstructs the full lattice model, and leads to relations between the auxiliary model couplings and the lattice model couplings.

1.6.1 Important features of the "tiling" auxiliary model approach

Local transition as a diagnostic of lattice transition As mentioned in the preceding paragraphs, the approach developed within this thesis maps the low-energy physics of the lattice model to that of an impurity model with local correlation, hence rendering the problem much more tractable. This, for example, means that the presence of local Fermi liquid excitations on the impurity site marked by a broad central Kondo resonance in the local density of states corresponds to a Landau Fermi liquid phase for the lattice model. By inducing a gap in the impurity site density of states and stabilising the local moment states, one can also model a Mott insulator on the lattice model. More exotic phases such as non-Fermi liquids can be engineered by enhancing the impurity model with competing interactions that lead to a pseudogap in the impurity density of states. Taken together, this maps the renormalised groups flows of the impurity model to that of the lattice model.

Restoring translation symmetry using a many-body Blochs' theorem At the heart of the impurity-lattice model mapping are manybody translation operators T. Indeed, they serve to restore translation invariance lost by the introduction of the impurity site into the lattice: $H_{\text{lat}} = \sum_i T^\dagger(\mathbf{r}_i) H_{\text{aux}} T(\mathbf{r}_i)$. In a similar fashion, the translation operators serve to "extend" the auxiliary model eigenstates into those of the lattice model: $\Psi = \sum_i T^\dagger(\mathbf{r}_i)\psi_{\text{aux}}$. This constitutes a manybody version of Bloch's theorem [447], where local impurity model states are superposed to form delocalised wavefunctions that span the lattice. This step plays the role of the self-consistent loop in DMFT by restoring the discrete translational invariance of the lattice model. This is also seen from the fact that translation operation leads to effective renormalised relations for the lattice model couplings in terms of the auxiliary model quantities. These translation operators also introduce non-local inter-auxiliary model contributions into various correlation functions.

Mapping across Greens functions and entanglement measures One of the main results of our tiling approach is the set of mappings between real-space and momentum-space Greens functions of the auxiliary model and the lattice model. This makes it much more tractable to characterise phases on the lattice model. The mappings take into account correlations within a single impurity model as well as contributions from inter-auxiliary model connections, through the translation operators. In the absence of long-range critical fluctuations, these expressions can be truncated to terms up to a few sites away from the impurity complex, and this makes the practical computation possible. Near the transition, it becomes necessary to work with much larger number of sites.

Lattice-embedding as a route to encoding lattice geometry and band topology Another important aspect of the tiling approach is the ability to incorporate lattice effects such as momentum-space differentiation and topological features. We accomplish this by considering auxiliary model directly on the appropriate lattice. This ensures that the Hamiltonian couplings of the auxiliary model inherit the symmetries of the lattice, and the auxiliary model quantities are then sensitive to lattice effects. This, for example, has allowed us to study a momentum-space anisotropic pseudogap phase in the context of an extended Hubbard model on a 2D square lattice.

1.7 Summary Of Chapters

For the readers' convenience, we now briefly describe the content and layout of the following chapters.

After the introduction, the *Methods and Preliminaries* chapter describes the various techniques and some preliminary ideas employed throughout the thesis. The chapter begins by deriving the unitary renormalisation group formalism and describes it's various properties and a practical protocol for implementing it. That is followed by demonstrations of the unitary RG on three simple models

- a star graph model of spins in a magnetic field, the two-channel Kondo model, and the problem of a particle in a periodic potential. Through these problems, various aspects of the method are clarified. In order to provide more insights, the chapter also details how the unitary RG method is related to other RG-based methods that rely on canonical transformations, such as Poor Man's scaling, Schrieffer-Wolff transformation and CUT (continuous unitary transformation) RG. Apart from the URG, the chapter also describes an iterative diagonalisation method that has been used within this thesis to compute various correlation functions and entanglement measures from the hierarchy of Hamiltonians obtained from the URG. Since various entanglement measures have been used throughout the later chapters, we have also added a section on various entanglement measures here - what they mean, what are their inter-relations, why they are important and how one can compute some of them. The chapter concludes by describing some of the other strategies that have been employed to reduce computational costs in various projects.

In Chapter 3, *Kondo Frustration via Charge Fluctuations: A Route to Mott Localisation in Infinite Dimensions*, we study our first impurity model, the extended Anderson model with local correlation U_b in the conduction bath. We demonstrate that this model hosts a local version of the Mott metal-insulator transition on the Bethe lattice in infinite dimensions, as seen from DMFT. At a critical value of the ratio of U_b and the impurity-bath hybridisation, the effective impurity model shows a transition from a Kondo screened phase into an unscreened local moment phase. For the case of attractive local bath correlations, the model sheds new light on several aspects of the DMFT phase diagram. For example, the T = 0 metal-to-insulator quantum phase transition (QPT) is preceded by an excited state QPT (ESQPT) where the local moment eigenstates are emergent in the low-lying spectrum. Long-ranged fluctuations are observed near both the QPT and ESQPT, suggesting that they are the origin of the quantum critical scaling observed recently at high temperatures in DMFT simulations. The $T = 0$ gapless excitations at the quantum critical point display particle-hole interconversion processes, and exhibit power-law behaviour in self-energies and two-particle correlations. These are signatures of non-Fermi liquid behaviour that emerge from the partial breakdown of the Kondo screening. A many-body perturbation theoretic treatment of the Hubbard sidebands reveals that they are comprised of holon-doublon excitations created by the hybridisation of the impurity site with the conduction bath. These excitations consist of decoupled local Fermi liquids for the holons and doublons at the lowest order, which, at higher orders, become coupled via correlated holon-doublon scattering between impurity and bath.

In Chapter 4, *A New Auxiliary Model Approach for Interacting Electrons: Mapping Impurity Models To Lattice Models*, we lay out the details of the auxiliary model method developed to map impurity models to lattice models. We first motivate the general idea behind auxiliary model methods. This is followed by a construction of the manybody translation operators. We use this, along with a manybody Bloch's theorem, to construct lattice models from quantum impurity models and relate their eigenstates. This then allows us to map correlations functions, self-energies and entanglement measures across models. In particular, we demonstrate how the presence of the conduction bath introduces $k-$space resolution in various quantities.

In Chapter 5, *Mott Criticality as the Confinement Transition of of a Pseudogap-Mott Metal: Insights on Mott Transition and Pseudogap in Two Dimensions*, we apply the auxiliary model tiling method to a lattice-embedded extended SIAM. Applying a many-body tiling scheme to the fixed-point impurity model uncovers a lattice model with electron interactions and Kondo physics. At half-filling, the interplay between Kondo screening and bath charge fluctuations in the impurity model leads to Fermi liquid breakdown. This reveals a pseudogap phase characterized by a non-Fermi liquid (the Mott metal) residing on nodal arcs, gapped antinodal regions of the Fermi surface (Luttinger surfaces), and an anomalous scaling of the electronic scattering rate with frequency. The eventual confinement of holon-doublon excitations of this exotic metal obtains a continuous transition into the Mott insulator. Our results identify the pseudogap as a distinct long-range entangled quantum phase, and offer a new route to Mott criticality beyond the paradigm of local quantum criticality. The phase appears to encode multipartite entanglement, as seen from a large value of the quantum Fisher information. The critical point at the pseudogap-Mott insulator boundary is found to be described by the exactly-solvable Hatsugai-Kohmoto model. The emergence of Luttinger surfaces alters the anomaly structure associated with the generalized symmetry of the FS [44–46]. The corresponding topological charge is defined via the Luttinger–Ward functional. This charge is modified by Green's function zeros, signalling a shift in the underlying anomaly. Within our framework, this reorganization grants topological protection to the RG flows terminating in the PG regime, ensuring the stability of its NFL excitations.

In Chapter 6, *Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation*, we study a periodic Anderson model where we treat both the $c-$ and $f-$layers as extended Hubbard models, and analyse it using the tiling approach developed and implemented in the earlier chapters. We first study the case of half-filling and obtain a Kondo insulator at large values of the inter-band hybridisation. We also study the more interesting case of a Mott metal in the $f-$layer hybridising with the $c-$electrons.

In Chapter 7, *Holography of Entanglement in 2D Free Fermions: Emergent Geometry and Fermionic Criticality*, we provide an explicit demonstration of the holographic principle in the form of a simple and tractable exact holographic mapping (EHM) for a free fermion system with and without a mass gap. Our program yields analytic relations between the RG flow parameters and their dual bulk quantities such as distances and curvature. This also allows us to study the consequences of a critical Fermi surface (lying precisely at the quantum phase transition) on the nature of the holographic space.

We conclude in the final chapter by summarising the main results obtained within the thesis, in terms of the methods developed as well as insights obtained into the behaviour of various models. We describe the impact of these results on the field, the broad conclusions emerging from them and some open questions resulting from the work.

Chapter 2

Methods and Preliminaries

2.1 The Unitary renormalisation group method

The URG method was introduced and formalised in refs. [174,183,346,347]. This section is adapted from those references and expanded wherever required.

Description of the problem

We are given a Hamiltonian $\mathcal{H}$ which is not completely diagonal in the occupation number basis of the electrons, $\hat{n}_k$: $[\mathcal{H}, n_k] \neq 0$. k labels any set of quantum numbers depending on the system. For spin-less Fermions it can be the momentum of the particle, while for spin-full Fermions it can be the set of momentum and spin. There are terms that scatter electrons from one quantum number k to another quantum number k'.

We take a general Hamiltonian,

$$\mathcal{H} = H_e \hat{n}_{q\beta} + H_h \left(1 - \hat{n}_{q\beta}\right) + c^\dagger_{q\beta} T + T^\dagger c_{q\beta} \tag{1.1}$$

Formally, we can decompose the entire Hamiltonian in the subspace of the electron we want to decouple ($q\beta$).

$$\mathcal{H} = \begin{array}{c} \begin{array}{cc} |1\rangle & |0\rangle \end{array} \\ \begin{pmatrix} H_1 & T \\ T^\dagger & H_0 \end{pmatrix} \end{array} \tag{1.2}$$

The basis in which this matrix is written is $\{|1\rangle, |0\rangle\}$ where $|i\rangle$ is the set of all states where $\hat{n}_{q\beta} = i$. The aim of one step of the URG is to find a unitary transformation U such that the new Hamiltonian $U\mathcal{H}U^\dagger$ is diagonal in this already-chosen basis.

$$\tilde{\mathcal{H}} \equiv U\mathcal{H}U^\dagger = \begin{array}{c} \begin{array}{cc} |1\rangle & |0\rangle \end{array} \\ \begin{pmatrix} \tilde{H}_1 & 0 \\ 0 & \tilde{H}_0 \end{pmatrix} \end{array} \tag{1.3}$$

U_q is defined by

$$\tilde{\mathcal{H}} = U_q \mathcal{H} U_q^\dagger \text{ such that } \left[\tilde{\mathcal{H}}, n_q\right] = 0 \tag{1.4}$$

It is clear that U is the diagonalizing matrix for $\mathcal{H}$. Hence we can frame this problem as an eigenvalue equation as well. Let $|\psi_1\rangle, |\psi_0\rangle$ be the basis in which the original Hamiltonian $\mathcal{H}$ has no off-diagonal terms corresponding to $q\beta$. Hence, we can write

$$\mathcal{H} |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle, i \in \{0, 1\} \tag{1.5}$$

Since $|\psi_i\rangle$ is the set of eigenstates of $\mathcal{H}$ and $|i\rangle$ is the set of eigenstates in which $U\mathcal{H}U^\dagger$ has no off-diagonal terms corresponding to $q\beta$, we can relate $|\psi_i\rangle$ and $|i\rangle$ by the same transformation : $|\psi_i\rangle = U^\dagger |i\rangle$. We can expand the state $|\psi_i\rangle$ in the subspace of $q\beta$:

$$|\psi_i\rangle = \sum_{j=0,1} |j\rangle \langle j| |\psi_i\rangle \equiv |1\rangle |\phi_1^i\rangle + |0\rangle |\phi_0^i\rangle \tag{1.6}$$

where $|\phi_j^i\rangle = \langle j| |\psi_i\rangle$. If we substitute the expansion 1.2 into the eigenvalue equation 1.5, we get

$$\left[H_e \hat{n}_{q\beta} + H_h \left(1 - \hat{n}_{q\beta}\right) + c_{q\beta}^\dagger T + T^\dagger c_{q\beta}\right] |\psi_i\rangle = \tilde{H}_i |\psi_i\rangle \tag{1.7}$$

The diagonal parts $H_e = \text{tr}\left[\mathcal{H}\hat{n}_{q\beta}\right]$ and $H_e = \text{tr}\left[\mathcal{H}\left(1 - \hat{n}_{q\beta}\right)\right]$ can be separated into a purely diagonal part $\mathcal{H}^d$ that contains the single-particle energies and the multi-particle correlation energies or Hartree-like contributions, and an off-diagonal part $\mathcal{H}^i$ that scatters between the remaining degrees of freedom $k\sigma \neq q\beta$. That is,

$$H_e \hat{n}_{q\beta} + H_h \left(1 - \hat{n}_{q\beta}\right) = \mathcal{H}^d + \mathcal{H}^i$$

This gives

$$\left[c_{q\beta}^\dagger T + T^\dagger c_{q\beta}\right] |\psi_i\rangle = \left(\tilde{H}_i - \mathcal{H}^i - \mathcal{H}^d\right) |\psi_i\rangle \tag{1.8}$$

Obtaining the decoupling transformation

We now define a new operator $\hat{\omega}_i = \tilde{H}_i - \mathcal{H}^i$, such that

$$\left[c_{q\beta}^\dagger T + T^\dagger c_{q\beta}\right] |\psi_i\rangle = \left(\hat{\omega}_i - \mathcal{H}^d\right) |\psi_i\rangle \tag{1.9}$$

From the definition of $\hat{\omega}_i$, we can see that it is Hermitian and has no term that scatters in the subspace of $q\beta$, so it is diagonal in $q\beta$ and we can expand it as $\hat{\omega}_i = \hat{\omega}_i^1 \hat{n}_{q\beta} + \hat{\omega}_i^0 \left(1 - \hat{n}_{q\beta}\right)$. Using the expansion 1.6, we can write

$$\hat{\omega}_i |\psi_i\rangle = \hat{\omega}_i^1 |1\rangle |\phi_1^i\rangle + \hat{\omega}_i^0 |0\rangle |\phi_0^i\rangle \tag{1.10}$$

Since the only requirement on $|\psi_i\rangle$ is that it diagonalize the Hamiltonian in the subspace of $q\beta$, there is freedom in the choice of this state. We can exploit this freedom and choose the $|\phi_{0,}^i\rangle$ to be an eigenstates of $\hat{\omega}_i^{1,0}$ corresponding to real eigenvalues $\omega_i^{1,0}$:

$$\left[\mathcal{H}^d + c_{q\beta}^\dagger T + T^\dagger c_{q\beta}\right] |\psi_i(\omega_i)\rangle = \left(\omega_i^1 - \mathcal{H}^d\right) |1\rangle |\phi_1^i\rangle + \left(\omega_i^0 - \mathcal{H}^d\right) |0\rangle |\phi_0^i\rangle \tag{1.11}$$

If we now substitue the expansion 1.6 and gather the terms that result in $\hat{n}_{q\beta} = 1$, we get

$$c^\dagger_{q\beta} T \left|0\right\rangle \left|\phi^i_0\right\rangle = \left(\omega^1_i - \mathcal{H}^d\right) \left|1\right\rangle \left|\phi^i_1\right\rangle \tag{1.12}$$

Similarly, gathering the terms that result in $\hat{n}_{q\beta} = 0$ gives

$$T^\dagger c_{q\beta} \left|1\right\rangle \left|\phi^i_1\right\rangle = \left(\omega^0_i - \mathcal{H}^d\right) \left|0\right\rangle \left|\phi^i_0\right\rangle \tag{1.13}$$

We now define two many-particle transition operators:

$$\begin{aligned} \eta^\dagger(\omega^1_i) &= \frac{1}{\omega^1_i - \mathcal{H}^d} c^\dagger_{q\beta} T \equiv G_1 c^\dagger_{q\beta} T \\ \eta(\omega^0_i) &= \frac{1}{\omega^0_i - \mathcal{H}^d} T^\dagger c_{q\beta} \equiv G_0 T^\dagger c_{q\beta} \end{aligned} \tag{1.14}$$

wher G_j is the propagator $\frac{1}{\omega^j_i - \mathcal{H}^d}$. We can write this compactly as

$$\eta(\hat{\omega}) = G T^\dagger c_{q\beta} = \frac{1}{\hat{\omega}_i - \mathcal{H}^d} T^\dagger c_{q\beta} \tag{1.15}$$

where $\hat{\omega}_i = \omega^0_i \left(1 - \hat{n}_{q\beta}\right) + \omega^1_i \hat{n}_{q\beta} = \begin{pmatrix} \omega^1_i & \\ & \omega^0_i \end{pmatrix}$ is a 2x2 matrix and $\mathcal{H}^d = \mathcal{H}^d_0 \left(1 - \hat{n}_{q\beta}\right) + \mathcal{H}^d_1 \hat{n}_{q\beta}$ and $G = \left(\hat{\omega} - \mathcal{H}^d\right)^{-1}$. It is easy to check that this reproduces the previous forms of η_0 and $\eta^\dagger_1$. We will later find that it is important to demand that these two be Hermitian conjugates of each other; that constraint is imposed on the denominators:

$$\eta^\dagger(\omega^0_i) = \eta^\dagger(\omega^1_i) \implies \frac{1}{\omega^1_i - \mathcal{H}^d} c^\dagger_{q\beta} T = c^\dagger_{q\beta} T \frac{1}{\omega^0_i - \mathcal{H}^d} \tag{1.16}$$

Henceforth we will assume that this constraint has been imposed.

In terms of these operators, eq. 1.13 becomes

$$\left|1\right\rangle \left|\phi^i_1\right\rangle = \eta^\dagger \left|0\right\rangle \left|\phi^i_0\right\rangle, \quad \left|0\right\rangle \left|\phi^i_0\right\rangle = \eta \left|1\right\rangle \left|\phi^i_1\right\rangle \tag{1.17}$$

These allow us to write

$$\left|\psi_1\right\rangle = \left|1\right\rangle \left|\phi^i_1\right\rangle + \left|0\right\rangle \left|\phi^i_0\right\rangle = \left(1 + \eta\right) \left|1\right\rangle \left|\phi^i_1\right\rangle, \quad \left|\psi_0\right\rangle = \left(1 + \eta^\dagger\right) \left|0\right\rangle \left|\phi^i_0\right\rangle \tag{1.18}$$

Recalling that $\left|\psi_i\right\rangle = U^\dagger \left|i\right\rangle$, we can read off the required transformation:

$$U_1 = 1 + \eta \tag{1.19}$$

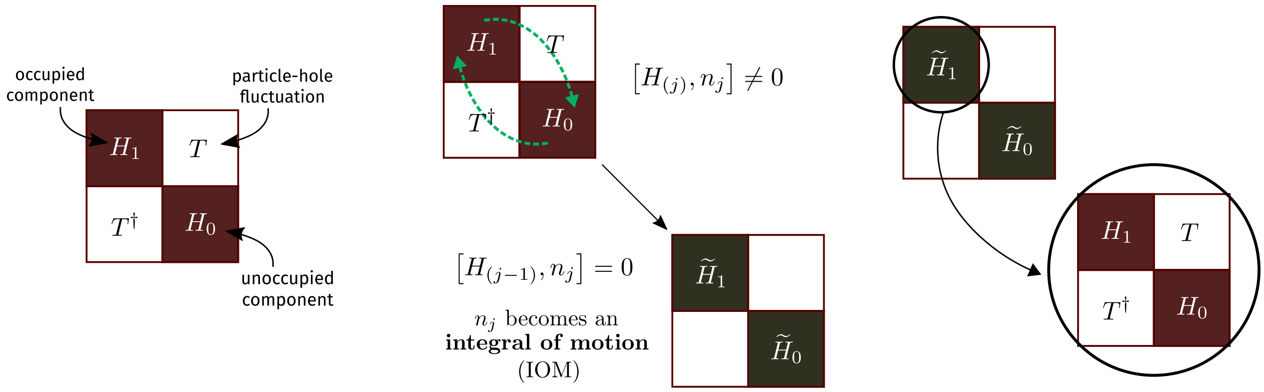

Figure 2.1: *Three steps of the URG: Decompose the Hamiltonian in a 2×2 matrix, apply the unitary operator to rotate it, then repeat these steps with one of the rotated blocks.*

Properties of the many-body transition operators

The operators η have some important properties. First is the Fermionic nature:

$$\eta^2 = \eta^{\dagger 2} = 0 \quad \left[c^{\dagger 2} = c^2 = 0\right] \tag{1.20}$$

Second is:

$$\begin{aligned} |1\rangle |\phi_1^i\rangle = \eta^\dagger |0\rangle |\phi_0^i\rangle = \eta^\dagger \eta |1\rangle |\phi_1^i\rangle &\implies \eta^\dagger \eta = \hat{n}_{q\beta} \\ |0\rangle |\phi_0^i\rangle = \eta |1\rangle |\phi_1^i\rangle = \eta\eta^\dagger |\phi_0^i\rangle &\implies \eta\eta^\dagger = 1 - \hat{n}_{q\beta} \end{aligned} \tag{1.21}$$

and hence the anticommutator

$$\implies \left\{\eta, \eta^\dagger\right\} = 1 \tag{1.22}$$

Note that the three equations in **1.21** work only when applied on the eigenstate $|\psi_i\rangle$ and not any arbitrary state.

$$\begin{gathered} \eta^\dagger \eta |\psi_i\rangle = |1\rangle |\phi_1^i\rangle = \hat{n}_{q\beta} |\psi_i\rangle \\ \eta\eta^\dagger |\psi_i\rangle = |0\rangle |\phi_0^i\rangle = \left(1 - \hat{n}_{q\beta}\right) |\psi_i\rangle \\ \left\{\eta^\dagger, \eta\right\} |\psi_i\rangle = |\psi_i\rangle \end{gathered}$$

Form of the unitary operators

Although we have found the correct similarity transformations U_i (eqs. **1.19**), we need to convert them into a unitary transformation. Say we are trying to rotate the eigenstate $|\psi_1\rangle$ into the state $|1\rangle$. We can then work with the transformation

$$U_1 = 1 + \eta \tag{1.23}$$

In this form, this transformation is not unitary. It can however be written in an exponential form:

$$U_1 = e^\eta \tag{1.24}$$

using the fact that $\eta^2 = 0$. It is shown in ref. [448] that corresponding to a similarity transformation e^{ω},there exists a unitary transformation e^G where

$$G = \tanh^{-1}\left(\omega - \omega^\dagger\right) \tag{1.25}$$

Applying that to the problem at hand gives

$$U_1^\dagger = \exp\left\{\tanh^{-1}\left(\eta - \eta^\dagger\right)\right\} \tag{1.26}$$

Let $x = \tanh y$. Then,

$$x = \frac{e^{2y}+1}{e^{2y}-1} \implies y = \frac{1}{2}\log\frac{1+x}{1-x} \implies e^y = e^{\tanh^{-1}x} = \sqrt{\frac{1+x}{1-x}} \tag{1.27}$$

Therefore,

$$\exp\left\{\tanh^{-1}\left(\eta - \eta^\dagger\right)\right\} = \frac{1+\eta-\eta^\dagger}{\sqrt{\left(1+\eta^\dagger-\eta\right)\left(1-\eta^\dagger+\eta\right)}} = \frac{1+\eta-\eta^\dagger}{\sqrt{1+\left\{\eta,\eta^\dagger\right\}}} = \frac{1}{\sqrt{2}}\left(1+\eta-\eta^\dagger\right) \tag{1.28}$$

The *unitary* operator that transforms the entangled eigenstate $|\psi_1\rangle$ to the state $|1\rangle$ is thus

$$U_1 = \frac{1}{\sqrt{2}}\left(1+\eta^\dagger-\eta\right) \tag{1.29}$$

It can also be written as $\exp\left\{\frac{\pi}{4}\left(\eta^\dagger-\eta\right)\right\}$ because

$$\begin{aligned}\exp\left\{\frac{\pi}{4}\left(\eta^\dagger-\eta\right)\right\} &= 1+\left(\eta^\dagger-\eta\right)\frac{\pi}{4}+\frac{1}{2!}\left(\eta^\dagger-\eta\right)^2\left(\frac{\pi}{4}\right)^2+\frac{1}{3!}\left(\eta^\dagger-\eta\right)^3\left(\frac{\pi}{4}\right)^3+\ldots\\ &= 1+\left(\eta^\dagger-\eta\right)\frac{\pi}{4}-\frac{1}{2!}\left(\frac{\pi}{4}\right)^2-\frac{1}{3!}\left(\eta^\dagger-\eta\right)\left(\frac{\pi}{4}\right)^3+\frac{1}{4!}\left(\frac{\pi}{4}\right)^4+\ldots\\ &= \cos\frac{\pi}{4}+\left(\eta^\dagger-\eta\right)\sin\frac{\pi}{4}\\ &= \frac{1}{\sqrt{2}}\left(1+\eta^\dagger-\eta\right)\end{aligned} \tag{1.30}$$

There we used

$$\left(\eta^\dagger-\eta\right)^2 = \eta^{\dagger 2}+\eta^2-\left\{\eta^\dagger,\eta\right\} = -1 \quad \left[\because \eta^2 = \eta^{\dagger 2} = 0\right] \tag{1.31}$$

and hence

$$\left(\eta^\dagger-\eta\right)^3 = -1\left(\eta^\dagger-\eta\right) \tag{1.32}$$

and so on.

Effective Hamiltonian

We can now compute the form of the effective Hamiltonian that comes about when we apply U_1 - that is - when we rotate one exact eigenstate $|\psi_1\rangle$ into the occupied Fock space basis $|1\rangle$. From eq. 1.29,

$$\begin{aligned}
U_1 \mathcal{H} U_1^\dagger &= \frac{1}{2}\left(1+\eta^\dagger-\eta\right)\mathcal{H}\left(1+\eta-\eta^\dagger\right) \\
&= \frac{1}{2}\left(1+\eta^\dagger-\eta\right)\left(\mathcal{H}+\mathcal{H}\eta-\mathcal{H}\eta^\dagger\right) \\
&= \frac{1}{2}\left(\mathcal{H}+\mathcal{H}\eta-\mathcal{H}\eta^\dagger+\eta^\dagger\mathcal{H}+\eta^\dagger\mathcal{H}\eta-\eta^\dagger\mathcal{H}\eta^\dagger-\eta\mathcal{H}-\eta\mathcal{H}\eta+\eta\mathcal{H}\eta^\dagger\right) \\
&= \frac{1}{2}\left(\mathcal{H}^d+\mathcal{H}^i+\mathcal{H}^I+\mathcal{H}\eta-\mathcal{H}\eta^\dagger+\eta^\dagger\mathcal{H}+\eta^\dagger\mathcal{H}\eta-\eta^\dagger\mathcal{H}\eta^\dagger-\eta\mathcal{H}-\eta\mathcal{H}\eta+\eta\mathcal{H}\eta^\dagger\right) \\
&= \frac{1}{2}\left(\mathcal{H}^d+\mathcal{H}^i+\mathcal{H}^I+\left[\eta^\dagger-\eta,\mathcal{H}\right]+\eta^\dagger\mathcal{H}\eta-\eta^\dagger\mathcal{H}\eta^\dagger-\eta\mathcal{H}\eta+\eta\mathcal{H}\eta^\dagger\right)
\end{aligned} \tag{1.33}$$

In the last two lines, we expanded the Hamiltonian into the three parts $\mathcal{H}^d, \mathcal{H}^i$ and a third piece $\mathcal{H}^I \equiv c^\dagger_{q\beta}T + T^\dagger c_{q\beta}$.

For reasons that will become apparent, we will split the terms into two groups:

$$\tilde{\mathcal{H}} = \frac{1}{2}\left(\underbrace{\mathcal{H}^d+\mathcal{H}^i+\left[\eta^\dagger-\eta,\mathcal{H}\right]+\eta^\dagger\mathcal{H}\eta+\eta\mathcal{H}\eta^\dagger}_{\text{group 1}}+\overbrace{\mathcal{H}^I-\eta^\dagger\mathcal{H}\eta^\dagger-\eta\mathcal{H}\eta}^{\text{group 2}}\right) \tag{1.34}$$

Group 2 can be easily shown to be 0. Note that terms that have two η or two $\eta^\dagger$ sandwiching a $\mathcal{H}$ can only be nonzero if the intervening $\mathcal{H}$ has an odd number of creation or destruction operators.

$$\eta\mathcal{H}\eta = \eta c_q^\dagger T\eta \tag{1.35}$$

and

$$\eta^\dagger\mathcal{H}\eta^\dagger = \eta^\dagger T^\dagger c_q\eta^\dagger \tag{1.36}$$

Group 2 becomes

$$\text{group 2} = \mathcal{H}^I - \eta^\dagger T^\dagger c_q\eta^\dagger - \eta c_q^\dagger T\eta = c_q^\dagger T + T^\dagger c_q - \eta^\dagger T^\dagger c_q \eta^\dagger - \eta c_q^\dagger T\eta \tag{1.37}$$

To simplify this, we use the relation

$$\begin{aligned}
\eta c_q^\dagger T\eta &= \frac{1}{\omega_i^0-\mathcal{H}^d}T^\dagger c_q c_q^\dagger T\eta \\
&= T^\dagger c_q\frac{1}{\omega_i^1-\mathcal{H}^d}c_q^\dagger T\eta \quad && \text{[eq. 1.16]} \\
&= T^\dagger c_q\eta^\dagger\eta && \text{[eq. 1.15]} \\
&= T^\dagger c_q\hat{n}_q && \text{[eq. 1.21]}
\end{aligned} \tag{1.38}$$

which gives

$$\eta c_q^\dagger T \eta = T^\dagger c_q \tag{1.39}$$

Taking the Hermitian conjugate of eq. 1.39 gives

$$\eta^\dagger T^\dagger c_q \eta^\dagger = c_q^\dagger T \tag{1.40}$$

Substituting the expressions 1.39 and 1.40 into the expression for group 2, 1.37, shows that is vanishes. This leaves us only with group 1:

$$\tilde{\mathcal{H}} = \frac{1}{2}\left(\mathcal{H}^d + \mathcal{H}^i + \overbrace{\eta^\dagger \mathcal{H}\eta + \eta\mathcal{H}\eta^\dagger}^{\text{group A}} + \underbrace{\left[\eta^\dagger - \eta, \mathcal{H}\right]}_{groupB}\right) \tag{1.41}$$

Group A simplifies in the following way. First note that $\eta^\dagger \mathcal{H}^I \eta = \eta^\dagger \mathcal{H}^I \eta = 0$ must be 0 because it will involve consecutive $c_{q\beta}$ or consecutive $c_{q\beta}^\dagger$. We are therefore left with the diagonal part of $\mathcal{H}$, which is $H_e \hat{n}_{q\beta} + H_h\left(1 - \hat{n}_{q\beta}\right)$.

$$\eta^\dagger\left[H_e \hat{n}_{q\beta} + H_h\left(1 - \hat{n}_{q\beta}\right)\right]\eta + \eta\left[H_e \hat{n}_{q\beta} + H_h\left(1 - \hat{n}_{q\beta}\right)\right]\eta^\dagger = \eta^\dagger H_h \eta + \eta H_e \eta^\dagger \tag{1.42}$$

This can be shown to be equal to the diagonal part:

$$\text{group A} = \eta^\dagger H_h \eta + \eta H_e \eta^\dagger = H_e \hat{n}_{q\beta} + H_h\left(1 - \hat{n}_{q\beta}\right) = \mathcal{H}^d + \mathcal{H}^i \tag{1.43}$$

It can also be shown that

$$\text{group B} = \left[\eta^\dagger - \eta, \mathcal{H}\right] = 2\left[c_{q\beta}^\dagger T, \eta\right] \tag{1.44}$$

Putting it all together,

$$\tilde{\mathcal{H}} = \mathcal{H}^d + \mathcal{H}^i + \left[c_{q\beta}^\dagger T, \eta\right] \tag{1.45}$$

The renormalizing in the Hamiltonian is

$$\Delta\mathcal{H} = \tilde{\mathcal{H}} - \mathcal{H}^d - \mathcal{H}^i = \left[c_{q\beta}^\dagger T, \eta\right] \tag{1.46}$$

Because of eq. 1.44, it can also be written as

$$\Delta\mathcal{H} = \frac{1}{2}\left[\eta^\dagger - \eta, \mathcal{H}_X\right] = \frac{1}{2}\left[\eta^\dagger - \eta, \mathcal{H}\right] \tag{1.47}$$

This form will be useful later when we make the connection with one-shot Schrieffer-Wolff transformation and CUT RG.

To check that the renormalised Hamiltonian indeed commutes with $\hat{n}_{q\beta}$,

$$\begin{aligned}\left[\tilde{\mathcal{H}}, \hat{n}_{q\beta}\right] &= \left[\left[c_{q\beta}^\dagger T, \eta\right], \hat{n}_{q\beta}\right] = \left[c_{q\beta}^\dagger T\eta, \hat{n}_{q\beta}\right] - \left[\eta c_{q\beta}^\dagger T, \hat{n}_{q\beta}\right] \\ &= c_{q\beta}^\dagger T\eta \hat{n}_{q\beta} - \hat{n}_{q\beta} c_{q\beta}^\dagger T\eta && \left[2^{\text{nd}}\ [\ .\ ]\text{ is }0,\ \because\ c_{q\beta}^\dagger \hat{n}_{q\beta} = \hat{n}_{q\beta}\eta = 0\right] \\ &= c_{q\beta}^\dagger T\eta - c_{q\beta}^\dagger T\eta = 0\end{aligned} \tag{1.48}$$

Fixed point condition

Within the URG, it is a prescription that the fixed point is reached when the denominator of the RG equation vanishes. This is equivalent to either $\omega_i^1 = \mathcal{H}_1^d$ or $\omega_i^0 = \mathcal{H}_0^d$. This shows that at the fixed point, one of the eigenvalues of $\hat{\omega}_i$ matches the corresponding eigenvalue of the diagonal blocks. This also leads to the vanishing of the off-diagonal block, because eqs. 1.12 and 1.13 gives

$$c_{q\beta}^\dagger T \left|0\right\rangle \left|\phi_0^i\right\rangle = \left(\omega_i^1 - \mathcal{H}_1^d\right) \left|1\right\rangle \left|\phi_1^i\right\rangle = 0 \implies c_{q\beta}^\dagger T = 0 \tag{1.49}$$

Multiple off-diagonal terms

There is a subtle assumption in the definitions eq. 1.14. In order for η to be the Hermitian conjugate of $\eta^\dagger$, $\mathcal{H}_d$ cannot have any information that relates to the structure of T. To see why, say the total off-diagonal term is composed of two parts: $T = T_1 + T_2$.

$$\begin{aligned} \eta &= \frac{1}{\omega_0 - \mathcal{H}_d}\left(T_1^\dagger + T_2^\dagger\right) c = \left[\frac{1}{\omega^0 - E_1^0} T_1^\dagger c + \frac{1}{\omega^0 - E_2^0} T_2^\dagger c\right] \\ \eta^\dagger &= \frac{1}{\omega^1 - \mathcal{H}_d} c^\dagger \left(T_1 + T_2\right) = \left[\frac{1}{\omega^1 - E_1^1} c^\dagger T_1 + \frac{1}{\omega^1 - E_2^1} c^\dagger T_2\right] \end{aligned} \tag{1.50}$$

where $\mathcal{H}_d T_i^\dagger c = E_i^0 T_i^\dagger c$ and $\mathcal{H}_d c^\dagger T_i = E_i^1 c^\dagger T_i$. We can now see that in order for $\eta = \left(\eta^\dagger\right)^\dagger$ to hold, two conditions must be met:

$$\omega^0 - E_1^0 = \omega^1 - E_1^1, \quad \omega^0 - E_2^0 = \omega^1 - E_2^1 \tag{1.51}$$

This will not hold generally. The correct solution is to realize that each such off-diagonal term T_i will come with its own quantum fluctuation scale ω_i.

$$\begin{aligned} \eta &= \sum_i \frac{1}{\omega_i^0 - E_i^0} T_i^\dagger c \\ \eta^\dagger &= \sum_i \frac{1}{\omega_i^1 - E_1^1} c^\dagger T_i \end{aligned} \tag{1.52}$$

If we now impose the condition that $\eta = \left(\eta^\dagger\right)^\dagger$, we get the relations

$$\omega_i^0 - \omega_i^1 = E_i^0 - E_1^1 \tag{1.53}$$

and so

$$\eta^\dagger - \eta = \sum_i \frac{1}{\omega_i^0 - E_i^0}\left(c^\dagger T_i - T_i^\dagger c\right) \tag{1.54}$$

The expression for the renormalization will not be just $\left[c^\dagger T, \eta\right]$ in this case. That form will be non-Hermitian. The correct form is obtained from the more general form $\left[\eta^\dagger - \eta, \mathcal{H}_X\right]$:

$$\begin{aligned}\Delta\mathcal{H} &= \frac{1}{2}\left[\eta^\dagger - \eta, c^\dagger T + T^\dagger c\right] = \frac{1}{2}\sum_{ij}\frac{1}{\omega_i^0 - E_i^0}\left[c^\dagger T_i - T_i^\dagger c, c^\dagger T_j + T_j^\dagger c\right] \\ &= \frac{1}{2}\sum_{ij}\frac{1}{\omega_i^0 - E_i^0}\left[\hat{n}\left(T_i T_j^\dagger + T_j T_i^\dagger\right) - (1-\hat{n})\left(T_i^\dagger T_j + T_j^\dagger T_i\right)\right] \\ &= \frac{1}{2}\sum_{ij}\left(\frac{1}{\omega_i^0 - E_i^0} + \frac{1}{\omega_j^0 - E_j^0}\right)\left[\hat{n} T_i T_j^\dagger - (1-\hat{n}) T_i^\dagger T_j\right]\end{aligned} \tag{1.55}$$

Prescription

Given a Hamiltonian

$$\mathcal{H} = \mathcal{H}_1 + \mathcal{H}_0 + c^\dagger T + T^\dagger c \tag{1.56}$$

the goal is to look at the renormalization of the various couplings in the Hamiltonian as we decouple high energy electron states. Typically we have a shell of electrons at some energy D. During the process, we make one simplification. We assume that there is only one electron on that shell at a time, say with quantum numbers q, σ, and calculate the renormalization of the various couplings due to this electron. We then sum the momentum q over the shell and the spin β, and this gives the total renormalization due to decoupling the entire shell.

From eq. 1.45, the first two terms in the rotated Hamiltonian are just the diagonal parts of the bare Hamiltonian; they are unchanged in that part. The renormalization comes from the third term. For one electron $q\beta$ on the shell, the renormalization is

$$\Delta\mathcal{H} = \left[c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right), \eta\right] = c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right)\eta - \eta c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right) \tag{1.57}$$

Since this assumes we have obtained this from U_1, it is fair to tag the η with a suitable label:

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right)\eta_1 - \eta_1 c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right) \tag{1.58}$$

It is clear that the first term takes into account virtual excitations that start from a filled state ($\hat{n}_{q\beta} = 1$ initially) - such a term is said to be a part of the *particle sector*.

$$\Delta_1\mathcal{H} = c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right)\eta_1 \tag{1.59}$$

The second term, on the other hand, considers excitations that start from an empty state. They constitute the *hole sector*.

$$\Delta_0\mathcal{H} = -\eta_1 c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right) \tag{1.60}$$

To write the total renormalization in a particle-hole symmetric form, we can use the relation $\eta_0 = -\eta_1$, such that both the terms will now come with a positive sign:

$$\Delta\mathcal{H} = c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right)\eta_1 + \eta_0 c_{q\beta}^\dagger \mathrm{Tr}\left(\mathcal{H} c_{q\beta}\right) \tag{1.61}$$

We can make one more manipulation: using eq. 1.16, we get

$$\Delta\mathcal{H} = c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)\eta_1 + \mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta}\eta^\dagger_0 \tag{1.62}$$

This form of the total renormalization is identical to the one we use in the "Poor Man's scaling"-type of renormalization that was used to get the scaling equations in the Kondo and Anderson models [449, 450]. Writing down the forms of η and $\eta^\dagger$ explicitly, we get

$$\Delta\mathcal{H} = c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)\frac{1}{\omega^0_1 - \mathcal{H}^d_0}\mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta} + \mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta}\frac{1}{\omega^1_0 - \mathcal{H}^d_1}c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right) \tag{1.63}$$

The renormalization due to the entire shell is obtained by summing over all states on the shell.

$$\Delta\mathcal{H} = \sum_{q\beta}\left[c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)\frac{1}{\omega^0_1 - \mathcal{H}^d_0}\mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta} + \mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta}\frac{1}{\omega^1_0 - \mathcal{H}^d_1}c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)\right] \tag{1.64}$$

These equations will now need to be simplified. For example, in the particle sector, we can set $\hat{n}_{q\beta} = 0$ in the numerator, because there is no such excitation in the initial state. Similarly, in the hole sector, we can set $\hat{n}_{q\beta} = 1$ because that state was occupied in the initial state. Another simplification we typically employ is that $\mathcal{H}^d_{0,1}$ will, in general, have the energies of all the electrons. But we consider only the energy of the on-shell electrons in the denominator. After integrating out these electrons, we can rearrange the remaining operators to determine which term in the Hamiltonian it renormalizes and what is the renormalization.

At first sight, one might think that we must evaluate lots of traces to obtain the terms in $\Delta\mathcal{H}$. A little thought reveals that the terms in the numerator are simply the off-diagonal terms in the Hamiltonian; $\mathrm{Tr}\left(c^\dagger_{q\beta}\mathcal{H}\right)c_{q\beta}$ is the off-diagonal term that has $c_{q\beta}$ in it, and $c^\dagger_{q\beta}\mathrm{Tr}\left(\mathcal{H}c_{q\beta}\right)$ is the off-diagonal term that has $c^\dagger_{q\beta}$ in it. $\mathcal{H}^D$ is just the diagonal part of the Hamiltonian.

2.2 Example: Unitary RG analysis of the Kondo model

We now describe the implementation of the unitary RG method in a simple paradigmatic model, the single-channel Kondo model [42]. It describes the scattering of itinerant non-interacting electrons off a magnetic impurity $\mathbf{S}_d$ [33, 444, 451], and is defined by the Hamiltonian

$$H_K = \sum_{k,\sigma}\epsilon_k n_{k,\sigma} + J\mathbf{S}_d\cdot\frac{1}{2}\sum_{\alpha,\beta}\sigma_{\alpha,\beta}\sum_{k_1,k_2}c^\dagger_{k_1,\alpha}c_{k_2,\beta}\ . \tag{2.1}$$

Note that the Kondo term is effectively of the form $\mathbf{S}_1\cdot\mathbf{S}_2$ with the local spin density of the conduction bath making up the second spin:

$$\frac{1}{2}\sum_{\alpha,\beta}\sigma_{\alpha,\beta}\sum_{k_1,k_2}c^\dagger_{k_1,\alpha}c_{k_2,\beta} = \frac{1}{2}\sum_{\alpha,\beta}\sigma_{\alpha,\beta}c^\dagger_{0,\alpha}c_{0,\beta} = \mathbf{s}_0\ . \tag{2.2}$$

As a result, the Kondo term can be written in the form

$$S_d^z s_0^z + \frac{1}{2} S_d^+ s_0^- + \frac{1}{2} S_d^- s_0^+ \ . \tag{2.3}$$

We will make use of this form.

Because of spin-flip scattering processes, the problem is manybody in nature. In order to obtain the low-energy physics, we will progressively integrate out high-energy modes in the conduction bath and approach the low-energy degrees of freedom. Such an approach is needed because the high and low-energy states interact with each other through the impurity spin and cannot be trivially removed from the problem. The way this works is that the process of integrating out high-energy states generates additional scattering processes within the low-energy states; if these processes already exist in the Hamiltonian, the new contributions *renormalise* the existing couplings, but if the processes don't exist, then the Hamiltonian generates new terms as low energies are approached.

Consider an arbitrary j^{th} step of the process where we wish to integrate out the conduction bath states within a narrow shell about the energies $\pm\epsilon_j$ and retain only the states at $|\epsilon| < |\epsilon_j|$. We designate the states on this shell by the momentum index q, and the states below it by indices k_1 and k_2. We need to first separate the Hamiltonian into diagonal and off-diagonal terms from the perspective of the shell being integrated out:

$$\begin{aligned} H_D &= \sum_{q,\sigma} \epsilon_q n_{q,\sigma} + J\mathbf{S}_d^z \frac{1}{2} \sum_q \left(n_{q,\uparrow} - n_{q,\downarrow}\right) \ , \\ H_X &= J\mathbf{S}_d \cdot \frac{1}{2} \sum_{\alpha,\beta} \sigma_{\alpha,\beta} \sum_{k,q} \left(c^\dagger_{k,\alpha} c_{q,\beta} + c^\dagger_{q,\alpha} c_{k,\beta}\right) \ . \end{aligned} \tag{2.4}$$

In order to make the particle-hole symmetry in the kinetic energy term manifest, we replace the number operators with pseudospin operators $\tau = n - 1/2$ and drop the constant term:

$$\begin{aligned} H_D &= \sum_{q,\sigma} \epsilon_q \tau_{q,\sigma} + J\mathbf{S}_d^z \sum_q \left(n_{q,\uparrow} - n_{q,\downarrow}\right) \ , \\ H_X &= J\mathbf{S}_d \cdot \sum_{\alpha,\beta} \sigma_{\alpha,\beta} \sum_{k,q} \left(c^\dagger_{k,\alpha} c_{q,\beta} + c^\dagger_{q,\alpha} c_{k,\beta}\right) \ . \end{aligned} \tag{2.5}$$

This ensures that particle and hole excitations both cost the same energy. We now describe how to obtain the renormalisation in the Hamiltonian arising from unitarily decoupling the states on the shell. As mentioned earlier, this happens because the process of integrating out the states q generates interactions between the other states $\{k\}$ and this modifies the Hamiltonian couplings.

To keep things brief, we focus on obtaining the renormalisation of the following spin-flip term in the Kondo Hamiltonian:

$$\frac{1}{2} J S_d^+ \sum_{k_1,k_2} c^\dagger_{k_1,\downarrow} c_{k_2,\uparrow} \tag{2.6}$$

For that, we need to figure out which combinations of off-diagonal terms in H_X can result in such terms, when the $q-$states are integrated out. One possibility is

$$\frac{1}{4} \sum_{q,k_1,k_2} J S_d^z c^\dagger_{q,\uparrow} c_{k_2,\uparrow} \frac{1}{\omega - H_D} J S_d^+ c^\dagger_{k_1,\downarrow} c_{q,\uparrow} \ . \tag{2.7}$$

The factor of $1/4$ in front appears due to the two factors of $1/2$ coming from the zz and $+-$ terms in the Kondo terms in H_X.

In order evaluate H_D in the denominator, we note that the hole excitation of $q\uparrow$ at the right (i) indicates that ϵ_q is positive, and the configuration $n_{q,\uparrow} = 0$ in the intermediate state means that $\epsilon_q \tau_{q,\uparrow}$ works out to be $|\epsilon_j|/2$, and (ii) the impurity and bath spins are antiparallel in the intermediate state because of the opposite spins of their excitations, and so the $J-$term in the denominator evaluates to $J(1/2)(-1/2) = -J/4$. Substituting these in the expression (and noting the minus sign in front of H_D) gives

$$\frac{1}{4}\sum_{q,k_1,k_2} JS_d^z c^\dagger_{q,\uparrow}c_{k_2,\uparrow}\frac{1}{\omega - |\epsilon_j|/2 + J/4}JS_d^+ c^\dagger_{k_1,\downarrow}c_{q,\uparrow} = -\frac{J^2/4}{\omega - |\epsilon_j|/2 + J/4}\sum_{q,k_1,k_2}\left(\frac{1}{2}S_d^+\right)c^\dagger_{k_1,\downarrow}c_{k_2,\uparrow}n_{q,\uparrow} , \tag{2.8}$$

where we used the identity $S_d^z S_d^+ = \frac{1}{2}S_d^+$.

In order to truly integrate out the mode q, we set $n_{q,\uparrow} = 1$, guided by the fact that the mode was initially occupied. The sum over q then simply evaluates to the number of momentum states available at the energy ϵ_j, which is the density of states $\rho(\epsilon_j)$:

$$-\frac{1}{2}\frac{\rho(\epsilon_j)J^2/4}{\omega - |\epsilon_j|/2 + J/4}\sum_{k_1,k_2} S_d^+ c^\dagger_{k_1,\downarrow}c_{k_2,\uparrow} . \tag{2.9}$$

A partner process can be constructed by exchanging the roles of q and k:

$$-\frac{1}{4}\sum_{q,k_1,k_2} JS_d^z c^\dagger_{k_1,\downarrow}c_{q,\downarrow}\frac{1}{\omega - H_D}JS_d^+ c^\dagger_{q,\downarrow}c_{k_2,\uparrow} , \tag{2.10}$$

where the minus sign in front arises from the fact that the term before the Greens function is the spin down component of s_0^z. In this case, the intermediate state has a particle excitation of the $q-$ state, which means $\tau_{q.\downarrow} = \frac{1}{2}$, but the energy ϵ_q of the state must also flip sign because the state was initially occupied, so the net contribution again works out to $|E_j|/2$. The $J-$term is also unchanged from the previous process, because the spin orientations are again antiferromagnetic. Together, we get

$$\sum_{q,k_1,k_2} JS_d^z c^\dagger_{k_1,\downarrow}c_{q,\downarrow}\frac{-J^2/4}{\omega - |\epsilon_j|/2 + J/4}S_d^+ c^\dagger_{q,\downarrow}c_{k_2,\uparrow} = -\frac{1}{2}\frac{\rho(\epsilon_j)J^2/4}{\omega - |\epsilon_j|/2 + J/4}\sum_{k_1,k_2} S_d^+ c^\dagger_{k_1,\downarrow}c_{k_2,\uparrow} . \tag{2.11}$$

Additional contributions are obtained by considering the time-reversed forms of these processes. From the first process above (eq. 2.7), we have:

$$\frac{1}{4}\sum_{q,k_1,k_2} JS_d^+ c^\dagger_{k_1,\downarrow}c_{q,\uparrow}\frac{1}{\omega - H_D}JS_d^z c^\dagger_{q,\uparrow}c_{k_2,\uparrow} . \tag{2.12}$$

In this case, the intermediate state has a particle excitation of the $q-$ state, which means $\tau_{q.\uparrow} = \frac{1}{2}$, but the energy ϵ_q of the state must also flip sign because the state was initially occupied, so the net contribution again works out to $|\epsilon_j|/2$. The $J-$term is also of the same value, because the spin orientations are again antiferromagnetic. Together, we get

$$\frac{1}{4}J\sum_{q,k_1,k_2} S_d^+ c^\dagger_{k_1,\downarrow}c_{q,\uparrow}\frac{1}{\omega - |\epsilon_j|/2 + J/4}JS_d^z c^\dagger_{q,\uparrow}c_{k_2,\uparrow} = \frac{-\frac{1}{2}\rho(\epsilon_j)J^2/4}{\omega - |\epsilon_j|/2 + J/4}\sum_{k_1,k_2} S_d^+ c^\dagger_{k_1,\downarrow}c_{k_2,\uparrow} . \quad (2.13)$$

In this case, we used the identity $S_d^+ S_d^z = -\frac{1}{2}S_d^+$.

We finally consider the time-reversed partner of eq. 2.10:

$$\frac{1}{4}\sum_{q,k_1,k_2} JS_d^+ c^\dagger_{k_1,\downarrow}c_{q,\uparrow}\frac{1}{\omega - H_D}JS_d^z c^\dagger_{q,\uparrow}c_{k_2,\uparrow} . \quad (2.14)$$

Evaluating this similarly gives

$$\frac{-\frac{1}{2}\rho(\epsilon_j)J^2/4}{\omega - |\epsilon_j|/2 + J/4}\sum_{k_1,k_2} S_d^+ c^\dagger_{k_1,\downarrow}c_{k_2,\uparrow} \quad (2.15)$$

Combining all four contributions, the total renormalisation to the $S_d^+ s_0^-$ term comes out to be

$$-\rho(\epsilon_j)J^2\frac{1}{\omega - |\epsilon_j|/2 + J/4}\frac{1}{2}\sum_{k_1,k_2} S_d^+ c^\dagger_{k_1,\downarrow}c_{k_2,\uparrow} . \quad (2.16)$$

By comparing with the term in the original Hamiltonian (eq. 2.6), we can read off the renormalisation in the coupling J for the j^{th} step:

$$\Delta J^{(j)} = -\frac{\rho(\epsilon_j)J^2}{\omega - |\epsilon_j|/2 + J/4} . \quad (2.17)$$

This is the URG equation for the Kondo coupling J. It differs from the perturbative "Poor Man's scaling" equations [451] through the appearance of the Kondo coupling in the denominator. Expanding the URG equation in powers of J/ϵ_j recovers the perturbative form.

2.3 Comparison of renormalisation group approaches based on canonical transformations

We have considered three canonical transformations in this section: the Poor Man's scaling (PMS), the Schrieffer-Wolff transformation (SWT) and the continuous unitary transformation renormalization group (CUT-RG). PMS and SWT are more or less identical; they differ in the context in which they are used. PMS is used when there is an entire spectrum of energies in the model, ranging from a low energy limit to a high energy; it is then employed to decouple the highest energy modes in an iterative fashion. SWT is used when the Hamiltonian can be split into one high and one low energy part, and we need to decouple these two modes. It is clear when seen from this perspective that SWT is like a one-shot PMS; it decouples the UV from the IR in one step, compared to the iterative approach of PMS. However, as has been shown in the previous subsections, both PMS and SWT can be formalized in an identical fashion, so that one can be switched for the other in both contexts.

Now that we have established that PMS and SWT are formally identical, we can relate them to URG. URG is similar in philosophy to PMS in the fact that URG also successively decouples high energy modes from the low energy ones. The difference is that URG takes care of the quantum fluctuations, at least in principle, by introducing a new set of energy scales ω. These ω then parameterise the RG flows of URG, compared to the single RG flow of PMS. Since SWT is formally the same as PMS, these is also a comparison between SWT and URG. Both PMS and SWT trivialize the quantum fluctuation scales of URG by replacing it with the bare energy values.

CUT-RG is philosophically different from the other transformations. Its goal is to gradually reduce the contributions of the off-diagonal terms by making them decay along a certain scale l. Off-diagonal terms that connect states with large energy differences decay faster. In this sense, there is no sequential dropping out of off-diagonal terms; all off-diagonal terms disappear at $l = \infty$. In this sense, it is like a continuous version of SWT. While SWT strives to remove an entire off-diagonal term in one-shot, CUT RG does this gradually by introducing a scale l. This separation of scales is absent in SWT. It does exist in URG and PMS, albeit in a different fashion. There, the separation comes in when we decouple single electron states starting from the Brillouin zone boundary Λ_N and come down to the Fermi surface Λ_0. Each Λ provides a natural energy scale for separating the high and low energy physics.

If one integrates the continuum generator η over all the scales, one should recover something analogous to the SWT generator. This generator is however necessarily perturbative in the off-diagonal term, since by definition η will only be of first order in $\mathcal{H}_X$. This is in contrast to the non-perturbative generators in URG and, at least in principle, PMS. This non-perturbative form is encapsulated in the presence of a second completely different set of energy scales ω (or E in PMS). This second energy scale is absent in CUT RG because it takes care of at most the second order term.

Another point to note is that since SWT keeps the entire off-diagonal piece in the generator, new terms will almost certainly by generated at every step, and they have to be truncated. This is not the case with URG or PMS, becasue in those methods, we decouple just a single-electron at each step, and so those electrons become integrals of motion at that step, leading to their removal from the off-diagonal piece, and very often the Hamiltonian retains the same form as the bare model. This makes the philosophy of URG and PMS easier to work with. Tied to this is the fact that CUT RG often takes a certain type of interaction in the generator part, and not the entire off-diagonal piece. Hence, at the limit of the flow parameter going to ∞, the chosen off-diagonal piece goes to zero but the remaining off-diagonal pieces still remain. As a result, the Hamiltonian is atmost block diagonal at this stage. On the other hand, URG progressively decouples single electrons, meaning all scattering terms corresponding to that electron vanish at each step.

One last thing to note is that URG, being unitarily implemented with a well-defined generator, does not accomodate for spontaneous symmetry breaking (SSB). In order to see SSB, the symmetry-breaking term has to be added to the bare model explicitly; if this term grows under the RG, then the ground state will be symmetry-broken. CUT RG, however, does allow for SSB through the idea that the generator is not uniquely defined. If the generator commutes with a particular symmetry, then the family of Hamiltonians will also have the symmetry [452]. However, if the generator is replaced by something that is normal ordered w.r.t a particular expectation value, then the CUT RG flow will usually be towards either the symmetry-preserved state or the symmetry-broken state, depending on whichever is stable [453].

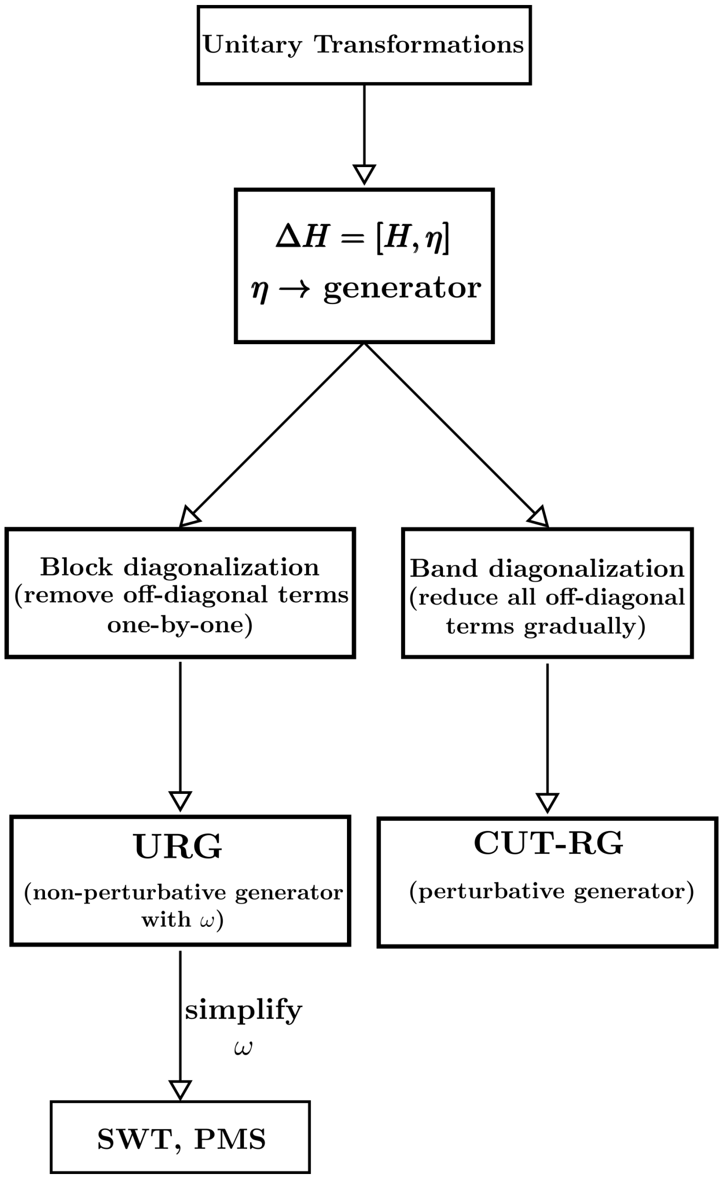

Figure 2.2: *Comparison of the various unitary transformations and their relationships to each other.*

2.4 Numerical Implementation of Unitary RG Equations

Since the bath interaction doesn't renormalise, it's functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_\alpha}{\Delta D} = -J_\alpha \mathcal{G}_\alpha J_\alpha^\dagger - 4\mathcal{W}_\alpha \mathrm{Tr}\left[\mathcal{G}'_\alpha\right] , \tag{4.1}$$

where all objects are matrices. J_α is the Kondo coupling matrix. $\mathcal{G}_\alpha$ is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q,q] = \begin{cases} \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})+\mu/2-J/4}\right), \ \mathbf{q}\in \mathrm{PS} , \\ \rho(q)\left(\frac{1}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2+3J/4} + \frac{3}{8}\frac{1}{1/G_\alpha^{(0)}(\mathbf{q})-\mu/2-J/4}\right), \ \mathbf{q}\in \mathrm{HS} , \end{cases} \tag{4.2}$$

if q is in the UV subspace and zero otherwise. $\mathcal{W}_\alpha$ is the bath interaction matrix whose matrix elements are given by $\mathcal{W}_\alpha[k,k'] = \frac{W}{2}\left[\cos(k_x-k'_x)+\cos(k_y-k'_y)\right]$. Finally, $\mathcal{G}'_\alpha$ is another diagonal matrix whole matrix elements are defined as $\mathcal{G}'_\alpha[q,q] = \mathcal{G}[q,q]J_\alpha[q,\bar{q}]$.

The RG equation for the inter-layer coupling J can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2}\mathbf{J}\cdot\mathbf{G} , \tag{4.3}$$

where $\mathbf{J}$ and $\mathbf{G}$ are vectors defined by the elements $\mathbf{J}[q] = J_f(\bar{\mathbf{q}},\mathbf{q})J_d(\mathbf{q},\bar{\mathbf{q}})$ and

$$\mathbf{G}[q] = \rho(\mathbf{q})\left[\frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2}J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4}J} + \frac{1}{\omega - \frac{1}{2}\bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2}J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4}J}\right] . \tag{4.4}$$

2.5 Renormalised perturbation theory about fixed points

In this section, we will describe the technique of carrying out renormalised theory about a gapless fixed point theory towards obtaining a low-energy effective Hamiltonian for the excitations above the ground state. This is useful for characterising metallic phases, and detecting whether the metal is, say, a Landau Fermi liquid or a non-Fermi liquid.

We consider here the case of the single-channel Kondo model. We would like to demonstrate the method by deriving a local Fermi liquid theory for the gapless excitations above the ground state, as obtained by Nozières [40]. In its simplest form, the Kondo model features a spin-$1/2$ local moment undergoing antiferromagnetic spin-exchange scattering with the local spin-density of a non-interacting electronic conduction bath:

$$H_\mathrm{K} = J\mathbf{S}_d\cdot\mathbf{s} + \sum_{k,\sigma}\epsilon_k n_{k,\sigma} , \tag{5.1}$$

where S_d and s are the spins of the impurity and the conduction bath. For any non-zero value of the Kondo coupling J (the coupling for the impurity-bath spin exchange processes), it can be shown that J is relevant under a renormalisation group flow from high energy to low-energy, and the coupling effectively diverges as the fixed point is approached [444]. At the fixed point, the kinetic energy of the conduction bath can be ignored relative to the Kondo coupling, and the ground state is a spin-singlet between the impurity $\mathbf{S}_d$ and the conduction bath local spin density $\mathbf{s}$:

$$|\Psi\rangle_0 = \frac{1}{\sqrt{2}}\left(|\uparrow_d, \Downarrow\rangle - |\downarrow_d, \Uparrow\rangle\right) , \tag{5.2}$$

where $\uparrow_d$ represents the configuration of the impurity and the thick arrow is the configuration of the conduction bath spin.

In order to obtain the excitations, we now note that because of the RG-relevance of J, the first term in the Hamiltonian in eq. 5.1 can only have finite energy excitations. Since we are looking for excitations that cost vanishingly small energy, we focus on the second term. With an eye towards developing a perturbation theory over $1/J$, we recast the kinetic energy term in real-space as a tight-binding Hamiltonian connected to the local density that hybridises with the impurity:

$$H_{\text{KE}} = -t^* \sum_{i,\sigma}(c^\dagger_{i,\sigma}c_{i+1,\sigma} + \text{h.c.}) . \tag{5.3}$$

Hopping processes from the zeroth site modify the singlet ground state and lead to virtual excitations. We will now obtain the effect of this Hamiltonian by obtaining its renormalisation effects within the ground state subspace.

To better frame the perturbation theory which we will use to obtain the renormalisation effects, we clarify the zeroth Hamiltonian H_0 and perturbation V:

$$\begin{aligned} H_0 &= J^* \mathbf{S}_d \cdot \mathbf{s}, \\ V &= -t^* \sum_{i,\sigma}(c^\dagger_{i,\sigma}c_{i+1,\sigma} + \text{h.c.}) . \end{aligned} \tag{5.4}$$

The goal is to obtain the shifts in the energies of H_0 arising from the perturbation term. The eigenstates of H_0 are

$$\begin{aligned} |\Psi\rangle_{0,0,2} &= \frac{1}{\sqrt{2}}\left(|\uparrow_d, \Downarrow\rangle - |\downarrow_d, \Uparrow\rangle\right) : & E &= -3J^*/4 \\ |\Psi\rangle_{\frac{1}{2},\frac{\sigma}{2},1} &= |\sigma_d, 0\rangle : & E &= 0 \\ |\Psi\rangle_{\frac{1}{2},\frac{\sigma}{2},3} &= |\sigma_d, 2\rangle : & E &= 0 \\ |\Psi\rangle_{1,0,2} &= \frac{1}{\sqrt{2}}\left(|\uparrow_d, \Downarrow\rangle + |\downarrow_d, \Uparrow\rangle\right) : & E &= J^*/4 \\ |\Psi\rangle_{1,\sigma,2} &= |\sigma_d, \sigma\rangle : & E &= J^*/4, \end{aligned} \tag{5.5}$$

where the quantum numbers in the subscript are the total spin $S = S_d + s$, the $z-$component of the total spin $S^z = S^z_d + s^z$, and the total occupancy $N = n_d + n$. The second, third and fifth states are doubly degenerate with σ taking values ± 1. The ground state is of course $|\Psi\rangle_{0,0,2}$. However, this only takes into account the zeroth site in the conduction bath. In order to consider perturbative processes where an electron hops from the zeroth site to the next (first) site, we need to involve the first site as well. Since the first site does not have any energy in H_0, the eigenstates simply become four-fold degenerate, due to the four possible configurations of the first site. The eigenstates are then labelled as $\Psi_{S,S^z,N,c_1} = |\Psi_{S,S^z,N}\rangle \otimes |c_1\rangle$ where c_1 can be $0, \uparrow, \downarrow$ or 2 depending on the configuration of the first site.

To obtain the excitations, we will calculate the perturbative shifts in the four-fold degenerate ground state subspace spanned by the states $\{\Psi_{0,0,2,0}, \Psi_{0,0,2,\uparrow}, \Psi_{0,0,2,\downarrow}, \Psi_{0,0,2,2}\}$. We will do this order by order, stopping when we obtain the lowest order non-trivial Hamiltonian. At first and third order, the energy shifts are given by

$$\Delta E_n^{(1)} = \langle\Psi_n|V|\Psi_n\rangle \ ,$$
$$\Delta E_n^{(3)} = \sum_{k_2\neq n, k_3\neq n} \frac{\langle\Psi_n|V|\Psi_{k_3}\rangle\langle\Psi_{k_3}|V|\Psi_{k_2}\rangle\langle\Psi_{k_2}|V|\Psi_n\rangle}{\left(E_n - E_{k_3}\right)\left(E_n - E_{k_2}\right)} - \Delta E_n^{(1)} \sum_{k_3\neq n} \frac{|\langle\Psi_n|V|\Psi_{k_3}\rangle|^2}{\left(E_n - E_{k_3}\right)^2} \ . \tag{5.6}$$

Both these contributions vanish because of mismatched quantum numbers: upon applying the perturbation Hamiltonian V any odd number of times, the final resultant state will have a different occupancy on the impurity+zeroth site compared to the starting state. Since H_0 conserves the occupancy of the impurity+zeroth site complex, the overlap between such states is zero. More specifically, what this means for the first term of $\Delta E_n^{(3)}$ is the following:

- the excited state k_2 will have either 1 or 3 electrons on the $d+0$ complex (because the ground state hosts 2 electrons there, and V will either add an electron there or remove one electron from it).
- Another application of V means that k_3 will have 0, 2 or 4 electrons.
- A final application of V means that the terminal state will have an odd number of electrons – 1, 3 or 5,

The overlap of this state with the ground state – $\langle\Psi_n|V|\Psi_{k_3}\rangle$ – will therefore vanish.

We therefore focus on the second and fourth order effects. At second order, the renormalisation is given by

$$\Delta E_n^{(2)} = \sum_{k_2\neq n} \frac{|\langle\Psi_n|V|\Psi_{k_2}\rangle|^2}{E_n - E_{k_2}} \ . \tag{5.7}$$

We work with the shifts in the $(0,0,2,0)$ and $(0,0,2,\uparrow)$ states, because the shifts of $(0,0,2)$ and $(0,0,\downarrow)$ can be inferred, respectively, from them using particle-hole symmetry and spin-rotation symmetry. For the state $(0,0,2,0)$, we have

$$V|\Psi_{0,0,2,0}\rangle = -t^*\frac{1}{\sqrt{2}}\left(|\uparrow_d, 0, \downarrow_1\rangle - |\downarrow_d, 0, \uparrow_1\rangle\right) = \frac{-t^*}{\sqrt{2}}|\Psi\rangle_{\frac{1}{2},\frac{1}{2},1,\downarrow} + \frac{t^*}{\sqrt{2}}|\Psi\rangle_{\frac{1}{2},-\frac{1}{2},1,\uparrow} \ . \tag{5.8}$$

Summing k_2 over the states $\frac{1}{2},\frac{1}{2},1,\downarrow$ and $\frac{1}{2},-\frac{1}{2},1,\uparrow$ with energies $E_n - E_{k_2} = -3J^*/4$, the total shift comes out to be

$$\Delta E_{(0,0,2,0)}^{(2)} = 2\frac{t^{*2}}{2} \times \frac{1}{-3J^*/4} = -\frac{4t^{*2}}{3J^*} \ . \tag{5.9}$$

For the state $(0,0,2,\uparrow)$, we have

$$V|\Psi_{0,0,2,\uparrow}\rangle = t^*\frac{1}{\sqrt{2}}\left(|\uparrow_d, 0, 2\rangle + |\uparrow_d, 2, 0\rangle\right) = \frac{t^*}{\sqrt{2}}|\Psi\rangle_{\frac{1}{2},\frac{1}{2},1,2} + \frac{t^*}{\sqrt{2}}|\Psi\rangle_{\frac{1}{2},\frac{1}{2},1,0} \ . \tag{5.10}$$

Summing k_2 over the appropriate eigenstates and using $E_n - E_{k_2} = -3J^*/4$ gives the same shift:

$$\Delta E^{(2)}_{(0,0,2,\uparrow)} = -\frac{4t^{*2}}{3J^*} \ . \tag{5.11}$$

Using symmetries, we also have

$$\Delta E^{(2)}_{(0,0,2,\downarrow)} = \Delta E^{(2)}_{(0,0,2,2)} = -\frac{4t^{*2}}{3J^*} \ . \tag{5.12}$$

W also note that off-diagonal matrix elements are zero in such a setup, because any two states in the ground state subspace will differ by at least one of the two conserved quantum numbers s_1^z and n_1.

We now need to convert these shifts into a renormalised Hamiltonian that describes the excitations arising from such virtual scattering processes. In the absence of shift of any off-diagonal matrix element, the diagonal shifts will only lead to the generation of number-conserving operators. For example, the shift $\Delta E^{(2)}_{(0,0,2,0)} = -\frac{4t^{*2}}{3J^*}$ means that the energy of the null-occupied state of the first site is lowered by that amount. This can be represented through the operator

$$-\frac{4t^{*2}}{3J^*}(1 - n_{1\uparrow})(1 - n_{1\downarrow}) \ . \tag{5.13}$$

Writing down operator forms for the other shifts as well, we get

$$\Delta H^{(2)} = -\frac{4t^{*2}}{3J^*}\left[(1 - n_{1\uparrow})(1 - n_{1\downarrow}) + n_{1\uparrow}(1 - n_{1\downarrow}) + (1 - n_{1\uparrow})n_{1\downarrow} + n_{1\uparrow}n_{1\downarrow}\right] \ . \tag{5.14}$$

However, because the shifts in all the states are identical, adding up all the operators reduces the renormalisation to a constant shift:

$$\Delta H^{(2)} = -\frac{4t^{*2}}{3J^*} \ . \tag{5.15}$$

So we conclude that there's no non-trivial renormalisation at second order.

We now extend the perturbation theory to fourth order. Since the first and third order corrections vanish, and the second order correction only produces a constant shift, the first potentially non-trivial renormalisation arises at fourth order.

For a non-degenerate state, the fourth-order energy shift is

$$\Delta E_n^{(4)} = \sum_{k_1,k_2,k_3\neq n} \frac{\langle\Psi_n|V|\Psi_{k_3}\rangle\langle\Psi_{k_3}|V|\Psi_{k_2}\rangle\langle\Psi_{k_2}|V|\Psi_{k_1}\rangle\langle\Psi_{k_1}|V|\Psi_n\rangle}{(E_n - E_{k_1})(E_n - E_{k_2})(E_n - E_{k_3})} - \Delta E_n^{(2)} \sum_{k_2\neq n} \frac{|\langle\Psi_n|V|\Psi_{k_2}\rangle|^2}{(E_n - E_{k_2})^2} \ . \tag{5.16}$$

We first calculate it for the state $\Psi_{0,0,2,0}$, where the first site is unoccupied. We initially focus on the first term. One application of V gives

$$V|\Psi\rangle_{0,0,2,0} = -\frac{t}{\sqrt{2}}\left(|\uparrow_d, 0, \downarrow_1\rangle - |\downarrow_d, 0, \uparrow\rangle\right) \ . \tag{5.17}$$

This allows two possible states in k_2: $|\uparrow_d, 0, \downarrow_1\rangle$ and $|\downarrow_d, 0, \uparrow\rangle$, with overlaps $\pm t/\sqrt{2}$. They are linked by particle-hole symmetry, so we will calculate the subsequent contribution from only the first and then double the result.

With $k_2 = |\uparrow_d, 0, \downarrow_1\rangle$, applying V once more gives

$$V\,|k_2\rangle = -t\,|\uparrow_d, \downarrow, 0_1\rangle \ . \tag{5.18}$$

This has overlap with the triplet state with zero spin, leading to $k_3 = \Psi_{1,0,2,0}$, with an overlap $-t/\sqrt{2}$. A third application of V gives

$$V\,|k_3\rangle = -\frac{t}{\sqrt{2}}\left(|\uparrow_d, 0, \downarrow_1\rangle + |\downarrow_d, 0, \uparrow_1\rangle\right) \ . \tag{5.19}$$

The final excited states k_4 are therefore the same as k_2. A final application of V on the two possible k_4 gives

$$\begin{aligned} V\,|\uparrow_d, 0, \downarrow_1\rangle &= -t\,|\uparrow_d, \downarrow, 0_1\rangle \,, \\ V\,|\downarrow_d, 0, \uparrow_1\rangle &= -t\,|\downarrow_d, \uparrow, 0_1\rangle \ . \end{aligned} \tag{5.20}$$

The two final states have overlaps with opposite sign with the ground state. Since the excitation energies in the denominator in the first term of eq. 5.16 are the same for each of the paths, the sum over k_4 reduces to a sum over it's overlap with the ground state. As obtained above, this overlap is of different signs for the two possibilities of k_4, leading to this vanishing of the first term of eq. 5.16.

For the second term, we have already calculated the overlap $\langle\Psi_n|V|k_2\rangle$ as well as the second order shift. Plugging those in gives

$$\Delta E^{(4)}_{0,0,2,0} = -\frac{-4t^{*2}}{3J^*} \times 2\frac{t^{*2}/2}{(-3J^*/4)^2} = \frac{64t^{*4}}{27J^{*3}} \ . \tag{5.21}$$

This is the total fourth order shift for the state $\Psi_{0,0,2,0}$.

We next look at the state $\Psi_{0,0,2,\uparrow}$, starting with the first term. One application of the perturbation Hamiltonian gives

$$V\,|\Psi_{0,0,2,\uparrow}\rangle = \frac{-t^*}{\sqrt{2}}\left(-|\uparrow_d, 0, 2_1\rangle - |\uparrow, 2, 0_1\rangle\right) \ . \tag{5.22}$$

Proceeding exactly as before, the possible excited states k_2 are $|\uparrow_d, 0, 2_1\rangle$ and $|\uparrow, 2, 0_1\rangle$. The overlaps and denominators for both states are $t^*/\sqrt{2}$ and $-3J^*/4$ respectively. We now focus henceforth on the state $|\uparrow_d, 0, 2_1\rangle$ because the contribution from the other will be identical. Applying the perturbation on this state gives

$$V\,|\uparrow_d, 0, 2_1\rangle = -t^*\left(|\uparrow_d, \uparrow, \downarrow_1\rangle - |\uparrow_d, \downarrow, \uparrow_1\rangle\right) \ . \tag{5.23}$$

These gives two choices for k_3: the $S^z = 1$ triplet state for $d + 0$ $|\uparrow_d, \uparrow, \downarrow_1\rangle$, and the $S^z = 0$ triplet state $\Psi_{1,0,2,\uparrow}$. While the denominators are $-J^*$ for both the states, the overlaps are $-t^*$ and $t^*/\sqrt{2}$ respectively.

Following the first state and applying V on it, we get

$$V\,|\uparrow_d, \uparrow, \downarrow_1\rangle = -t^*\left(|\uparrow, 0, 2_1\rangle + |\uparrow, 2, 0_1\rangle\right) \ . \tag{5.24}$$

The final set of excited states, k_4, therefore work out to be $|\uparrow, 0, 2_1\rangle$ and $|\uparrow, 2, 0_1\rangle$, giving denominators $-3J^*/4$ and overlaps $-t^*$. The final application of V on either of these states gives

$-t^*\left(|\uparrow_d, \uparrow, \downarrow_1\rangle - |\uparrow_d, \downarrow, \uparrow_1\rangle\right)$, and the final overlap with the ground state is $t^*/\sqrt{2}$. The total contribution from the excitation branch $\Psi_{0,0,2,\uparrow} \to |\uparrow_d, 0, 2_1\rangle \to |\uparrow_d, \uparrow, \downarrow_1\rangle \to \ldots$ therefore comes out to be

$$S_1 = 2(t^*/\sqrt{2})(-t^*)^2(t^*/\sqrt{2})/((-3J^*/4)^2(-J^*)) = -\frac{16t^{*4}}{9J^{*3}} . \tag{5.25}$$

We now focus on the other excitation branch, $\Psi_{0,0,2,\uparrow} \to |\uparrow_d, 0, 2_1\rangle \to \Psi_{1,0,2,\uparrow} \to \ldots$ Applying V on this excited state gives

$$V\Psi_{1,0,2,\uparrow} = \frac{t^*}{\sqrt{2}}\left(|\uparrow_d, 0, 2_1\rangle + |\uparrow_d, 2, 0_1\rangle\right) . \tag{5.26}$$

This gives the same k_3 as the contribution from the other branch, but with a reduced overlap $t^*/\sqrt{2}$ instead of t^*. Since we have already carried out the previous calculation, we can directly write down the result:

$$S_2 = S_1/2 = -\frac{8t^{*4}}{9J^{*3}} . \tag{5.27}$$

The factor of $1/2$ arises because the second and third overlaps are suppressed by $1/\sqrt{2}$ compared to the first branch S_1. Combining S_1 and S_2, the total contribution to the first term in the fourth order expression arising from the branch $\Psi_{0,0,2,\uparrow} \to |\uparrow_d, 0, 2_1\rangle \to \ldots$ is

$$-\frac{8t^{*4}}{3J^{*3}} . \tag{5.28}$$

By particle-hole symmetry, the contribution from the other branch $\Psi_{0,0,2,\uparrow} \to |\uparrow_d, 2, 0_1\rangle \to \ldots$ is identical. Together, the total fourth order renormalisation from the first term is

$$-\frac{16t^{*4}}{3J^{*3}} . \tag{5.29}$$

We now compute the second term in the fourth order expression. Again, the components have already been calculated during the second order calculation, so we just plug them in:

$$-\frac{-4t^{*2}}{3J^*} \times 2\frac{t^{*2}/2}{(-3J^*/4)^2} = \frac{64t^{*4}}{27J^{*3}} . \tag{5.30}$$

Adding the two parts, we get for the total fourth order shift:

$$\Delta E^{(4)}_{0,0,2,\uparrow} = -\frac{80t^{*4}}{27J^{*3}} \tag{5.31}$$

Using particle-hole and spin rotation symmetries, we can now write down the fourth order shifts for all four states in the low-energy manifold:

$$\begin{aligned} \Delta E^{(4)}_{0,0,2,0} &= \Delta E^{(4)}_{0,0,2,2} = \frac{64t^{*4}}{27J^{*3}} \\ \Delta E^{(4)}_{0,0,2,\uparrow} &= \Delta E^{(4)}_{0,0,2,\downarrow} = -\frac{80t^{*4}}{27J^{*3}} . \end{aligned} \tag{5.32}$$

We can already see that the singly-occupied states experience a lowering of energy due to virtual excitations of the singlet state, while the doubly-occupied state has its energy raised. This effectively means if an electron is already present on the first site, a second incoming electron experiences a repulsion due to it. Replacing each state with the appropriate operator, we can convert this intuition into an effective Hamiltonian for the excitations:

$$\begin{aligned}\Delta H^{(4)} &= \frac{16t^{*4}}{27J^{*3}} \left|\Psi_{0,0,2}\right\rangle \left\langle\Psi_{0,0,2}\right| \left[4(1-n_{1,\uparrow})(1-n_{1,\downarrow}) + 4n_{1,\uparrow})n_{1,\downarrow} - 5n_{1,\uparrow}(1-n_{1,\downarrow}) - 5n_{1,\downarrow}(1-n_{1,\uparrow})\right] \\ &= \frac{16t^{*4}}{27J^{*3}} \left|\Psi_{0,0,2}\right\rangle \left\langle\Psi_{0,0,2}\right| \left[-u\sum_\sigma n_{1\sigma} + un_{1\uparrow}n_{1\downarrow}\right] ,\end{aligned} \tag{5.33}$$

where $u = 9$. This is the local Fermi liquid Hamiltonian for excitations above the Kondo screened ground state; the repulsive $n_\uparrow n_\downarrow$ term ensures that the excitations are electronic in nature.

2.6 Iterative Diagonalisation of Quantum Impurity Models

The broad idea is the following. We have a Hamiltonian describing an interacting set of states (in real or momentum space). We can express the total Hamiltonian in an incremental fashion: $H = \sum_i H_i$, where H_{i+1} involves more number of states than H_i. For example, a tight-binding model can be written in that form, with the definition $H_i = c^\dagger_{i+1}c_i$ + h.c.. The iterative diagonalisation method obtains the low-energy spectrum of this problem in the following manner: We first diagonalise the Hamiltonian H for a smaller value of i, small enough such that this can be done exactly. We then truncate the spectrum to a predefined size, and rotate all existing operators to this truncated basis, including the Hamiltonian H. We then consider the "bonding Hamiltonian" ΔH between the existing sites and the new sites, and rotate the same into the truncated basis. Adding the previous rotated Hamiltonian H and the rotated increment Hamiltonian ΔH gives us a truncated but effective Hamiltonian for the increased number of sites. We again diagonalise this, and again retain only a fixed number of states in the spectrum. We keep repeating this until we reach the required number of sites.

Jordan-Wigner Matrix Representation of Fermionic Operators

For a single qubit, the creation/annihilation matrices are

$$c = \begin{pmatrix}0 & 1\\0 & 0\end{pmatrix}, c^\dagger = \begin{pmatrix}0 & 0\\1 & 0\end{pmatrix} , \tag{6.1}$$

given the convention that $|1\rangle = (0\ \ 1)$ is the occupied state. For a many-body system, these must be replaced with field operators that have the canonical fermionic algebra. For a system of N 1-particle levels, this can be accomplished through a *Jordan-Wigner*-like transformation

$$c_j = \left(\otimes_1^j \sigma_z\right) \otimes c \otimes \left(\otimes_1^{N-j}\mathbb{I}\right) , j \in [0, N-1] , \tag{6.2}$$

where $\sigma_z = \begin{pmatrix}1 & 0\\0 & -1\end{pmatrix}$ and $\mathbb{I} = \begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$. For example, for spinless electrons on a three-site lattice, we can have three fermionic operators: c_1,c_2, and c_3. Following the above expression, their matrices are

$$c_1 = \begin{bmatrix}0&0&1&0&0&0&0&0\\0&0&0&1&0&0&0&0\\0&0&0&0&0&0&0&0\\0&0&0&0&0&0&0&0\\0&0&0&0&0&0&-1&0\\0&0&0&0&0&0&0&-1\\0&0&0&0&0&0&0&0\\0&0&0&0&0&0&0&0\end{bmatrix}, c_2 = \begin{bmatrix}0&0&1&0&0&0&0&0\\0&0&0&1&0&0&0&0\\0&0&0&0&0&0&0&0\\0&0&0&0&0&0&0&0\\0&0&0&0&0&0&-1&0\\0&0&0&0&0&0&0&-1\\0&0&0&0&0&0&0&0\\0&0&0&0&0&0&0&0\end{bmatrix}, c_3 = \begin{bmatrix}0&1&0&0&0&0&0&0\\0&0&0&0&0&0&0&0\\0&0&0&-1&0&0&0&0\\0&0&0&0&0&0&0&0\\0&0&0&0&0&-1&0&0\\0&0&0&0&0&0&0&0\\0&0&0&0&0&0&0&1\\0&0&0&0&0&0&0&0\end{bmatrix},$$

For later use, we define the *antisymmetriser matrix* $S(j)$ and identity matrix $\mathbb{I}(j)$,

$$\begin{aligned} S(j) &= \otimes_1^j \sigma_z = \sigma_z \otimes \ldots j \text{ times } \ldots \otimes \sigma_z, \\ \mathbb{I}(j) &= \otimes_1^{N-j} \mathbb{I} = \mathbb{I} \otimes \ldots j \text{ times } \ldots \otimes \mathbb{I} , \end{aligned} \tag{6.3}$$

to express the fermionic representation compactly:

$$c_j = S(j) \otimes c \otimes \mathbb{I}(N-j) . \tag{6.4}$$

Structure of Hamiltonian

We consider a general impurity model, of L number of sites (excluding the impurity site):

$$H_\text{sys}(L) = H_\text{imp} + H_\text{imp-bath} + -t \sum_{\sigma, j=1}^{L-1} \left(c^\dagger_{j,\sigma} c_{j+1,\sigma} + \text{h.c.}\right) , \tag{6.5}$$

where H_imp is the Hamiltonian for the decoupled impurity site and $H_\text{imp-bath}$ is the impurity-bath coupling. Such a system has $2L + 2$ single-particle levels in the full system (2 spin levels for each of the $L + 1$ sites).

Iteration scheme (the initial number of states is L_0):

$$\begin{aligned} H_0(L_0) &= H_\text{sys}(L_0) , \\ H_{r+1}(L_0) &= H_r(L_0) + \Delta H_r(L_0),\ r \geq 0 , \\ \Delta H_r(L_0) &= -t \sum_\sigma \left(c^\dagger_{L_0+r,\sigma} c_{L_0+r+1,\sigma} + \text{h.c.}\right) . \end{aligned} \tag{6.6}$$

H_0 is the initial Hamiltonian, consisting of L_0 lattice sites in the bath, $H_{r+1}(L_0)$ is the Hamiltonian after $r+1$ iterations having started with L_0 sites, and $\Delta H_r(L_0)$ is the increment term that gets added to $H_r(L_0)$ during the $(r + 1)^\text{th}$ iteration to give rise to $H_{r+1}(L_0)$.

Iterative Diagonalization

Let the starting Hamiltonian be $H_0(L_0)$, consisting of L_0 single-particle levels (hence a Hilbert space dimension of 2^{L_0}). We start with a single-particle computational basis and construct the L_0 fermionic operators $c_1, c_2, \ldots, c_{L_0}$ in this basis, using eq. 6.4. We also keep track of a large antisymmetriser matrix $S(L_0)$ that will be used to attach new sites when we expand the system. Let M_s be the maximum number of eigenstates we retain in the spectrum at any given step. The value of M_s should be chosen so that a $M_s \times M_s$ matrix can be diagonalised in reasonable time.

S1. Construct the complete Hamiltonian matrix H_0 in our present basis using the field operators $\left\{c_j\right\}$. Diagonalise the Hamiltonian (of size $D_0 \times D_0$) and obtain the eigenvalues E_n and eigenstates X_n. Each X_n is a column vector of size D_0.

S2. Retain at most M_s number of eigenstates, preferring the ones with lower energy. The reduced basis for this rotated truncated subspace is constructed by stacking the column vectors X_n horizontally:

$$R = [X_1 X_2 \ldots X_{M_s}]_{D_0 \times M_s} \ . \tag{6.7}$$

This matrix also acts as the transformation to rotates and truncate all operators from the old basis into the new one.

S3. We rotate our Hamiltonian H_0, our fermionic operators $\left\{c_j\right\}$ and the large antisymmetrizer matrix $S(L_0)$ into the new reduced basis, using the transformation $O \to R^\dagger O R$.

S4. We now need to expand our system by adding the increment Hamiltonian ΔH_0. Let the number of *new* 1-particle levels in ΔH_0 be L_Δ. These new levels will be indexed as $L_0+1, \ldots, L_0+L_\Delta$. We need to define antisymmetrized fermionic operators for the new sites:

$$\begin{aligned} c_{L_0+1} &= c \otimes \mathbb{I}(2^{L_\Delta - 1}), \\ c_{L_0+2} &= \sigma_z \otimes c \otimes \mathbb{I}(2^{L_\Delta - 2}), \\ &\ldots \\ c_{L_0+L_\Delta} &= (\otimes_1^{L_\Delta - 1} \sigma) \otimes c \ . \end{aligned} \tag{6.8}$$

These have to be calculated in the local computational basis (of size 2^{L_Δ}) of the new levels.

S5. Combining the new sites with the old sites leads to a combined Hilbert space dimension of $(M_s + 2^{L_\Delta}) \times (M_s + 2^{L_\Delta})$. To allow all operators to act on the enlarged Hilbert space, we expand both sets of operators:

$$\begin{aligned} c_j &= S(L_0) \otimes c_j; \quad j = L_0 + 1, \ldots, L_\Delta \ , \\ c_j &= c_j \otimes \mathbb{I}(2^{L_\Delta}); \quad j = 1, 2, \ldots, L_0 \ , \\ H_0 &\to H_0 \otimes \mathbb{I}(2^{L_\Delta}), \\ S(L_1) &= S(L_0) \otimes \mathbb{I}(2^{L_\Delta}) \ . \end{aligned} \tag{6.9}$$

where $\mathbb{I}(2^{L_\Delta})$ is an identity matrix of dimension $2^{L_\Delta} \times 2^{L_\Delta}$. Note that the operators in the last three equations are the rotated ones (following Step 3).

S6. Using the transformed operators $c_{L_0+1}, \ldots, c_{L_0+L_\Delta}$ for the new sites, construct the difference Hamiltonian matrix ΔH_0 and hence the updated Hamiltonian $H_1 = H_0 + \Delta H_0$ for the next step. Repeat the process starting from step 2 with the new Hamiltonian H_1, the new operators c_j and the new matrix $S(L_1)$ replacing the old counterparts.

2.6.1 Static correlations

We are in general interested in $n-$point correlations of the form $\langle O_1 O_2 \ldots O_n \rangle$, where O_i are operators that act on 1-particle Hilbert spaces. Let the earliest step of the iterative diagonalisation procedure at which all these operators have entered the system be r. At this step r, construct the correlation operator $O_1 O_2 \ldots O_n$ using the fermionic matrices c_j at that step (recall that these matrices will be highly rotated versions of the matrices we started with). Having constructed the operator O, we expand and rotate it after the completion of every future step: $O_{n+1} = (R_n^\dagger O_n R_n) \otimes \mathbb{I}(2^{L_\Delta})$, where R_n is the rotation matrix for the n^{th} step. The last step n^* of the iterative diagonalisation consists of only a diagonalisation and no expansion, resulting in a final set of eigenstates $\{X_i\}$. The form of the correlation operator in this basis is $O_{n^*} = R_{n^*-1}^\dagger O_{n^*-1} R_{n^*}$. The expectation value can now be calculated using the matrix O_{n^*} and the ground state of $\{X_i\}$.

2.6.2 Entanglement measures

In Section 2.8.1, we show how the matrix elements of reduced density matrices can be expressed as expectation values of certain local operators. Using such relations, the computation of reduced density matrix effective boils down to the calculation of specific correlations, which is covered in the previous section.

2.7 Optimising Fermi Surface Calculations

The below is an algorithm for reducing computational cost of carrying out iterative diagonalisation processes focused on the Fermi surface of a quantum impurity model, by rotating the interaction matrix to a sparser form. It is useful then the Hamiltonian has certain symmetry properties arising from specific lattice geometries.

Rotated basis for simplified representation

At the renormalisation group fixed point, we have a renormalised theory for the interaction of the impurity spin S_f with the $f-$layer Fermi surface:

$$H^* = \sum_{\alpha,\beta} \mathbf{S}_f \cdot \boldsymbol{\sigma}_{\alpha,\beta} \sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k},\mathbf{q}) f_{\mathbf{k},\alpha}^\dagger f_{\mathbf{q},\beta} \ . \tag{7.1}$$

We will now obtain a more minimal representation of this interaction. Each momentum label is summed over all four quadrants Q_1 through Q_4. We assume the lattice structure is such that the following equation relates Q_1 with Q_3 and Q_2 with Q_4:

$$J_{\mathbf{k},\mathbf{k}'} = -J_{\mathbf{k}+\mathbf{Q},\mathbf{k}'} = -J_{\mathbf{k},\mathbf{k}'+\mathbf{Q}}, \quad \mathbf{Q} = (\pi,\pi). \tag{7.2}$$

With this, we can write

$$\begin{aligned}
\sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} f_{\mathbf{q},\beta} &= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k}\in Q_1,Q_2} J_f^*(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} + \sum_{\mathbf{k}\in Q_3,Q_4} J_f^*(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} \right] f_{\mathbf{q},\beta} \\
&= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k}\in Q_1,Q_2} J_f^*(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} + \sum_{\mathbf{k}\in Q_1} J_f^*(\mathbf{k}-\mathbf{Q_1},\mathbf{q}) f^\dagger_{\mathbf{k}-\mathbf{Q_1},\alpha} + \sum_{\mathbf{k}\in Q_2} J_f^*(\mathbf{k}+\mathbf{Q_2},\mathbf{q}) f^\dagger_{\mathbf{k}+\mathbf{Q_2},\alpha} \right] f_{\mathbf{q},\beta} \\
&= \sum_{\mathbf{q}} \left[\sum_{\mathbf{k}\in Q_1} J_f^*(\mathbf{k},\mathbf{q}) \left(f^\dagger_{\mathbf{k},\alpha} - f^\dagger_{\mathbf{k}-\mathbf{Q_1},\alpha} \right) + \sum_{\mathbf{k}\in Q_2} J_f^*(\mathbf{k},\mathbf{q}) \left(f^\dagger_{\mathbf{k},\alpha} - f^\dagger_{\mathbf{k}+\mathbf{Q_2},\alpha} \right) \right] f_{\mathbf{q},\beta} \ .
\end{aligned} \tag{7.3}$$

For ease of notation, we define new fermionic operators $A_{\mathbf{k},\sigma}$:

$$A_{\mathbf{k},\sigma,\pm} = \begin{cases} \frac{1}{\sqrt{2}} \left(f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}-\mathbf{Q_1},\sigma} \right), \mathbf{k} \in Q_1 \ , \\ \frac{1}{\sqrt{2}} \left(f_{\mathbf{k},\sigma} \pm f_{\mathbf{k}+\mathbf{Q_2},\sigma} \right), \mathbf{k} \in Q_2 \ , \end{cases} \tag{7.4}$$

which satisfy the appropriate algebra: $\left\{ A_{\mathbf{k},\sigma,p}, A_{\mathbf{k}',\sigma',p'} \right\} = \delta_{\mathbf{k},\mathbf{k}'}\delta_{\sigma,\sigma'}\delta p, p'$, with $p = \pm$ denoting the flavours of the A fields. Note that only the $p = -1$ flavour enters the Hamiltonian. Henceforth, we drop the label $\pm$ and it is implied that A refers to the $p = -1$ variant.

Decomposing the sum over $\mathbf{q}$ in a similar fashion, we get

$$\sum_{\mathbf{k},\mathbf{q}} J_f^*(\mathbf{k},\mathbf{q}) f^\dagger_{\mathbf{k},\alpha} f_{\mathbf{q},\beta} = 2 \sum_{\mathbf{k},\mathbf{q}\in Q_1,Q_2} J_f^*(\mathbf{k},\mathbf{q}) A^\dagger_{\mathbf{k},\alpha} A_{\mathbf{q},\beta} \ . \tag{7.5}$$

Correlations in truncated representation

In order to calculate equal-time correlations (such as $\langle S_d^+ f^\dagger_{\mathbf{k}\downarrow} f_{\mathbf{k}'\uparrow} \rangle$), we use eq. 7.4 to express the bare fields $f_{\mathbf{k},\sigma}$ in terms of the new fields $A_{\mathbf{k},\sigma,\pm}$ and $B_{\mathbf{k},\sigma,\pm}$:

$$\begin{aligned}
f_{\mathbf{k},\sigma} &= \frac{1}{\sqrt{2}} \left(A_{\mathbf{k},\sigma,+} + A_{\mathbf{k},\sigma,-} \right) \ , \mathbf{k} \in Q_1, Q_2 \\
f_{\mathbf{k}-\mathbf{Q}_1,\sigma} &= \frac{1}{\sqrt{2}} \left(A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-} \right) \ , \mathbf{k} \in Q_1 \\
f_{\mathbf{k}+\mathbf{Q}_2,\sigma} &= \frac{1}{\sqrt{2}} \left(A_{\mathbf{k},\sigma,+} - A_{\mathbf{k},\sigma,-} \right) \ , \mathbf{k} \in Q_2
\end{aligned} \tag{7.6}$$

where the four relations act on operators in four quadrants. These relations allow us to express correlations $F_{k_i,k_j} = \langle S_d^+ c^\dagger_{k_i,\uparrow} c_{k_j,\downarrow} \rangle$ for the bare fields f in terms of the rotated fields. Using the fact that the Hamiltonian does not couple the fields B_+ and A_+, we have

$$\begin{aligned} F_{k_1,k_1} = F_{k_3,k_3} = -F_{k_1,k_3} &= \frac{1}{2}\langle S_d^+ A^\dagger_{k_1,\downarrow,-} A_{k_1,\uparrow,-})\rangle \ , \\ F_{k_2,k_2} = F_{k_4,k_4} = -F_{k_2,k_4} &= \frac{1}{2}\langle S_d^+ A^\dagger_{k_2,\downarrow,-} A_{k_2,\uparrow,-})\rangle \ , \\ F_{k_1,k_2} = F_{k_3,k_4} = -F_{k_1,k_4} = -F_{k_2,k_3} &= \frac{1}{2}\langle S_d^+ A^\dagger_{k_1,\downarrow,-} A_{k_1,\uparrow,-})\rangle \ , \end{aligned} \tag{7.7}$$

where the momentum state k_i belongs to the quadrant Q_i.

Effective Star Graph Representation of Kondo Interaction

To make computation less intensive, we want to make the Hamiltonian in eq. 7.5 sparser, by going to a star graph representation where each node of the graph couples only with the impurity spin (central node) and not to the other nodes of the graph. We therefore consider a rotated basis of excitations

$$\gamma^\dagger_{i,\sigma} = \sum_k \Lambda_{i,k} A^\dagger_{k,\sigma} \ , \text{ where } \sum_k \Lambda^*_{i,k}\Lambda_{j,k} = \delta_{i,j} \ . \tag{7.8}$$

These fermions of course satisfy the appropriate algebra:

$$\left\{\gamma_{i,\sigma}, \gamma^\dagger_{j,\sigma'}\right\} = \sum_{k,q} \Lambda^*_{i,k}\Lambda_{j,q} \left\{A_{k,\sigma}, A^\dagger_{q,\sigma'}\right\} = \delta_{i,j}\delta_{\sigma,\sigma'} \ . \tag{7.9}$$

In order to calculate the coefficients $\Lambda_{i,k}$, we impose the star graph simplification condition described above:

$$\sum_{\mathbf{k},\mathbf{q}} J^*_f(\mathbf{k},\mathbf{q}) A^\dagger_{\mathbf{k},\alpha} A_{\mathbf{q},\beta} = \sum_i K_i \gamma^\dagger_{i,\alpha}\gamma_{i,\beta} \ , \tag{7.10}$$

where K_i are the eigenvalues of the Kondo coupling matrix $J^*_f(\mathbf{k},\mathbf{q})$. Substituting the definition of the γ fields (eq. 7.8) leads to the equation

$$J^*_f(\mathbf{k},\mathbf{q}) = \sum_i \Lambda_{k,i} K_i \Lambda^\dagger_{i,q} \implies \mathcal{J} = \Lambda K \Lambda^\dagger \ , \tag{7.11}$$

where $\mathcal{J}$ is the Kondo coupling matrix with matrix elements $J^*_f(\mathbf{k},\mathbf{q})$, K is a diagonal matrix with the eigenvalues (K_i) of $\mathcal{J}$ along its diagonal, and Λ is the matrix $\Lambda_{i,k}$ (the rotated indices i along the row, and the earlier momentum indices along the column). From eq. 7.11, it is clear that since $\mathcal{J}$ is Hermitian, the matrix Λ is simply the unitary matrix which diagonalises the Kondo matrix $\mathcal{J}$:

$$\Lambda^\dagger \mathcal{J} \Lambda = K \ . \tag{7.12}$$

The prescription is therefore as follows. We compute the unitary transformation Λ that diagonalises $\mathcal{J}$ and obtain the eigenvalues $\{K_i\}$. Using the eigenvalues, we work with the sparser Hamiltonian

$$\vec{S}_f \cdot \sum_{\alpha,\beta} \vec{\sigma}_{\alpha,\beta} \sum_i K_i \gamma^\dagger_{i,\alpha}\gamma_{i,\beta} \ . \tag{7.13}$$

If we have to calculate a certain correlation $C_{k,q}$, we instead calculate the correlation matrix $\tilde{C}_{i,j}$, and then transform back using the unitary transformation:

$$C = \Lambda \tilde{C} \Lambda^\dagger \ . \tag{7.14}$$

2.8 Measures of entanglement and their computation

The present millennium has seen a rapid surge in the use of quantum entanglement towards the study of quantum and condensed matter systems [241–246]. By tracking quantum correlations among particles across both small and large distances, entanglement measures offer valuable insights into the nature of the ground state and low-energy excitations. This, for example, allows us to distinguish strongly-correlated topologically ordered states characterised by long-ranged entanglement (in the form of a sub-leading topological term in their entanglement entropy [247, 248, 251, 255]) from weakly-interacting topologically trivial phases that have short-ranged area-law entanglement [243, 268, 269].

2.8.1 Reduced density matrix elements as equal-time correlation functions

A density matrix plays the role of a probability density function (PDF) for a quantum system. A classical variable X that can take values $x \in [-X^*, X^*]$ has a PDF $F(X = x)$, which is used, for example, for calculating the average value (expectation value) of the variable:

$$\langle X \rangle = \int_{-X^*}^{X^*} x F(x) dx \ . \tag{8.1}$$

In quantum mechanics, the variables we are interested in are operators O, and a quantum PDF ρ will then give the expectation value of the operator, in a fashion similar to its classical counterpart:

$$\langle O \rangle = \sum_\phi \langle \phi | \rho O | \phi \rangle \ ; \tag{8.2}$$

here, the integral over the classical values x is replaced by a sum over the all possible quantum states $|\phi\rangle$ (complete orthonormal basis) of the system. However, we already know that the expectation value of a quantum operator is given by

$$\langle O \rangle = \langle \Psi | O | \Psi \rangle \ . \tag{8.3}$$

To bring this form closer to the one in terms of ρ (eq. 2), we insert an identity resolution: $1 = \sum_\phi |\phi\rangle \langle \phi|$:

$$\langle O \rangle = \sum_\phi \langle \Psi | O | \phi \rangle \langle \phi | \Psi \rangle = \sum_\phi \langle \phi | \Psi \rangle \langle \Psi | O | \phi \rangle \ . \tag{8.4}$$

A comparison of this equation to eq. (2) gives

$$\rho = |\Psi\rangle \langle \Psi| \ ; \tag{8.5}$$

a density matrix (also called density operator) is therefore formally defined as the outer product of the wavefunction with its conjugate. It encodes all information about the state; for example, the probability of measuring the system to be at a position x (hence in a state $|x\rangle$) is given by $P(x) = \langle x|\rho|x\rangle$. From its definition in eq. (5), the DM has the very important property that it has unity trace:

$$\mathrm{Tr}[\rho] = 1 \ , \tag{8.6}$$

where $\mathrm{Tr}[\cdot]$ is the trace of an operator. This is seen very easily be taking the trace in an orthonormal basis in which the state $|\Psi\rangle$ is one of the basis states. For example, consider the DM

$$\rho = \frac{1}{\sqrt{2}}\left[|\Psi_1\rangle\langle\Psi_1| + |\Psi_2\rangle\langle\Psi_2|\right] \ , \tag{8.7}$$

where $|\Psi_{1,2}\rangle$ are orthogonal normalised states. The trace of ρ^2 for such a state is only $1/2$; such states satifying $\mathrm{Tr}[\rho^2] < 1$ are termed *mixed states*; the DM describing such states cannot be written in the form $\rho = |\Psi\rangle\langle\Psi|$ for any wavefunction $|\Psi\rangle$. On the other hand, the DMs that *can* be expressed like that are termed *pure states*, and have $\mathrm{Tr}[\rho^2] = 1$.

One way in which a pure state can be realistically converted into a mixed state is through a measurement process. Consider the zero total spin state formed by two spins:

$$|\Psi\rangle = \frac{1}{2}\left[|\uparrow,\downarrow\rangle + |\downarrow,\uparrow\rangle\right] \ , \rho = |\Psi\rangle\langle\Psi| \ . \tag{8.8}$$

The state $|\uparrow,\downarrow\rangle$ has the first spin pointing up and the second one down, and this is in a superposition with the its flipped counterpart $|\downarrow,\uparrow\rangle$. This is clearly a pure state.

If we now measure the state of the second spin, it will collapse into one of it's two states; the other spin will collapse into the opposite state (given the antiparallel configurations of the spins). What this means is that until one checks what the result of the measurement is, the first spin remains in a statistical superposition of both outcomes (up and down). This is formally expressed through a partial trace operation: the state of the first spin after a measurement of the second spin is obtained by taking the full density matrix ρ and computing its trace over the states $|\sigma_B\rangle$ of the second spin, leaving out the states of the first spin from the trace:

$$\rho_1 = \mathrm{Tr}_2[\rho] = \sum_{\sigma_2}\langle\sigma_2|\rho|\sigma_2\rangle \ ; \tag{8.9}$$

where $|\sigma_2\rangle$ sums over the states $|\uparrow\rangle$ and $|\downarrow\rangle$ of the second spin. this partial tracing operation Tr_2 codifies the measurement process, and the partially traced density matrix ρ_1 is termed a reduced density matrix (RDM) of the first spin, because it only captures information about the first spin (the second spin has been traced out).

To see the connection with mixedness, let us compute the RDM for the first spin explicitly:

$$\rho_1 = \langle\uparrow_2|\rho|\uparrow_2\rangle + \langle\downarrow_2|\rho|\downarrow_2\rangle = \frac{1}{2}\left[|\uparrow_1\rangle\langle\uparrow_1| + |\downarrow_1\rangle\langle\downarrow_1|\right] \ , \tag{8.10}$$

where the numbered subscripts on the kets and bras indicate that they only act on the respective Hilbert spaces. To see that ρ_1 describes a mixed state, we adopt a specific representation:

$$\begin{aligned} |\uparrow_1\rangle &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}, |\downarrow_1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \\ \implies \rho_1 &= \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \rho_1^2 = \frac{1}{4}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} . \end{aligned} \tag{8.11}$$

The trace of ρ_1^2 is half, hence it is a mixed state.

The deviation of trace of ρ^2 from its maximum value of unity can be thought of as a measure of the lost information. In the example above, when we traced out the states of the second spin, we lost some information about the state of the first spin as well. This is because the two spins were *entangled*; they had quantum correlations between them. The entanglement arises from the fact that the states of the two spins are anti-correlated.

If instead we had chosen a non-entangled parent state to start from, the reduced density matrix would not have been mixed. For example, consider the state

$$|\Psi\rangle = \frac{1}{2}\left(|\uparrow_1\rangle + |\downarrow_1\rangle\right) \otimes \left(|\uparrow_2\rangle + |\downarrow_2\rangle\right) . \tag{8.12}$$

It is easy to see that this state does not contain any correlations between the two spins - no matter the configuration of the second spin, the first spin is always in the state $|\uparrow_1\rangle + |\downarrow_1\rangle$. The RDM for the first spin comes out to be

$$\rho_1 = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \rho_1^2 = \frac{1}{4}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} , \tag{8.13}$$

revealing that ρ_1 is not mixed.

The above method of calculating an RDM works for simple cases but can fail if (i) we have fermion signs to account for, or (ii) the full basis cannot be expressed as a direct product of local basis sets. The former arises if we are dealing with fermions, and the latter arises if we are working in a rotated basis. We now derive an alternative method for calculating the RDM, one that automatically takes care of the fermion signs and works even if the basis is rotated and complicated to write down. We start by defining the RDM more precisely. Given a state $|\Psi\rangle$ and a bipartition into subsystems S_1 and its complement S_2, the density matrix for the state is $\rho = |\Psi\rangle\langle\Psi|$. A basis state of the full system can be written as $|A,\alpha\rangle = |A\rangle \otimes |\alpha\rangle$, where $\{A\}$ and $\{\alpha\}$ are complete sets of basis states describing S_1 and S_2 respectively. The RDM for S_2 is then defined as

$$\rho_1 = \text{Tr}_\alpha \rho = \sum_\alpha \langle\alpha|\rho|\alpha\rangle . \tag{8.14}$$

The matrix elements of ρ_1 in the basis $\{A\}$ can be expressed as

$$\langle A|\rho_1|B\rangle = \sum_\alpha \langle A,\alpha|\,\rho\,|B,\alpha\rangle . \tag{8.15}$$

We next define an operator $O_{A,B}$ that represents scattering processes between the various states of subsystem S_1: $O_{A,B} = |A\rangle \langle B| \otimes I_2$, but acts as identity on S_2. The expectation value of this operator is

$$\langle O_{A,B}\rangle = \mathrm{Tr}(\rho O_{A,B}) = \sum_{C,\beta} \langle C,\beta|\rho|A\rangle \langle B|C,\beta\rangle = \sum_{\beta} \langle B,\beta|\rho|A,\beta\rangle \ . \tag{8.16}$$

Comparing with eq. 8.15, we obtain the relation

$$\langle A|\rho_1|B\rangle = \langle O_{B,A}\rangle \ . \tag{8.17}$$

This result is significant because it guarantees that in order to calculate the RDM for a subsystem, it is enough to calculate the expectation values of all inter-state off-diagonal transition operators. For example, we take subsystem S_1 to be a two fermion system represented by the operators c_1 and c_2. Assuming a basis $\{|0,0\rangle, |0,1\rangle, |1,0\rangle, |1,1\rangle\}$, where the two entries represent the occupancies of the two fermions, the RDM for this subsystem is determined by the following matrix:

$$\rho_1 = \begin{pmatrix} \langle h_1 h_2\rangle & \langle h_1 c_2^\dagger\rangle & \langle c_1^\dagger h_2\rangle & \langle c_1^\dagger c_2^\dagger\rangle \\ & \langle h_1 n_2\rangle & \langle c_1^\dagger c_2\rangle & \langle c_1^\dagger n_2\rangle \\ & & \langle n_1 h_2\rangle & \langle n_1 c_2^\dagger\rangle \\ & & & \langle n_1 n_2\rangle \end{pmatrix} , \tag{8.18}$$

where the blank entries can be determined from hermiticity. To see how the operators are determined, take the example of the top right element. The two basis states for that matrix element are $|0,0\rangle$ and $|1,1\rangle$.

2.8.2 Quantum Fisher information: a witness of multipartite entanglement

The quantum Fisher information (QFI) F_Q [454,455] quantifies how sensitive a state ρ is to unitary transformations generated by an observable $\hat{O}$. For a pure state $\rho = |\psi\rangle \langle\psi|$, $F_Q(\psi, \hat{O})$ is defined in terms of the variance of the operator $\hat{O}$ (i.e., the connected correlation function):

$$F_Q(\psi, \hat{O}) = 4\Delta(\hat{O})^2 = 4\left(\langle\psi|\hat{O}^2|\psi\rangle - \langle\psi|\hat{O}|\psi\rangle^2\right) , \tag{8.19}$$

and it provides an upper bound on the precision that can be achieved in measuring the parameter θ that is dual to the observable $\hat{O}$: $\mathcal{N}(\Delta\theta)^2 \geq F_Q^{-1}$, $\mathcal{N}$ being the number of independent measurements [454,455]. $F_Q(\psi, \hat{O})$ is thus a measure of the entanglement arising from quantum fluctuation content in $|\psi\rangle$ (considered with respect to an eigenstate of $\hat{O}$). This can be made more manifest by considering a traceless operator $\hat{M}$ with eigenstates $\{|m\rangle\}$. Without loss of generality, we can rescale the operator such that $\sum m^2 = 1/2$. If $\varepsilon_m(\psi) \equiv 1 - |\langle m|\psi\rangle|^2$ is the geometric entanglement between a given state $|\psi\rangle$ and the eigenstates of $\hat{M}$, the QFI corresponding to $\hat{M}$ in the state ψ can be written as

$$F_Q(\psi, \hat{M}) = 4\left[\frac{1}{2} - \sum_m m^2 \varepsilon_m - \left(\sum_m m\varepsilon_m\right)^2\right] . \quad (8.20)$$

The total geometric entanglement $\sum_m \varepsilon_m$ is constrained to be $d_M - 1$, where d_M is the dimension of the operator $\hat{M}$. Because each eigenvalue m has a magnitude of at most $1/\sqrt{2}$ (all m^2 must add up to $1/2$), the maximum magnitude of the last two terms in eq. (8.20) (and hence the minimum value of the QFI) is attained for the case when some of the ε_m are zero. However, because the eigenstates are orthogonal, only one of them can have perfect overlap with the state $|\psi\rangle$ and only one ε_m can be zero at a time. The case of minimum QFI therefore corresponds to $\varepsilon_{m^*} = 0, \varepsilon_{m\neq m^*} = 1$, leading to $F_Q = 0$. The opposite situation arises when all the geometric entanglement measures are non-zero and equal, $\varepsilon_m = 1/d_M$, leading to a maximum QFI value of $F_Q = 2/d_M$. A uniformly spread geometric entanglement distribution, therefore, leads to a larger QFI, while a more focused distribution leads to a smaller QFI. This sums up the relation between the quantum Fisher information and geometric entanglement.

If the operator $\hat{M}$ is such that its eigenstates are separable states from the perspective of a certain party, the vanishing QFI arises when $|\psi\rangle$ has zero geometric entanglement with respect to one of the separable states, and the QFI saturates when $|\psi\rangle$ has a finite geometric entanglement with all of the states. A large QFI can thus be associated with more entanglement. Moreover, if the eigenstates of $\hat{M}$ are symmetry-broken states, a vanishing QFI indicates that $|\psi\rangle$ is very close to such a symmetry-broken state, while a large QFI indicates that $|\psi\rangle$ is a uniform superposition of such symmetry-broken states and therefore preserves the symmetry as a whole. A larger QFI points towards the presence of more quantum fluctuations and a reduced susceptibility towards symmetry-breaking.

We will now relate the QFI to some other measures of correlation. The QFI corresponding to an observable can be directly related to the static lesser Greens functions associated with that observable:

$$F_Q(\psi, \hat{O}_2) = 4\left(\langle\hat{O}_2\hat{O}_1^\dagger\rangle\Big|_{\hat{O}_1^\dagger=\hat{O}_2} - \langle\hat{O}_2\hat{O}_1^\dagger\rangle^2\Big|_{\hat{O}_1=1}\right) = 4\left[G^<_{O_2,O_2}(t\to\infty) - \left(G^<_{1,O_2}(t\to\infty)\right)^2\right] \quad (8.21)$$

Further, the retarded Greens function defined above in eq. (??) can very generally be related to the QFI. By using the simplified forms $\frac{i}{2}G^R_{O_2,O_2}(t\to\infty) = \langle(O_2)^2\rangle$ and $\frac{i}{2}G^R_{1,O_2}(t\to\infty) = \langle O_2\rangle$, we get

$$F_Q(\psi, \hat{O}_2) = 2iG^R_{O_2,O_2}(t\to\infty) + \left(G^R_{1,O_2}(t\to\infty)\right)^2 . \quad (8.22)$$

For the sake of completeness, we generalise the above relations to finite temperatures by adopting the approach of Hauke et al. [455]. For a mixed state at inverse temperature β characterised by the density matrix $\rho = \sum_n f_n |n\rangle\langle n|$ where $(E_n, |n\rangle)$ are the eigenvalue-eigenstate pairs and $f_n = e^{-\beta E_n}/\sum_n e^{-\beta E_n}$ are the Boltzmann weights, the FQI can be assigned a spectral representation of the form [455]

$$F_Q(\rho, \hat{O}) = 2\sum_{m,n}\frac{(f_m - f_n)^2}{f_m + f_n}|\langle m|\hat{O}|n\rangle|^2 . \quad (8.23)$$

By using the identity $\frac{f_m - f_n}{f_m + f_n} = \int_{-\infty}^{\infty} d\omega \delta(\omega + E_m - E_n) \tanh(\beta\omega/2)$, the finite temperature FQI can be written in the form

$$F_Q(\rho, \hat{O}) = 2 \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \sum_{m,n} (f_m - f_n) \delta(\omega + E_m - E_n) |\langle m|\hat{O}|n\rangle|^2 . \tag{8.24}$$

In order to relate this with a correlation function, we now point out that the lesser Greens function $G^{<}$ and the greater Greens function $G^{>}$ that we will now define admit spectral representations in terms of the eigenstates $|n\rangle$ of the full problem:

$$G^{<}_{O^\dagger,O}(t) \equiv i \langle \hat{O}^\dagger \hat{O}(t) \rangle \implies G^{<}_{O^\dagger,O}(\omega) = 2\pi i \textstyle\sum_{m,n} f_n \delta(\omega + E_m - E_n) |\langle m|\hat{O}^\dagger|n\rangle|^2 , \tag{8.25}$$

$$G^{>}_{O^\dagger,O}(t) \equiv -i \langle \hat{O}(t) \hat{O}^\dagger \rangle \implies G^{>}_{O^\dagger,O}(\omega) = -2\pi i \textstyle\sum_{m,n} f_m \delta(\omega + E_m - E_n) |\langle m|\hat{O}|n\rangle|^2 . \tag{8.26}$$

If $\hat{O}$ is a Hermitian operator, eqs. (8.25) can be used to rewrite the QFI in terms of $G^{\lessgtr}_{O^\dagger,O}(\omega)$. Indeed, by combining eqs. (8.25) and eq. (8.24) with the assumption $\hat{O} = \hat{O}^\dagger$, we get

$$F_Q(\rho, \hat{O}) = -\frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \left(G^{>}_{O,O}(\omega) + G^{<}_{O,O}(\omega) \right) . \tag{8.27}$$

This constitutes the general relation between the finite temperature QFI for a general Hermitian many-particle operator $\hat{O}$ and the corresponding many-particle correlations at zero and finite frequencies.

The Kubo linear response function associated with the operator $\hat{O}(t)$ due to a perturbation from the same operator is associated with a correlation function/susceptibility $C_{\hat{O}}(t, t') = i\theta(t - t') \left\langle \left[\hat{O}(t), \hat{O}(t') \right] \right\rangle$. By going to the frequency domain, the imaginary part of such a function can be related to the sum of the lesser and greater Greens functions:

$$\begin{aligned}
C_{\hat{O}}(\omega) &= i \int_0^{\infty} d(t-t') e^{i\omega(t-t')} \left(\left\langle \hat{O}(t)\hat{O}(t') \right\rangle - \left\langle \hat{O}(t')\hat{O}(t) \right\rangle \right) \\
\implies C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^* &= i \int_0^{\infty} d(t-t') \left(e^{i\omega(t-t')} - e^{-i\omega(t-t')} \right) \left(\left\langle \hat{O}(t)\hat{O}(t') \right\rangle - \left\langle \hat{O}(t')\hat{O}(t) \right\rangle \right) \\
&= i \int_{-\infty}^{\infty} d(t-t') e^{i\omega(t-t')} \left(\left\langle \hat{O}(t)\hat{O}(t') \right\rangle - \left\langle \hat{O}(t')\hat{O}(t) \right\rangle \right) \\
&= -\left(G^{>}_{O,O}(\omega) + G^{<}_{O,O}(\omega) \right) .
\end{aligned}$$

This allows us to connect the QFI with the imaginary part of the Kubo correlation function/susceptibility obtained in Ref. [455]:

$$F_Q(\rho, \hat{O}) = \frac{1}{\pi i} \int_{-\infty}^{\infty} d\omega \tanh(\beta\omega/2) \left(C_{\hat{O}}(\omega) - C_{\hat{O}}(\omega)^* \right) . \tag{8.28}$$

2.8.3 Decomposition of entanglement entropy in terms of multipartite information I_n

We first define a partitioning of the entire system into N disjoint regions, $\mathcal{S}_N = \{\nu_1, \nu_2, \ldots \nu_N\}$. Going beyond bipartite mutual information, one can define the $n-$partite information I_n between the members of $\mathcal{S}_n = \{v_1, v_2, \ldots, v_n\}$, which is itself a subset of the exhaustive set of regions $\mathcal{S}_N$:

$$I_n(v_1, \ldots, v_n) = \sum_{\sigma \in \mathcal{P}(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{\mathrm{EE}}(\sigma) , \tag{8.29}$$

where σ is a subset drawn from the power set of $\mathcal{S}_n$, and $|\sigma|$ is the number of members in the subset σ. The 4-partite information can be written as

$$\begin{aligned} & I_4(v_1, v_2, v_3, v_4) \\ & = S_{\mathrm{EE}}(v_1) + S_{\mathrm{EE}}(v_2) + S_{\mathrm{EE}}(v_3) + S_{\mathrm{EE}}(v_4) - S_{\mathrm{EE}}(v_1 \cup v_2) - S_{\mathrm{EE}}(v_1 \cup v_3) \\ & - S_{\mathrm{EE}}(v_1 \cup v_4) - S_{\mathrm{EE}}(v_2 \cup v_3) - S_{\mathrm{EE}}(v_2 \cup v_4) - S_{\mathrm{EE}}(v_3 \cup v_4) + S_{\mathrm{EE}}(v_1 \cup v_2 \cup v_3) \\ & + S_{\mathrm{EE}}(v_1 \cup v_2 \cup v_4) + S_{\mathrm{EE}}(v_1 \cup v_3 \cup v_4) + S_{\mathrm{EE}}(v_2 \cup v_3 \cup v_4) + S_{\mathrm{EE}}(v_1 \cup v_2 \cup v_3 \cup v_4) . \end{aligned} \tag{8.30}$$

One can also define the average $n-$partite information $\bar{I}_n$ by considering all sets of n regions:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n) = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{\sigma \in \mathcal{P}(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{\mathrm{EE}}(\sigma) , \tag{8.31}$$

where $C_n(\mathcal{S}_N)$ represents all ways of choosing n distinct elements out of the full set $\mathcal{S}_N$ of subsystems.

The expression for $\bar{I}_n$ can be simplified by splitting the summation over the elements of the power set $\mathcal{P}$ into sets of definite number:

$$\bar{I}_n = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n) = \frac{(N-n)!n!}{N!} \sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{m=1}^{n} \sum_{\sigma_m \in \mathcal{P}_m(\mathcal{S}_n)} (-1)^{1+|\sigma|} S_{\mathrm{EE}}(\sigma_m) , \tag{8.32}$$

where $\mathcal{P}_m$ represents all subsets of the power set that have m elements. With some combinatorial magic, it can be shown that

$$\sum_{S_n \in C_n(\mathcal{S}_N)} \sum_{\sigma_m \in \mathcal{P}_m(\mathcal{S}_n)} = \binom{N-m}{n-m} \sum_{\sigma_m \in C_m(\mathcal{S}_N)} . \tag{8.33}$$

To see how this comes about, note that what this equation says is that summing over combinations of subsets of length n of the super set S_N and further summing over all subsets of length $m \leq n$ for each set S_n is equivalent to summing just once over subsets of length m of the super set S_N but now each subset of length m appears with a frequency $f_m = \binom{N-m}{n-m}$. The value for the frequency f_m is simply the number of subsets S_n in which the smaller subset σ_m appears. This can be calculated by allowing m of the n slots in S_n to be filled by the members of σ_m, and the remaining $n-m$ slots then have to be filled by choosing from the remaining $N-m$ members of the super set S_N; there's obviously $\binom{N-m}{n-m}$ ways of carrying this out.

Implementing this into the expression for $\bar{I}_n$ gives

$$\bar{I}_n = \sum_{m=1}^{n} (-1)^{1+m} \frac{\binom{n}{m}}{\binom{N}{m}} \sum_{\sigma_m \in C_m(\mathcal{S}_N)} S_{\mathrm{EE}}(\sigma_m) = \sum_{m=1}^{n} (-1)^{1+m} \binom{n}{m} S_{\mathrm{EE}}^{(m)} , \tag{8.34}$$

where we have define the average entanglement entropy $S_{\text{EE}}^{(m)}$ for all $m-$member sets of regions:

$$S_{\text{EE}}^{(m)} = \frac{1}{\binom{N}{m}} \sum_{\sigma_m \in C_m(\mathcal{S}_N)} S_{\text{EE}}(\sigma_m) \ . \tag{8.35}$$

With most of the machinery in place, we would now like to express the entanglement entropy S_{EE} purely in terms multipartite measures I_n, for the reasons stated above. For a system that is invariant under real-space translation (as our lattice model is), the entanglement entropy of any real-space interval should only depend on the extent of the interval. It therefore suffices to calculate the entanglement entropy averaged over all such intervals, and the corresponding measure in our formulation is $S_{\text{EE}}^{(1)}$.

We will now express $S_{\text{EE}}^{(1)}$ in terms of $\bar{I}_n$, by eliminating $S_{\text{EE}}^{(m)} (m > 1)$ from the equations. We adopt the ansatz

$$S_{\text{EE}}^{(1)} = \sum_{n=2}^{N} \lambda_n \bar{I}_n = \sum_{n=2}^{N} \lambda_n \binom{n}{m} \sum_{m=1}^{n} (-1)^{1+m} S_{\text{EE}}^{(m)} \ . \tag{8.36}$$

In order to separate $S_{\text{EE}}^{(1)}$ from the rest of the $S_{\text{EE}}^{(m)}$, we exchange the order of summation:

$$S_{\text{EE}}^{(1)} = \sum_{m=1}^{N} (-1)^{1+m} S_{\text{EE}}^{(m)} \sum_{n=m}^{N} \lambda_n \binom{n}{m} - S_{\text{EE}}^{(1)} \lambda_1 = S_{\text{EE}}^{(1)} \sum_{n=2}^{N} n\lambda_n - \sum_{m=2}^{N-1} S_{\text{EE}}^{(m)} (-1)^m \sum_{n=m}^{N} \binom{n}{m} \lambda_n \ , \tag{8.37}$$

where the constraint of $m \leq n$ in the previous form is now enforced by letting n run from m to N. The $N - 1$ coefficients $\{\lambda_n\}$ will be determined by requiring that the $S_{\text{EE}}^{(m)} (m > 1)$ vanish on the right-hand side, as well as by matching the coefficients of $S_{\text{EE}}^{(1)}$ on both sides. Note that m runs only upto $N - 1$ in the second term; this is because, $S_{\text{EE}}^{(N)}$ is the entanglement entropy of the full system and hence automatically zero. The equations are

$$\sum_{n=2}^{N} n\lambda_n = 1, \tag{8.38}$$

$$\sum_{n=m}^{N} \binom{n}{m} \lambda_n = 0; \ m = 2, \ldots, N - 1 \ . \tag{8.39}$$

We focus on the second equation first, since it is very short for m close to N. For $m = N - 1$ and $m = N - 2$, we get

$$\begin{aligned} m = N - 1 &: \lambda_{N-1} + N\lambda_N = 0 \implies \lambda_{N-1} = -N\lambda_N \\ m = N - 2 &: \lambda_{N-2} + (N-1)\lambda_{N-1} + \frac{N(N-1)}{2} \lambda_N = 0 \implies \lambda_{N-2} = \frac{N(N-1)}{2} \lambda_N \ . \end{aligned} \tag{8.40}$$

We extrapolate this to the general solution,

$$\lambda_m = (-1)^m \binom{N}{m} \lambda_N, \ m < N \ , \tag{8.41}$$

and then prove that if this is true for λ_{N-m}, it also solves the equation for λ_{N-m-1}, thus amounting to a proof by mathematical induction. The general equation for λ_{N-m-1} is

$$\sum_{n=N-m-1}^{N} \binom{n}{N-m-1}\lambda_n = 0 \implies \lambda_{N-m-1} + \sum_{m'=0}^{m} \binom{N-m'}{N-m-1}\lambda_{N-m'} = 0 \,. \tag{8.42}$$

Substituting the extrapolated solution gives

$$\begin{aligned}\lambda_{N-m-1} &= -\sum_{m'=0}^{m}(-1)^{N-m'}\binom{N-m'}{N-m-1}\binom{N}{m'}\lambda_N = \binom{N}{m+1}\lambda_N \sum_{m'=0}^{m}(-1)^{N-m'-1}\binom{m+1}{m'} \\ &= (-1)^{N-m-1}\binom{N}{m+1}\lambda_N \,,\end{aligned} \tag{8.43}$$

which is consistent with the general solution eq. 8.41. This determines all coefficients λ_n for $n < N$ in terms of λ_N. To determine λ_N itself, we use eq. 8.38. Substituting eq. 8.41 into it gives

$$\lambda_N \sum_{n=2}^{N} n(-1)^n \binom{N}{n} = 1 \implies \lambda_N = \frac{(-1)^N}{N} \,. \tag{8.44}$$

The decomposition of entanglement entropy into multipartite measures is now complete:

$$\begin{aligned} S^{(1)}_{\text{EE}} &= \frac{1}{N}\sum_{n=2}^{N}(-1)^n\binom{N}{n}\bar{I}_n, \\ \bar{I}_n &= \frac{(N-n)!n!}{N!}\sum_{S_n \in C_n(\mathcal{S}_N)} I_n(S_n),\end{aligned} \tag{8.45}$$

where we have assumed N is even in order to drop the global phase factor $(-1)^N$.

2.9 Wavefunction Renormalisation Group

2.9.1 General idea

A reverse URG analysis involves studying the evolution of wavefunctions from the IR to the UV, by implementing inverse renormalisation group transformations. This is made possible by the fact that the IR fixed point Hamiltonians are often tractable and allow the computation of wavefunctions. Such an analysis gives insight into how states change as high-energy quantum fluctuations are resolved during the Hamiltonian renormalisation from the UV to the IR. It also allows studying the evolution of entanglement measures and correlation functions.

We will start with a very simple IR ground state wavefunction, and then go back towards the UV ground state by applying the inverse unitary operator $U^\dagger$:

$$U : \underbrace{|1, 2, \ldots, N\rangle}_{\text{UV ground state}} \to |1, 2, \ldots, N-1\rangle\,|N\rangle \to \ldots \to \underbrace{|1, 2, \ldots, N^*\rangle\,|N^*+1\rangle \ldots |N\rangle}_{\text{IR ground state}}$$

$$U^\dagger : \underbrace{|1, 2, ..., N^*\rangle |N^* + 1\rangle ... |N\rangle}_{\text{IR ground state}} \to |1, 2, ..., N^* + 1\rangle |N^* + 2\rangle ... |N\rangle \to ... \to \underbrace{|1, , 2, ..., N\rangle}_{\text{UV ground state}}$$

The first process is the forward RG which we used to obtain the scaling equations. The second process is the reverse RG which we will undertake now. The method has already been applied to study the renormalisation of entanglement in various models of strong correlation. In general, we start with an IR wavefunction that consists a certain number of momentum states n_1 still entangled with the impurity, and the remaining momentum states n_2 disentangled from the impurity. The former are said to be part of the emergent Kondo cloud, while the latter are said to be part of the integrals of motion (IOMs) and appear in direct product with the cloud+impurity system in the ground state wavefunction. Each momentum state is tagged with two conduction bath levels, one above the Fermi surface and one below. These will be represented as $k, \pm$. If we represent the emergent cloud momentum states as $\{k_i\}$ and the IOM states as $\{q_i\}$, the ground state wavefunction can be written as

$$|\Psi\rangle_0 = \left(\otimes_{i=1}^{n_2} |q_i, -, \uparrow\rangle |q_i, -, \downarrow\rangle\right) |\Phi\rangle_{\text{cloud}} \left(\otimes_{i=1}^{n_2} |q_i, +, \uparrow\rangle |q_i, +, \downarrow\rangle\right) \tag{9.1}$$

Throughout, we have assumed that the IOMs below the Fermi surface are occupied while those above are unoccupied. This means:

$$|\Psi\rangle_0 = \left(\otimes_{i=1}^{n_2} |\hat{n}_{q_i,-,\uparrow} = 1\rangle |\hat{n}_{q_i,-,\downarrow} = 1\rangle\right) |\Phi\rangle_{\text{cloud}} \left(\otimes_{i=1}^{n_2} |\hat{n}_{q_i,+,\uparrow} = 0\rangle |\hat{n}_{q_i,+,\downarrow} = 0\rangle\right) \tag{9.2}$$

$|\Phi\rangle_{\text{cloud}}$ will be obtained by diagonalising a small system of n_1 conduction bath k states. This completes the construction of the IR ground state $|\Psi\rangle_0$. The algorithm of reverse RG then involves applying unitary transformations on this IR wavefunction; the unitary transformations will be the inverse of those that were applied during the forward RG algorithm. Since the forwards RG transformations decoupled k states and transformed them into the IOMS, the inverse transformations will take the momentum states in the IOMs and re-entangle them with the impurity+cloud system, one at a time. This is shown in fig. 2.3.

The next step is to write down the unitaries that will take us from the IR ground state to the UV ground state. In the forward RG, we used the following unitary for decoupling the shell ϵ_q and spin β is

$$U_{q\beta} = \frac{1}{\sqrt{2}}\left[1 + \eta_{0\beta} + \eta^\dagger_{1\beta}\right] . \tag{9.3}$$

Here, the subscripts 0 and 1 indicate it decouples an electron above and below the Fermi surface respectively. The inverse transformation for re-entangling $q\beta$ is

$$U^\dagger_{q\beta} = \frac{1}{\sqrt{2}}\left[1 + \eta^\dagger_{0\beta} + \eta_{1\beta}\right] . \tag{9.4}$$

The wavefunction after reversing one step of the RG will thus be

$$|\Psi\rangle_1 = U^\dagger_{q\uparrow} U^\dagger_{q\downarrow} |\Psi\rangle_0 \tag{9.5}$$

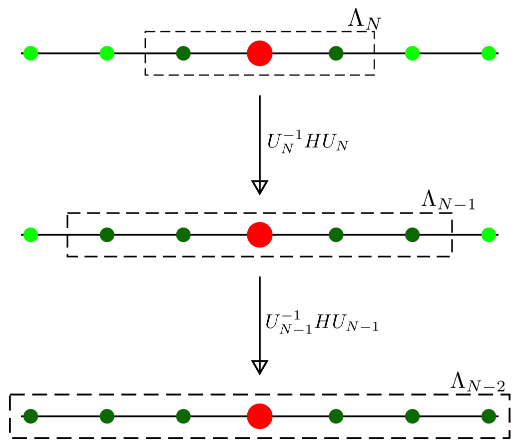

Figure 2.3: *We start with a Hamiltonian with an impurity site (red) coupled with two conduction electrons (dark green), with four other decoupled electrons (bright green). The dotted rectangle represents the emergent window $\left(-\Lambda_j, \Lambda_j\right)$ at each step; the electrons inside that rectangle are still entangled with the impurity, while the ones inside have been decoupled. The next step of reverse RG involves applying the inverse transformation on the Hamiltonian, which will couple two more electrons from the IOMS (hence four dark green circles in the second step), leading to an enlargement of the emergent window. The unitary varies for each step, hence the notation U_j.*

Here, q is the first momentum state immediately outside the emergent cloud. Re-entangling further momentum states using their respective unitaries lead to the sequence of wavefunctions $|\Psi\rangle_2, |\Psi\rangle_3$ and so on. A straightforward way of calculating entanglement and correlations across the RG flow is to create a second-quantised prototypical form of the IR state $|\Psi_0\rangle$, and then apply the inverse unitary operators $\{U_i\}$ to this wavefunction in order to generate the family of states $|\Psi_i\rangle = U_i |\Psi_{i-1}\rangle, i > 0$. Quantities like entanglement entropy and correlations can then be calculated by using this family of states. If the unitary is of a specific form, there is instead a more efficient method of doing this, which is what we will describe in the next subsection.

2.9.2 Evolution equation for state tensor coefficients

In the following, we restrict ourselves to the reverse RG analysis of an impurity problem hybridising with the local degree of freedom of a conduction bath. At every step, we allow two states from the IOMS to interact with the already-entangled states (one above the Fermi energy and one below, in order to maintain particle-hole symmetry). After the j^{th} application of the reverse RG transformation, $2j$ number of states are entangled with the impurity site, and the state can be represented as

$$|\Psi_j\rangle = \sum_{\nu_d, \nu_1, \ldots, \nu_{2j}} C^{(j)}_{\nu_d, \nu_1, \ldots, \nu_{2j}, I_{2j+1}} |\nu_d\rangle |\nu_1\rangle |\nu_2\rangle \ldots |\nu_{2j}\rangle \otimes |I_{2j+1}\rangle , \tag{9.6}$$

where $|\nu_d\rangle$ and $|\nu_i\rangle$ $(i \in [1, 2j])$ sum over the configurations of the impurity site and the effective zeroth site configuration at the j^{th} step. $|I_{2j+1}\rangle$ is the configuration of the integrals of motion from the $(2j+1)^{\text{th}}$ site and beyond. We define the product state $|\psi_{dz}(j)\rangle = |\nu_d\rangle |\nu_1\rangle \ldots |\nu_{2j}\rangle$ for future reference.

We consider a restricted form of RG evolution, where the unitary transformation from IR towards UV can be expressed at any step as

$$U_j^\dagger = 1 + \sum_{i\in[1,2j]} \sum_{\nu_d,\mu_d,\nu_i,\mu_i} \mathcal{T}_{\nu_d}^{\mu_d}\left(J_{\nu_d,\nu_i}^{\mu_d,\mu_i} c_{2j+1}\mathcal{T}_{\nu_i}^{\mu_i} + K_{\nu_d,\nu_i}^{\mu_d,\mu_i} c_{2j+2}^\dagger \mathcal{T}_{\nu_i}^{\mu_i}\right) , \tag{9.7}$$

where $\mathcal{T}_\alpha^\nu = |\nu\rangle\langle\alpha|$ is a transition operator and sums over all transition operators of the specific subspace (impurity or bath sites). We first obtain the action of the transformation $U_j^\dagger$ on the basis states. We have

$$\begin{aligned}
&\mathcal{T}_{\mu_d}^{\nu_d} c_{2j+2}^\dagger \mathcal{T}_{\mu_i}^{\nu_i} |\mu_d\rangle \ldots |\mu_i\rangle \ldots |2_{2j+1}\rangle |0_{2j+2}\rangle \\
&= \mathcal{T}_{\mu_d}^{\nu_d} c_{2j+2}^\dagger (-1)^{\mu_d+\sum_{k<i}\nu_i} |\mu_d\rangle \ldots |\nu_i\rangle \ldots |2_{2j+1}\rangle |0_{2j+2}\rangle \\
&= \mathcal{T}_{\mu_d}^{\nu_d} (-1)^{\mu_d+\sum_{k<i}\nu_i} (-1)^{\mu_d+\sum_k \nu_i} |\mu_d\rangle \ldots |\nu_i\rangle \ldots |2_{2j+1}\rangle |1_{2j+2}\rangle \\
&= (-1)^{\sum_{k\geq i}\nu_i} |\nu_d\rangle \ldots |\nu_i\rangle \ldots |2_{2j+1}\rangle |1_{2j+2}\rangle \ ,
\end{aligned} \tag{9.8}$$

and

$$\mathcal{T}_{\mu_d}^{\nu_d} c_{2j+1} \mathcal{T}_{\mu_i}^{\nu_i} |\mu_d\rangle \ldots |\mu_i\rangle \ldots |2_{2j+1}\rangle |0_{2j+2}\rangle = (-1)^{\sum_{k\geq i}\nu_i} |\nu_d\rangle \ldots |\nu_i\rangle \ldots |1_{2j+1}\rangle |0_{2j+2}\rangle \ . \tag{9.9}$$

Using these expressions, we construct the state at the $(j+1)^{\text{th}}$ step:

$$\begin{aligned}
|\Psi_{j+1}\rangle = &\sum_{\nu_d,\nu_1,\ldots,\nu_j} C_{\nu_d,\nu_1,\ldots,}^{(j)} |\psi_{dz}(j)\rangle |2_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\
&+ \sum_i \sum_{\mu_d,\mu_i,\nu_d,\nu_i} C_{\mu_d,\ldots,\mu_i,\ldots,}^{(j)} J_{\mu_d,\mu_i}^{\nu_d,\nu_i} (-1)^{\sum_{k\geq i}\nu_k} |\psi_{dz}(j)\rangle |1_{2j+1}\rangle |0_{2j+2}\rangle |I_{2j+3}\rangle \\
&+ \sum_i \sum_{\mu_d,\mu_i,\nu_d,\nu_i} C_{\mu_d,\ldots,\mu_i,\ldots}^{(j)} K_{\mu_d,\mu_i}^{\nu_d,\nu_i} (-1)^{\sum_{k\geq i}\nu_k} |\psi_{dz}(j)\rangle |2_{2j+1}\rangle |1_{2j+2}\rangle |I_{2j+3}\rangle
\end{aligned} \tag{9.10}$$

Gathering similar terms and normalising the state, we can write it as

$$\begin{aligned}
|\Psi_{j+1}\rangle = &\frac{1}{L_j} \sum_{\nu_d,\{\nu_i\}} |\psi_{dz}(j)\rangle \Big[C_{\nu_d,\{\nu_i\},1_{2j+1},0_{2j+2},I_{j+1}}^{(j)} |2_{2j+1}\rangle |0_{2j+2}\rangle + C_{+,\nu_d,\{\nu_i\}}^{(j)} |2_{2j+1}\rangle |1_{2j+2}\rangle \\
&+ C_{-,\nu_d,\{\nu_i\}}^{(j)} |1_{2j+1}\rangle |0_{2j+2}\rangle \Big] |I_{2j+3}\rangle \ ,
\end{aligned} \tag{9.11}$$

where $L_j^2 = \sum_{\nu_d,\{\nu_i\}} \left({C_{\nu_d,\{\nu_i\}}^{(j)}}^2 + {C_{+,\nu_d,\{\nu_i\}}^{(j)}}^2 + {C_{-,\nu_d,\{\nu_i\}}^{(j)}}^2 \right)$ defines the normalisation factor, and

$$\begin{aligned}
C_{+,\nu_d,\{\nu_i\}} &= \sum_{i,\mu_d,\mu_i} C_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}}^{(j)} K_{\mu_d,\mu_i}^{\nu_d,\nu_i} (-1)^{\sum_{k\geq i}\nu_k} \\
C_{-,\nu_d,\{\nu_i\}} &= \sum_{i,\mu_d,\mu_i} C_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}}^{(j)} J_{\mu_d,\mu_i}^{\nu_d,\nu_i} (-1)^{\sum_{k\geq i}\nu_k} \ .
\end{aligned} \tag{9.12}$$

We can now read off the renormalisation of the state tensor coefficients:

$$
\begin{aligned}
C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},1_{2j+1},0_{2j+2},I_{2j+3}} &= \frac{1}{L_j} C^{(j)}_{\nu_d,\nu_1,\ldots,\nu_{2j},2_{2j+1},0_{2j+2},I_{2j+3}} ~, \\
C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},2_{2j+1},1_{2j+2},I_{2j+3}} &= \frac{1}{L_j} \sum_i (-1)^{\sum_{k\geq i}\nu_k} \sum_{\mu_d,\mu_i} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} K^{\nu_d,\nu_i}_{\mu_d,\mu_i} ~, \\
C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},1_{2j+1},0_{2j+2},I_{2j+3}} &= \frac{1}{L_j} \sum_i (-1)^{\sum_{k\geq i}\nu_k} \sum_{\mu_d,\mu_i} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} J^{\nu_d,\nu_i}_{\mu_d,\mu_i} ~,
\end{aligned} \tag{9.13}
$$

We now make an important assumption: the coefficients $K^{\nu_d,\nu_i}_{\mu_d,\mu_i}$ and $J^{\nu_d,\nu_i}_{\mu_d,\mu_i}$ are uniform for the set of states $S(\nu_i)$ and zero for other states. That gives

$$
\begin{aligned}
C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},1_{2j+1},1_{2j+2},I_{2j+3}} &= \frac{1}{L_j} K \sum_i (-1)^{\sum_{k\geq i}\nu_k} \sum_{\substack{\mu_d\in S(\nu_d),\\ \mu_i\in S(\nu_i)}} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} ~, \\
C^{(j+1)}_{\nu_d,\nu_1,\ldots,\nu_{2j},1_{2j+1},0_{2j+2},I_{2j+3}} &= \frac{1}{L_j} J \sum_i (-1)^{\sum_{k\geq i}\nu_k} \sum_{\substack{\mu_d\in S(\nu_d),\\ \mu_i\in S(\nu_i)}} C^{(j)}_{\mu_d,\ldots,\mu_i,\ldots,\nu_{2j},1_{2j+1},0_{2j+2}} ~,
\end{aligned} \tag{9.14}
$$

One can also look at renormalisation of expectation values of operators; this is useful for tracking the evolution of correlations and entanglement measures. Consider an operator $O^{\nu_k}_{\mu_k}$ that acts on the k^{th} bath site. The expectation value of this operator at any step can be written as

$$
\begin{aligned}
\langle O^{\mu_k}_{\nu_k}\rangle_j &= \sum_{\nu_d,\nu_1,\ldots,\nu_{k-1},\nu_{k+1},\ldots,\nu_{2j}} C^{(j)}_{\nu_d,\ldots,\mu_k,\ldots} C^{(j)}_{\nu_d,\ldots,\nu_k,\ldots} ~, \\
\langle O^{\mu_k}_{\nu_k}\rangle_{j+1} &= \sum_{\nu_d,\nu_1,\ldots,\nu_{k-1},\nu_{k+1},\ldots,\nu_{2j+2}} C^{(j+1)}_{\nu_d,\ldots,\mu_k,\ldots} C^{(j+1)}_{\nu_d,\ldots,\nu_k,\ldots}
\end{aligned} \tag{9.15}
$$

2.10 Kondo model spectral function

2.10.1 Single-impurity Kondo model

For the Kondo model defined by the Hamiltonian

$$
H_K = \sum_{k,\sigma} \epsilon_k n_{k,\sigma} + J \sum_{k_1,k_2} \mathbf{S}_d \cdot \sigma_{\alpha,\beta} c^\dagger_{k_1,\alpha} c_{k_2,\beta} ~, \tag{10.1}
$$

the impurity spin S_d does not have any fermionic operator. In order to define a spectral function for the impurity, we therefore look at the scattering of conduction electrons off it. This is captured by the conduction bath Greens function $G_{k,k'}$, which can be expressed in terms of a $T-$matrix when there are two-particle scattering processes present in the system:

$$
G_{k,k'}(\omega) = G^{(0)}_k(\omega) + G^{(0)}_k(\omega) T_{k,k'}(\omega) G^{(0)}_{k'}(\omega) ~, \tag{10.2}
$$

where $T_{k,k'}$ is non-zero because of the presence of the impurity spin which scatters momentum states into each other. Because of the local nature of the impurity scattering, the $T-$matrix is also local: $T_{k,k'}(\omega) = T(\omega)$. We therefore use this to define a spectral function:

$$A_{\text{imp}} = -\frac{1}{\pi}\text{Im } T(\omega) \ . \tag{10.3}$$

In order to calculate $T(\omega)$, we evaluate $G_{k,k'}$ using the equation of motion method. It is define as

$$G_{k,k'}(t) = -i\theta(t)\langle\langle c_{k,\sigma}(t), c^\dagger_{k',\sigma}\rangle\rangle \ . \tag{10.4}$$

Differentiating against time gives

$$\partial_t G_{k,k'}(t) = -i\delta(t)\langle\langle c_{k,\sigma}, c^\dagger_{k',\sigma}\rangle\rangle - \theta(t)\langle\langle [c_{k,\sigma}(t), H_K], c^\dagger_{k',\sigma}\rangle\rangle \ . \tag{10.5}$$

Evaluating the commutator gives

$$[c_{k,\sigma}(t), H_K] = \epsilon_k c_{k,\sigma} + J\sum_{q,\beta} \mathbf{S}_d \cdot \sigma_{\sigma,\beta} c_{q\beta} \ . \tag{10.6}$$

For convenience we define the local bath operator $c_{0,\beta} = \sum_q c_{q,\beta}$. Substituting the commutator back into the expression gives

$$\partial_t G_{k,k'}(t) = -i\delta(t)\delta_{k,k'} - \theta(t)\epsilon_k\langle\langle c_{k,\sigma}(t), c^\dagger_{k',\sigma}\rangle\rangle - \theta(t)J\sum_\beta\langle\langle \mathbf{S}_d \cdot \sigma_{\sigma,\beta}c_{0,\beta}, c^\dagger_{k',\sigma}\rangle\rangle(t) \ . \tag{10.7}$$

Defining a new Greens function $\tilde{G}$,

$$\tilde{G}_{k'}(t) = -i\theta(t)\sum_\beta\langle\langle \mathbf{S}_d \cdot \sigma_{\sigma,\beta}c_{0,\beta}, c^\dagger_{k',\sigma}\rangle\rangle \ , \tag{10.8}$$

we have the compact form

$$\partial_t G_{k,k'}(t) = -i\delta(t)\delta_{k,k'} - \theta(t)\epsilon_k\langle\langle c_{k,\sigma}(t), c^\dagger_{k',\sigma}\rangle\rangle - i\tilde{G}_{k'}(t) \ . \tag{10.9}$$

We now transform to frequency space, using the definition $f(t) = \int d\omega e^{-i\omega t} g(\omega)$:

$$\omega G_{k,k'}(\omega) = \delta_{k,k'} + \epsilon_k G_{k,k'}(\omega) + J\tilde{G}_{k'}(\omega) \ . \tag{10.10}$$

We setup the equation of motion for $\tilde{G}$ in a similar fashion. We use the Hermitian conjugate $-i\theta(t)\sum_\beta\langle\langle c_{k',\sigma}(t), \mathbf{S}_d \cdot \sigma_{\sigma,\beta}c^\dagger_{0,\beta}\rangle\rangle$, because it is easier to work with:

$$\begin{aligned}
\partial_t \tilde{G}_{k'}(t) &= -i\delta(t)\sum_\beta \delta_{\sigma,\beta}\langle \mathbf{S}_d \cdot \sigma_{\sigma,\beta}\rangle - \theta(t)\langle\langle [c_{k',\sigma}(t), H_K], \sum_\beta \mathbf{S}_d \cdot \sigma_{\sigma,\beta}c^\dagger_{0,\beta}\rangle\rangle \\
&= -i\delta(t)\langle \mathbf{S}^z_d\rangle - \theta(t)\langle\langle (\epsilon_{k'}c_{k',\sigma}(t) + J\sum_\alpha \mathbf{S}_d \cdot \sigma_{\sigma,\alpha}c_{0,\alpha}), \sum_\beta \mathbf{S}_d \cdot \sigma_{\sigma,\beta}c^\dagger_{0,\beta}\rangle\rangle \\
&= -i\delta(t)\langle \mathbf{S}^z_d\rangle - i\epsilon_{k'}G_{k'}(t) - J\theta(t)\langle\langle \sum_\alpha \mathbf{S}_d \cdot \sigma_{\sigma,\alpha}c_{0,\alpha}, \sum_\beta \mathbf{S}_d \cdot \sigma_{\sigma,\beta}c^\dagger_{0,\beta}\rangle\rangle \ .
\end{aligned} \tag{10.11}$$

At this point, we define a Kondo excitation operator $O_\sigma = J\sum_\alpha \mathbf{S}_d \cdot \sigma_{\sigma,\alpha} c_{0,\alpha}$. Transforming to frequency space gives

$$\omega \tilde{G}_{k'}(\omega) = \langle S_d^z \rangle + \epsilon_{k'} G_{k'}(\omega) + \frac{1}{J}\langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle \ , \tag{10.12}$$

which gives an expression for $\tilde{G}_{k'}(\omega)$:

$$(\omega - \epsilon_{k'})\tilde{G}_{k'}(\omega) = \langle S_d^z \rangle + \frac{1}{J}\langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle \ . \tag{10.13}$$

Substituting this into eq. 10.10 and noting that the non-interacting Greens function of the conduction bath is $G_k^{(0)} = (\omega - \epsilon_k)^{-1}$, we get

$$(G_k^{(0)})^{-1} G_{k,k'}(\omega) = \delta_{k,k'} + J\left[\langle S_d^z \rangle + \frac{1}{J}\langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle\right] G_{k'}^{(0)} \ , \tag{10.14}$$

which can be finally compared with the general expression in eq. 10.2:

$$G_{k,k'}(\omega) = G_k^{(0)} + G_k^{(0)}\left[J\langle S_d^z \rangle + \langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle\right] G_{k'}^{(0)} \ . \tag{10.15}$$

The $T-$matrix for scattering of conduction electrons off the impurity can be read off:

$$T_\sigma(\omega) = J\langle S_d^z \rangle + \langle\langle O_\sigma, O_\sigma^\dagger \rangle\rangle \ , \tag{10.16}$$

2.10.2 Two-impurity Kondo model

We now generalise this calculation to a two-impurity model S_1, S_2, each with it's own conduction bath:

$$H_{2K} = \sum_{k,\sigma,l} \epsilon_k n_{k,\sigma,l} + J_l \sum_{k_1,k_2,l,\sigma_1,\sigma_2} \mathbf{S}_l \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{k_1,\sigma_1,l} c_{k_2,\sigma_2,l} + K\mathbf{S}_1 \cdot \mathbf{S}_2 \ , \tag{10.17}$$

where l sums over the two spins and respective conduction baths, and K is the inter-impurity spin-exchange coupling.

Because of the composite nature of the impurity in this model, we consider a Greens function matrix

$$G = \begin{pmatrix} G_{11} & G_{12} \\ G_{21} & G_{22} \end{pmatrix} \ , \tag{10.18}$$

consisting of the diagonal Greens functions G_{ll} on the two impurities as well as an off-diagonal Greens function G_{12}. This is associated with a similar structure in the $T-$matrices:

$$T(k,k') = \begin{pmatrix} T(k,k')_{11} & T(k,k')_{12} \\ T(k,k')_{21} & T(k,k')_{22} \end{pmatrix} \ , \tag{10.19}$$

where the $T_{l,l'}$ are defined similar to the one-impurity case, but now with the additional index:

$$G_{(k,l),(k',l')}(\omega) = G^{(0)}_{k,l}(\omega) + G^{(0)}_{k,l}(\omega) T_{(k,l),(k',l')}(\omega) G^{(0)}_{k',l'}(\omega) \ . \tag{10.20}$$

The additional physics here is the possibility of scattering from a momentum state k in the bath l to a momentum state k' in the other bath l'.

We can then use the $T-$matrices to define diagonal and off-diagonal spectral functions:

$$A_{\text{imp}}(l,l') = -\frac{1}{\pi}\text{Im } T_{l,l'}(\omega) \ . \tag{10.21}$$

In order to calculate $T(\omega)$, we evaluate $G_{(k,l),(k',l')}$ using the equation of motion method. It is define as

$$G(t) = -i\theta(t)\langle\langle c_{k,\sigma,l}(t), c^\dagger_{k',\sigma,l'}\rangle\rangle \ . \tag{10.22}$$

Differentiating against time gives

$$\partial_t G_{k,k'}(t) = -i\delta(t)\langle\langle c_{k,\sigma,l}, c^\dagger_{k',\sigma,l'}\rangle\rangle - \theta(t)\langle\langle [c_{k,\sigma,l}(t), H_{2K}], c^\dagger_{k',\sigma,l'}\rangle\rangle \ . \tag{10.23}$$

Evaluating the commutator gives

$$[c_{k,\sigma,l}(t), H_{2K}] = \epsilon_k c_{k,\sigma,l} + J_l \sum_{q,\beta} \mathbf{S}_l\cdot\sigma_{\sigma,\beta} c_{q,\beta,l} \ . \tag{10.24}$$

For convenience we define the local bath operator $c_{0,\beta,l} = \sum_q c_{q,\beta,l}$. Substituting the commutator back into the expression gives

$$\begin{aligned}\partial_t G_{(k,l),(k',l')}(t) = &-i\delta(t)\delta_{k,k'}\delta_{l,l'} - \theta(t)\epsilon_k\langle\langle c_{k,\sigma,l}(t), c^\dagger_{k',\sigma,l'}\rangle\rangle \\ &-\theta(t)J_l\sum_\beta\langle\langle \mathbf{S}_l\cdot\sigma_{\sigma,\beta}c_{0,\beta,l}, c^\dagger_{k',\sigma,l'}\rangle\rangle(t) \ .\end{aligned} \tag{10.25}$$

Defining a new Greens function $\tilde{G}$,

$$\tilde{G}_{(0,l),(k',l')}(t) = -i\theta(t)\sum_\beta\langle\langle \mathbf{S}_l\cdot\sigma_{\sigma,\beta}c_{0,\beta,l}, c^\dagger_{k',\sigma,l'}\rangle\rangle \ , \tag{10.26}$$

we have the compact form

$$\partial_t G_{(k,l),(k',l')}(t) = -i\delta(t)\delta_{k,k'}\delta_{l,l'} - \theta(t)\epsilon_k\langle\langle c_{k,\sigma,l}(t), c^\dagger_{k',\sigma,l'}\rangle\rangle - i\tilde{G}_{(0,l),(k',l')}(t) \ . \tag{10.27}$$

We now transform to frequency space, using the definition $f(t) = \int d\omega e^{-i\omega t} g(\omega)$:

$$\omega G_{(k,l),(k',l')}(\omega) = \delta_{k,k'}\delta_{l,l'} + \epsilon_k G_{(k,l),(k',l')}(\omega) + J_l\tilde{G}_{(0,l),(k',l')}(\omega) \ . \tag{10.28}$$

We setup the equation of motion for $\tilde{G}$ in a similar fashion. We use the Hermitian conjugate $-i\theta(t)\sum_\beta\langle\langle c_{k',\sigma,l'}(t), \mathbf{S}_l\cdot\sigma_{\sigma,\beta}c^\dagger_{0,\beta,l}\rangle\rangle$, because it is easier to work with:

$$\begin{aligned}\partial_t\tilde{G}_{(0,l),(k',l')}(t) &= -i\delta(t)\delta_{l,l'}\sum_\beta\delta_{\sigma,\beta}\langle\mathbf{S}_l\cdot\sigma_{\sigma,\beta}\rangle - \theta(t)\langle\langle[c_{k',\sigma,l'}(t), H_{2K}], \sum_\beta \mathbf{S}_l\cdot\sigma_{\sigma,\beta}c^\dagger_{0,\beta,l}\rangle\rangle \\ &= -i\delta(t)\delta_{l,l'}\langle\mathbf{S}^z_l\rangle - \theta(t)\langle\langle\epsilon_{k'}c_{k',\sigma,l'}(t) + J_{l'}\sum_\alpha\mathbf{S}_{l'}\cdot\sigma_{\sigma,\alpha}c_{0,\alpha,l'}, \sum_\beta\mathbf{S}_l\cdot\sigma_{\sigma,\beta}c^\dagger_{0,\beta,l}\rangle\rangle \\ &= -i\delta(t)\langle\mathbf{S}^z_l\rangle - i\epsilon_{k'}\tilde{G}_{(0,l),(k',l')}(t) - J_{l'}\theta(t)\langle\langle\sum_\alpha\mathbf{S}_{l'}\cdot\sigma_{\sigma,\alpha}c_{0,\alpha,l'}, \sum_\beta\mathbf{S}_l\cdot\sigma_{\sigma,\beta}c^\dagger_{0,\beta,l}\rangle\rangle \ .\end{aligned} \tag{10.29}$$

At this point, we define a Kondo excitation operator $O_{\sigma,l} = J_l \sum_\alpha \mathbf{S}_l \cdot \sigma_{\sigma,\alpha} c_{0,\alpha,l}$. Transforming to frequency space gives

$$\omega \tilde{G}_{(0,l),(k',l')}(\omega) = \delta_{l,l'} \langle S_l^z \rangle + \epsilon_{k'} \tilde{G}_{(0,l),(k',l')}(\omega) + \frac{1}{J_l} \langle\langle O_{\sigma,l'}, O^\dagger_{\sigma,l} \rangle\rangle \ , \tag{10.30}$$

which gives an expression for $\tilde{G}_{k'}(\omega)$:

$$(\omega - \epsilon_{k'}) \tilde{G}_{(0,l),(k',l')}(\omega) = \delta_{l,l'} \langle S_l^z \rangle + \frac{1}{J_l} \langle\langle O_{\sigma,l'}, O^\dagger_{\sigma,l} \rangle\rangle \ . \tag{10.31}$$

Substituting this into eq. 10.10 and noting that the non-interacting Greens function of the conduction bath is $G_k^{(0)} = (\omega - \epsilon_k)^{-1}$, we get

$$(G_k^{(0)})^{-1} G_{(k,l),(k',l')}(\omega) = \delta_{k,k'} \delta_{l,l'} + J_l \left[\delta_{l,l'} \langle S_l^z \rangle + \frac{1}{J_l} \langle\langle O_{\sigma,l}, O^\dagger_{\sigma,l'} \rangle\rangle \right] G_{k'}^{(0)} \ , \tag{10.32}$$

which can be finally compared with the general expression in eq. 10.2:

$$G_{(k,l),(k',l')}(\omega) = \delta_{k,k'} \delta_{l,l'} G_k^{(0)} + G_k^{(0)} \left[\delta_{l,l'} J_l \langle S_l^z \rangle + \langle\langle O_{\sigma,l}, O^\dagger_{\sigma,l'} \rangle\rangle \right] G_{k'}^{(0)} \ . \tag{10.33}$$

The $T-$matrix for scattering of conduction electrons off the impurity can be read off:

$$T_{(\sigma,l),(\sigma,l')}(\omega) = \delta_{l,l'} J_l \langle S_l^z \rangle + \langle\langle O_{\sigma,l}, O^\dagger_{\sigma,l'} \rangle\rangle \ , \tag{10.34}$$

2.11 Zero temperature Greens function in frequency domain

2.11.1 Spectral representation of Greens function

The impurity retarded Green's function is defined as

$$G_{d\sigma}(t) = -i\theta(t) \left\langle \left\{ O_\sigma(t), O_\sigma^\dagger \right\} \right\rangle \tag{11.1}$$

where the average $\langle\rangle$ is over a canonical ensemble at temperature T, and O_σ is the excitation whose spectral function we are interested in. What follows is a standard calculation where we write the Green's function in the spectral representation. The ensemble average for the operator O_σ can be written in terms of the exact eigenstates of the Hamiltonian:

$$H |n\rangle = E_n |n\rangle \ , \quad \langle O_\sigma \rangle \ \equiv \frac{1}{Z} \sum_n \langle n | O_\sigma | n \rangle \, e^{-\beta E_n} \tag{11.2}$$

where $Z = \sum_n e^{-\beta E_n}$ is the partition function and $\{|n\rangle\}$ is the set of eigenfunctions of the Hamiltonian. We can therefore write

$$\begin{aligned}
&\left\langle \left\{ O_\sigma(t), O_\sigma^\dagger \right\} \right\rangle \\
&= \frac{1}{Z} \sum_m e^{-\beta E_m} \langle m| \left\{ O_\sigma(t), O_\sigma^\dagger \right\} |m\rangle \\
&= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \langle m| \left(O_\sigma(t) |n\rangle \langle n| O_\sigma^\dagger + O_\sigma^\dagger |n\rangle \langle n| O_\sigma(t) \right) |m\rangle \qquad \left[\sum_n |n\rangle \langle n| = 1 \right] \\
&= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \langle m| \left(e^{iH^* t} O_\sigma e^{-iH^* t} |n\rangle \langle n| O_\sigma^\dagger + O_\sigma^\dagger |n\rangle \langle n| e^{iH^* t} O_\sigma e^{-iH^* t} \right) |m\rangle \\
&= \frac{1}{Z} \sum_{m,n} e^{-\beta E_m} \left(e^{i(E_m - E_n)t} \langle m| O_\sigma |n\rangle \langle n| O_\sigma^\dagger |m\rangle + e^{i(E_n - E_m)t} \langle m| O_\sigma^\dagger |n\rangle \langle n| O_\sigma |m\rangle \right) \\
&= \frac{1}{Z} \sum_{m,n} e^{i(E_m - E_n)t} || \langle m| O_\sigma |n\rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right)
\end{aligned} \tag{11.3}$$

The time-domain impurity Green's function can thus be written as (this is the so-called Lehmann-Kallen representation)

$$G_{d\sigma} = -i\theta(t) \frac{1}{Z} \sum_{m,n} e^{i(E_m - E_n)t} || \langle m| O_\sigma |n\rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) \tag{11.4}$$

We are interested in the frequency domain form.

$$\begin{aligned}
G_{dd}^\sigma(\omega) &= \int_{-\infty}^{\infty} dt e^{i(\omega)t} G_{dd}^\sigma(t) \\
&= \frac{1}{Z} \sum_{m,n} || \langle m| O_\sigma |n\rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n} \right) (-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega + E_m - E_n)t}
\end{aligned} \tag{11.5}$$

To evaluate the time-integral, we will use the integral representation of the Heaviside function:

$$\theta(t) = -\frac{1}{2\pi i} \lim_{\eta \to 0^+} \int_{-\infty}^{\infty} \frac{1}{x + i\eta} e^{-ixt} dx \tag{11.6}$$

With this definition, the integral in $G_{d\sigma}(\omega)$ becomes

$$\begin{aligned}
(-i) \int_{-\infty}^{\infty} dt \theta(t) e^{i(\omega + i0^+ + E_m - E_n)t} &= (-i) \frac{-1}{2\pi i} \lim_{\eta \to 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} \int_{-\infty}^{\infty} dt e^{i(\omega + i0^+ + E_m - E_n - x)t} \\
&= \frac{1}{2\pi} \lim_{\eta \to 0^+} \int_{-\infty}^{\infty} dx \frac{1}{x + i\eta} 2\pi\delta\left(\omega + i0^+ + E_m - E_n - x\right) \\
&= \frac{1}{\omega + E_m - E_n + i0^+} .
\end{aligned} \tag{11.7}$$

The frequency-domain Green's function is thus

$$G_{dd}^{\sigma}(\omega) = \frac{1}{Z}\sum_{m,n} || \langle m| O_\sigma |n\rangle ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) \frac{1}{\omega + E_m - E_n + i0^+} \tag{11.8}$$

The infinitesimally positive imaginary part in the denominator shifts the poles onto the lower half of the complex plane, leaving it analytic in the upper half: this is necessary to make the retarded Greens function causal.

2.11.2 Real and imaginary parts - The Sokhotski-Plemelj theorem

In order to write down the real and imaginary parts of the Greens function, we first prove the *Sokhotski-Plemelj* formula:

$$\lim_{\eta\to 0^+} \frac{1}{x + i\eta} = P\left(\frac{1}{x}\right) - i\pi\delta(x) \ . \tag{11.9}$$

where $P(f(x))$ is the Cauchy principal value, defined as

$$P\left[\int_{-\infty}^{\infty} \frac{dx}{x}\right] = \int_{-\infty}^{0^-} \frac{dx}{x} + \int_{0^+}^{\infty} \frac{dx}{x} \ . \tag{11.10}$$

To prove the above identity, we integrate the left-hand side using a test function $f(x)$:

$$\begin{aligned}\lim_{\eta\to 0^+} \int_{-\infty}^{\infty} \mathrm{dx}\ \frac{f(x)}{x + i\eta} &= \lim_{\eta\to 0^+} \int_{-\infty}^{\infty} \mathrm{dx}\ \frac{f(x)\,(x - i\eta)}{x^2 + \eta^2} \\ &= \lim_{\eta\to 0^+} \int_{-\infty}^{\infty} \mathrm{dx}\ f(x)\left[\frac{x}{x^2+\eta^2} - i\eta\frac{1}{x^2+\eta^2}\right]\end{aligned} \tag{11.11}$$

The first integral can be split into three parts:

$$\lim_{\eta\to 0^+} \int_{-\infty}^{\infty} \mathrm{dx}\ f(x)\frac{x}{x^2+\eta^2} = \lim_{\eta\to 0^+}\lim_{\varepsilon\to 0^+}\left[\int_{-\infty}^{-\varepsilon} \mathrm{dx}\ \frac{f(x)x}{x^2+\eta^2} + \int_{\varepsilon}^{\infty} \mathrm{dx}\ \frac{f(x)x}{x^2+\eta^2} + \int_{-\varepsilon}^{\varepsilon} \mathrm{dx}\ \frac{f(x)x}{x^2+\eta^2}\right] \tag{11.12}$$

For the first two parts, it is safe to take the limit of η, because x is always non-vanishing there. For the third part, we can approximate $f(x)$ as $f(0)$ in the neighbourhood $x \in [-\varepsilon, \varepsilon], \varepsilon \to 0$. This leaves an odd function $x/(x^2 + \eta^2)$ as the integrand, being integrated over a symmetric range. The third term therefore vanishes. In total, we get

$$\lim_{\eta\to 0^+} \int_{-\infty}^{\infty} \mathrm{dx}\ f(x)\frac{x}{x^2+\eta^2} = \int_{-\infty}^{-\varepsilon} \mathrm{dx}\ \frac{f(x)}{x} + \int_{\varepsilon}^{\infty} \mathrm{dx}\ \frac{f(x)}{x} = P\left(\int_{-\infty}^{\infty} \mathrm{dx}\ \frac{f(x)}{x}\right) \ . \tag{11.13}$$

To evaluate the second integral of eq. 11.11, we note that the only region in which the integrand is non-zero is when $|x| \simeq 0$; there, we again approximate $f(x)$ as $f(0)$. The remaining integral can then be evaluated easily:

$$-i\lim_{\eta\to 0^+} \eta \int_{-\infty}^{\infty} \mathrm{dx}\ \frac{f(x)}{x^2+\eta^2} = -i\lim_{\eta\to 0^+} \eta f(0) \int_{-\infty}^{\infty} \frac{\mathrm{dx}}{x^2+\eta^2} = -i\pi f(0), \tag{11.14}$$

where we used $\int \frac{dx}{x^2+\eta^2} = \frac{1}{\eta}\arctan(x/\eta)$. Combining eqs. 11.11,11.13 and 11.14, we get

$$\lim_{\eta\to 0^+}\int_{-\infty}^{\infty} \mathrm{dx}\ \frac{f(x)}{x+i\eta} = P\left(\int_{-\infty}^{\infty} \mathrm{dx}\ \frac{f(x)}{x}\right) - i\pi\int_{-\infty}^{\infty} \mathrm{dx}\ f(x)\delta(x)\ . \tag{11.15}$$

This proves eq. 11.9.

The Sokhotski-Plemelj formula allows us to split the spectral representation into a real and an imaginary part:

$$\begin{aligned} G'_{d\sigma} &= \frac{1}{Z}\sum_{m,n} ||\langle m|\, O_\sigma\, |n\rangle\, ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) P\left(\frac{1}{\omega + E_m - E_n}\right) \\ G''_{d\sigma} &= -\pi\frac{1}{Z}\sum_{m,n} ||\langle m|\, O_\sigma\, |n\rangle\, ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) \delta\left(\omega + E_m - E_n\right)\ . \end{aligned} \tag{11.16}$$

The spectral function, specifically, is defined as follows:

$$A_{d\sigma}(\omega) = -\frac{1}{\pi}G'_{d\sigma} = \frac{1}{Z}\sum_{m,n} ||\langle m|\, O_\sigma\, |n\rangle\, ||^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right) \delta\left(\omega + E_m - E_n\right) \tag{11.17}$$

2.11.3 Zero temperature spectral function

We specialise to zero temperature by taking the limit of $\beta \to \infty$. In both the partition function as well as inside the summation, the only term that will survive is the exponential of the ground state energy E_{GS}.

$$Z \equiv \sum_m e^{-\beta E_m} \implies \lim_{\beta\to\infty} Z = d_{\mathrm{GS}} e^{-\beta E_{\mathrm{GS}}},\quad E_{\mathrm{GS}} \equiv \min\{E_n\}$$

where d_{GS} is the degeneracy of the ground state. The spectral function then simplifies to

$$\begin{aligned} A_{d\sigma} &= \frac{1}{d_{\mathrm{GS}} e^{-\beta E_{\mathrm{GS}}}}\sum_{m,n} ||\langle m|\, O_\sigma\, |n\rangle\, ||^2 \left[e^{-\beta E_m}\delta_{E_m,E_{\mathrm{GS}}} + e^{-\beta E_n}\delta_{E_n,E_{\mathrm{GS}}}\right] \delta\left(\omega + E_m - E_n\right) \\ &= \frac{1}{d_{\mathrm{GS}}}\sum_{n,n_{\mathrm{GS}}} \left[||\langle n_{\mathrm{GS}}|\, O_\sigma\, |n\rangle\, ||^2 \delta\left(\omega + E_{\mathrm{GS}} - E_n\right) + ||\langle n|\, O_\sigma\, |n_{\mathrm{GS}}\rangle\, ||^2 \delta\left(\omega - E_{\mathrm{GS}} + E_n\right)\right] \end{aligned} \tag{11.18}$$

The label n_{GS} sums over all states $|n_{\mathrm{GS}}\rangle$ with energy E_{GS}. In practise, we evaluate this expression by replacing the formal Dirac delta functions with ”nascent” ones, such as a Lorentzian function with a sufficiently sharp peak.

2.11.4 Reconstructing full Green function from spectral function: Kramers-Kronig relations

As mentioned earlier, the retarded Greens function $G_R(\omega + i0^+)$ has a complex pole in the lower half of the complex plane. In the following, the frequency ω is assumed to be complex. Consider the integral

$$I(\omega) = \oint_C d\omega' \frac{G_R(\omega')}{\omega' - \omega},\ \omega \in \mathbb{R}\ . \tag{11.19}$$

The contour C encloses the entirety of the upper half of the complex plane as well as almost all points of the real line but avoids the pole at $\omega' = \omega$ by forming a semicircle about that point that extends into the upper half of the plane. Since the integrand is completely analytic in the region enclosed by the contour (recall again that G_R is analytic in the upper half), the integral evaluates to zero (by Cauchy's theorem):

$$I(\omega) = 0\ . \tag{11.20}$$

One can also evaluate the integral along each part of the contour. The semicircular part contributes zero: this is because while the length of the semicircular arc goes as $|\omega'|$, the integrand vanishes faster than $1/|\omega'|$ as $|\omega'| \to \infty$ (this assumes that the Green function vanishes in the limit of $\omega \to \infty$). The part of the contour along the real axis can now be evaluated. The two parts of the contour on either side of the pole contribute

$$\int_{-\infty}^{\omega-0^+} d\omega'\ \frac{G_R(\omega')}{\omega'-\omega} + \int_{\omega+0^+}^{\infty} d\omega'\ \frac{G_R(\omega')}{\omega'-\omega} = P\int_{-\infty}^{\infty} d\omega'\ \frac{G_R(\omega')}{\omega'-\omega} \tag{11.21}$$

To evaluate the part of the integral along the small semicircular around $\omega' = \omega$ (that extends into the upper half), we argue that this contribution should be equal to that obtained by taking the semi-circle that goes instead into the lower half but circulates in the opposite direction. Each of these contributions should, in turn, be equal to that obtained from joining these two contours. But joining these contours leads to a closed integral about the pole at $\omega' = \omega$, which is equal to $2\pi G_R(\omega)$.

$$\int_{\text{upper}} d\omega'\ \frac{G_R(\omega')}{\omega'-\omega} = \int_{\text{lower}} d\omega'\ \frac{G_R(\omega')}{\omega'-\omega} = \frac{1}{2}\oint_{C(\omega)} d\omega'\ \frac{G_R(\omega')}{\omega'-\omega} = i\pi G_R(\omega)\ . \tag{11.22}$$

Combining the last two equations, we get

$$I(\omega) = P\int_{-\infty}^{\infty} d\omega'\ \frac{G_R(\omega')}{\omega'-\omega} + i\pi G_R(\omega)\ . \tag{11.23}$$

Comparing with eq. 11.20 and separating into real part G_R' and imaginary part G_R'' gives

$$P\int_{-\infty}^{\infty} d\omega'\ \frac{\left[G_R'(\omega') + iG_R''(\omega')\right]}{\omega'-\omega} + \pi\left[-G_R''(\omega) + iG_R'(\omega)\right] = 0\ . \tag{11.24}$$

Comparing the imaginary parts gives

$$G_R'(\omega) = \frac{1}{\pi}P\int_{-\infty}^{\infty} d\omega'\ \frac{G_R''(\omega')}{\omega'-\omega} = -\mathcal{H}\left[G_R''(\omega)\right]\ , \tag{11.25}$$

where $\mathcal{H}\left[g(\omega)\right] = \frac{1}{\pi}P\int_{-\infty}^{\infty} d\omega'\ \frac{g(\omega')}{\omega-\omega'}$ is the Hilbert transform of the function $g(\omega)$. The spectral function was defined above as $-1/\pi$ times the imaginary part of the Greens function. Exchanging G_R'' for A gives

$$G_R'(\omega) = \pi\mathcal{H}\left[A(\omega)\right]\ . \tag{11.26}$$

This provides a method for calculating the real part of a Greens function if the spectral function is already known.

Chapter 3

Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation

As another application of the tiling approach, we will tackle the problem of heavy fermions in this chapter. The phenomenology of heavy fermion materials has been discussed in detail in the introduction. They involve localised $f-$electrons from rare earths and actinides hybridising with weakly correlated electrons from s, p or d orbitals. In the phase where this hybridisation is strong and the $f-$electrons are Kondo screened by the conduction layer, the narrow bandwidth of the $f-$electrons leads to highly increased masses for the carriers. Lowering the hybridisation leads to a transition into a small Fermi volume phase where the $f-$electrons do not participate in transport. For systems belonging to the so called Kondo breakdown QCP class, the Kondo screening disappears completely at the transition, leading to a quantum critical point at zero temperature and non-Fermi liquid behaviour emanating from the QCP at finite temperature.

Given the work carried out in the previous chapter, we will approach this problem as follows. In order to model the two orbital nature of the heavy fermion system, we study a bilayer extended Hubbard model. By tuning the correlation strength of one of the layers (the $f-$ layer), we will be able to interpolate between the heavy fermions limit where the $f-$layer is completely localised and other regimes where the $f-$layer has some itineracy. To get a feel for the various competing tendencies, we first study the infinite dimensional limit of the model through a two-impurity Kondo model. We then extend our investigation by considering a lattice-embedded version of the model that incorporates lattice geometries.

3.1 Bilayer extended Hubbard model: Impurity model

We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers (f and c). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer (S_f and S_d):

$$H_{\text{aux}} = H_{1\text{p}} + H_f + H_d + H_{fd} \ , \tag{1.1}$$

where $H_{1\text{p}}$ is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{1\text{p}} = -t\sum_{\sigma,\alpha}\sum_{\langle i,j\rangle}\left(c^\dagger_{i,\sigma,\alpha}c_{j,\sigma,\alpha} + \text{h.c.}\right) - \sum_{i,\sigma,\alpha}\mu_\alpha n_{i,\sigma,\alpha}\ , \tag{1.2}$$

such that α sums over the two layers f and d. H_f and H_d describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f = &-\frac{U_f}{2}(n_{f,\uparrow} - n_{f,\downarrow})^2 + \sum_{Z\in\text{NN}}\left[-V\sum_\sigma\left(c^\dagger_{f,\sigma}c_{Z,\sigma,f} + \text{h.c.}\right) + \frac{J}{4}\sum_{\alpha,\beta}\mathbf{S}_f\cdot\frac{1}{2}\boldsymbol{\sigma}_{\alpha\beta}c^\dagger_{Z,\alpha,f}c_{Z,\beta,f}\right. \\ &\left. -\frac{W_f}{4}\left(n_{Z,\uparrow,f} - n_{Z,\downarrow,f}\right)^2\right]\ , \\ H_d = &-\eta_d n_d - \frac{U_d}{2}(n_{d,\uparrow} - n_{d,\downarrow})^2 + \sum_{Z\in\text{NN}}\left[-V\sum_\sigma\left(c^\dagger_{d,\sigma}c_{Z,\sigma,d} + \text{h.c.}\right) + \frac{J}{4}\sum_{\alpha,\beta}\mathbf{S}_d\cdot\frac{1}{2}\boldsymbol{\sigma}_{\alpha\beta}c^\dagger_{Z,\alpha,d}c_{Z,\beta,d}\right]\ , \end{aligned} \tag{1.3}$$

where $c^\dagger_{\nu,\sigma,\alpha}$ can refer to creation operator for either the $f-$layer. Finally, H_{fd} represents the inter-layer hybridisation:

$$H_{fc} = -V_\perp\sum_\sigma\left(c^\dagger_{d,\sigma}c_{f,\sigma} + \text{h.c.}\right)\ . \tag{1.4}$$

3.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the "unit" cell of translation. As discussed in the text surrounding eq. **??**, a good choice for the unit cell is the impurity site and it's surrounding correlated environment. Placing the impurity sites at $\boldsymbol{r}_d$ in each layer, we get

$$H_{\text{clust}}(\boldsymbol{r}_d) = H_f(\boldsymbol{r}_d) + H_d(\boldsymbol{r}_d) + H_{fc}(\boldsymbol{r}_d)\ . \tag{2.1}$$

The details of tiling a cluster Hamiltonian were already shown in Sec. **??**; the general idea is to use manybody translation operators (details in Sec. **??**) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned} H_{\text{latt}} &= \sum_r T^\dagger(\boldsymbol{r})H_{\text{clust}}(\boldsymbol{r}_d)T(\boldsymbol{r}) \\ &= -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.}\right) - \sum_i\left[\eta_d n_{i,d} + \frac{U_d}{2}(n_{i,\uparrow,d} - n_{i,\downarrow,d})^2\right] + \frac{2J}{Z}\sum_{\langle i,j\rangle}\boldsymbol{S}_{i,d}\cdot\boldsymbol{S}_{j,d} \\ &\quad - \frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - \frac{U_f - W_f}{2}\sum_i\left(n_{i,\uparrow,f} - n_{i,\downarrow,f}\right)^2 + \frac{2J}{Z}\sum_{\langle i,j\rangle}\boldsymbol{S}_{i,f}\cdot\boldsymbol{S}_{j,f} \\ &\quad - V_\perp\sum_{i,\sigma}(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}) \end{aligned} \tag{2.2}$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of η_d defines the effective chemical potential of the $d-$layer; a non-zero value moves the layer away from half-filling. The $f-$layer is pinned at half-filling.

Following Sec. **??**, the retarded $k-$space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\boldsymbol{k},\sigma,\alpha;\omega) = \sum_{\nu} C_{\nu,\boldsymbol{k}\alpha}\mathcal{G}(\mathcal{T}_{\nu,\sigma,\alpha};\omega - \mathcal{E}_\nu(\boldsymbol{k})) \tag{2.3}$$

where $\alpha = f, d$ denotes the layer in which the Greens function is being calculated, the modes ν indicate the operators which diagonalise the Hamiltonian $H_{1\mathrm{p}}$, with energies $\mathcal{E}_\nu(\boldsymbol{k})$:

$$H_{1\mathrm{p}} = -\frac{2V}{Z}\sum_{\langle i,j\rangle,\sigma}\left(c^\dagger_{i,\sigma}c_{j,\sigma} + \text{h.c.} + f^\dagger_{i,\sigma}f_{j,\sigma} + \text{h.c.}\right) - V_\perp\sum_{i,\sigma}(f^\dagger_{i,\sigma}c_{i,\sigma} + \text{h.c.}) - \sum_i \eta_d n_{i,d}\ , \tag{2.4}$$

and the $\mathcal{T}-$matrices are defined in eq. **??**:

$$\mathcal{T}_{\nu,\sigma,\alpha} = \left[\left(\sum_{\sigma'} c^\dagger_{\boldsymbol{\alpha}\sigma} + \text{h.c.}\right) + \left(S^+_{\boldsymbol{\alpha}} + \text{h.c.}\right) + \left(C^+_{\boldsymbol{\alpha}} + \text{h.c.}\right)\right] c_{\nu,\sigma}\ , \tag{2.5}$$

In order to calculate $\mathcal{E}_\nu(\boldsymbol{k})$, we introduce the $k-$space modes via a Fourier transform:

$$c^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i}c^\dagger_{\boldsymbol{k},\sigma}\ ,\quad f^\dagger_{i,\sigma} = \frac{1}{\sqrt{N}}\sum_{\boldsymbol{k}} e^{-i\boldsymbol{k}\cdot\boldsymbol{r}_i}f^\dagger_{\boldsymbol{k},\sigma}\ . \tag{2.6}$$

In terms of these modes, the $1-$particle Hamiltonian decouples in $k-$space:

$$H_{1\mathrm{p}} = \sum_{\boldsymbol{k},\sigma}\left[(\epsilon_{\boldsymbol{k}} - \eta_d)(n_{\boldsymbol{k},\sigma,d} + n_{\boldsymbol{k},\sigma,f}) - V_\perp(c^\dagger_{\boldsymbol{k},\sigma}f_{\boldsymbol{k},\sigma} + \text{h.c.})\right]\ . \tag{2.7}$$

The diagonal modes are given by

$$\nu^\dagger_{\boldsymbol{k},\sigma,\pm} = C_{\pm,k}c^\dagger_{\boldsymbol{k},\sigma} + \sqrt{1 - C^2_{\pm,k}}f^\dagger_{\boldsymbol{k},\sigma}\ , \tag{2.8}$$

the coefficients obtained from diagonalising the matrix

$$H_{1\mathrm{p}}(\boldsymbol{k}) = \begin{pmatrix}\epsilon_{\boldsymbol{k}} & -V_\perp \\ -V_\perp & \epsilon_{\boldsymbol{k}} - \eta_d\end{pmatrix}\ . \tag{2.9}$$

The energies are given by $\mathcal{E}_\pm(\boldsymbol{k}) = \epsilon_k - \eta_d/2 \pm \Delta$, with ϵ_k being $-\frac{4V}{Z}(\cos k_x + \cos k_y)$ for a 2D square lattice and $\Delta = \sqrt{V_\perp^2 + \eta_d^2/4}$. Substituting into the expression for the lattice Greens function, we get

$$\begin{aligned}G(\boldsymbol{k},d;\omega) =& C_{+,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k}+,d},\mathcal{T}^\dagger_{\boldsymbol{k},d};\omega - \mathcal{E}_+(\boldsymbol{k})) + C_{-,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k}-,d},\mathcal{T}^\dagger_{\boldsymbol{k},d};\omega - \mathcal{E}_-(\boldsymbol{k})) \\ &+ C^*_{+,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},d},\mathcal{T}^\dagger_{\boldsymbol{k}+,d};\omega - \mathcal{E}_+(\boldsymbol{k})) + C^*_{-,\boldsymbol{k}}\mathcal{G}(\mathcal{T}_{\boldsymbol{k},d},\mathcal{T}^\dagger_{\boldsymbol{k}-,d};\omega - \mathcal{E}_-(\boldsymbol{k}))\ .\end{aligned} \tag{2.10}$$

Following eq. **??**, we can similarly write down an expression the local lattice Greens function for the $f-$ and $d-$electrons. For that, we define the local diagonal modes $\nu_\pm$ obtained from the local Hamiltonian

$$H_{\text{loc}} = \begin{pmatrix} 0 & -V_\perp \\ -V_\perp & 0 - \eta_d \end{pmatrix} , \tag{2.11}$$

and their associated energies $\varepsilon_\pm$. Within the impurity model, these local bonding/antibonding modes can be constructed on the impurity sites ($\nu_\pm$) or on the neighbouring bath sites, ($\nu^Z_\pm$). The local Greens function can be expressed in terms of these modes:

$$\begin{aligned} G_{dd}(\omega) =& \frac{1}{2}\sum_\nu C_{\nu,d}\left[\mathcal{G}(\nu\sigma, d\sigma; \omega - \varepsilon_\nu) + \mathcal{G}(d\sigma, \nu\sigma; \omega - \varepsilon_\nu)\right] \\ &+ \frac{1}{2}\sum_{\nu^Z} C_{\nu^Z,Z_d}\left[\mathcal{G}(\mathcal{T}_{Z_d,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z}) + \mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z})\right] \ . \end{aligned} \tag{2.12}$$

The presence of local hybridisation between the $d-$ and $f-$ orbitals makes the inter-orbital Greens function important as well. Following eq. **??**, we have

$$\begin{aligned} G_{fd}(\omega) =& \frac{1}{2}\sum_\nu \left[C_{\nu,f}\mathcal{G}(\nu\sigma, d\sigma; \omega - \varepsilon_\nu) + C_{\nu,d}\mathcal{G}(f\sigma, \nu\sigma; \omega - \varepsilon_\nu)\right] \\ &+ \frac{1}{2}\sum_{\nu^Z} \left[C_{\nu^Z,Z_f}\mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z}) + C_{\nu^Z,Z_d}\mathcal{G}(\mathcal{T}_{Z_f,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z})\right] \ . \end{aligned} \tag{2.13}$$

Using these, we can construct the hybridised Greens function associated with the states ν:

$$G_{\nu_+}(\omega) = C^2_{+,d}G_{dd}(\omega) + C^2_{+,f}G_{ff}(\omega) + C_{+,d}C_{+,f}\left[G_{df}(\omega) + G_{fd}(\omega)\right] , \tag{2.14}$$

3.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ESIAM

In the limit of large U_α, we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$\begin{aligned} H_{\text{aux}} =& \sum_{\mathbf{k},\sigma,\alpha}\epsilon_{\mathbf{k},\alpha}n_{\mathbf{k},\sigma,\alpha} + \sum_\alpha\sum_{Z\in\text{NN}}\left[\frac{1}{4}J_\alpha\sum_{\sigma,\sigma'}\mathbf{S}_\alpha\cdot\frac{1}{2}\boldsymbol{\sigma}_{\sigma\sigma'}c^\dagger_{Z,\sigma,\alpha}c_{Z,\sigma',\alpha} - \frac{W_\alpha}{4}\left(n_{Z,\uparrow,\alpha} - n_{Z,\downarrow,\alpha}\right)^2\right] \\ &+ J_\perp\mathbf{S}_f\cdot\mathbf{S}_d \ . \end{aligned} \tag{3.1}$$

As we will see, additional terms are generated within the Hamiltonian as we integrate out high-energy degrees of freedom in the conduction baths of the two impurity models. These terms connect the impurity of the $d-(f-)$layer to the nearest-neighbour conduction bath sites of the $f-(d-)$layer, It is therefore necessary to start with an enlarged impurity model Hamiltonian that contains these couplings as well:

$$
\begin{aligned}
H_{\text{aux}} = &\sum_{\mathbf{k},\sigma,\alpha} \epsilon_{\mathbf{k},\alpha} n_{\mathbf{k},\sigma,\alpha} + \sum_{\alpha} \sum_{Z\in\text{NN}} \left[\frac{1}{4} \sum_{\sigma,\sigma'} \left(J_\alpha \mathbf{S}_\alpha + K_{\bar{\alpha}} \mathbf{S}_{\bar{\alpha}} \right) \cdot \frac{1}{2} \boldsymbol{\sigma}_{\sigma\sigma'} c^\dagger_{Z,\sigma,\alpha} c_{Z,\sigma',\alpha} \right. \\
&\left. - \frac{W_\alpha}{4} \left(n_{Z,\uparrow,\alpha} - n_{Z,\downarrow,\alpha} \right)^2 \right] + J \mathbf{S}_f \cdot \mathbf{S}_d \ .
\end{aligned} \tag{3.2}
$$

Note the additional couplings $K_{\bar{\alpha}}$, where $\bar{\alpha}$ stands for the layer opposite to α.

The two Kondo couplings J and K and the bath interaction W acquire $k-$dependence:

$$
\begin{aligned}
J(\mathbf{k},\mathbf{q}) &= \frac{J}{2} \left[\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y) \right] \ , \\
K(\mathbf{k},\mathbf{q}) &= \frac{K}{2} \left[\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y) \right] \ , \\
W(\mathbf{k},\mathbf{q},\mathbf{k}',\mathbf{q}') &= \frac{W}{2} \left[\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y) \right] \ .
\end{aligned} \tag{3.3}
$$

The particle-hole transformed state $\bar{\mathbf{q}}$ associated with $\mathbf{q}$ is defined as $\bar{\mathbf{q}} = \mathbf{q} + (\pi,\pi)$ if $\mathbf{q}$ is in the first or third quadrant and $\mathbf{q} + (\pi,-\pi)$ is in the second or fourth quadrant.

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be $|\mathbf{q},\sigma,f\rangle$ and $|\mathbf{q},\sigma,d\rangle$ if they are above the Fermi energy (unoccupied in the low-energy configuration) and $|\bar{\mathbf{q}},\sigma,f\rangle$ and $|\bar{\mathbf{q}},\sigma,d\rangle$ if they are below the Fermi energy (occupied in the low-energy configuration) For every shell of conduction bath states of width ΔD integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states.

The full set of impurity-mediated scattering processes that lead to UV-IR mixing is:

Particle sector excitation These involve excited states above the Fermi energy:

$$
\begin{aligned}
P_{\text{dir}}(\alpha) &= \frac{1}{2} \sum_{k,q} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{q,\sigma_1,\alpha} c_{k,\sigma_2,\alpha} \ , \\
P_{\text{ind}}(\alpha) &= \frac{1}{2} \sum_{k,q} K_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{q,\sigma_1,\bar{\alpha}} c_{k,\sigma_2,\bar{\alpha}} \ .
\end{aligned} \tag{3.4}
$$

Hole sector These involve excited states below the Fermi energy:

$$
\begin{aligned}
H_{\text{dir}}(\alpha) &= \frac{1}{2} \sum_{k,q} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{k,\sigma_1,\alpha} c_{q,\sigma_2,\alpha} \ , \\
H_{\text{ind}}(\alpha) &= \frac{1}{2} \sum_{k,q} K_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{k,\sigma_1,\bar{\alpha}} c_{q,\sigma_2,\bar{\alpha}} \ .
\end{aligned} \tag{3.5}
$$

Mixed sector These involve simultaneous particle and hole excitations:

$$\begin{aligned} M_{\text{dir}}(\alpha) &= \frac{1}{2}\sum_{q,\bar{q}} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{q,\sigma_1,\alpha} c_{\bar{q},\sigma_2,\alpha} , \\ M_{\text{ind}}(\alpha) &= \frac{1}{2}\sum_{q,\bar{q}} K_{\bar{\alpha}} \mathbf{S}_{\bar{\alpha}} \cdot \sigma_{\sigma_1,\sigma_2} c^\dagger_{q,\sigma_1,\bar{\alpha}} c_{\bar{q},\sigma_2,\bar{\alpha}} . \end{aligned} \tag{3.6}$$

3.3.1 Particle/hole scattering processes: direct with direct

We already have the renormalisation group equations for the couplings J_α in the case of $J = K = 0$, arising from P_{dir} and H_{dir}:

$$\frac{\Delta J_f(\mathbf{k}_1, \mathbf{k}_2)}{\Delta D} = -\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_f(\mathbf{k}_1, \mathbf{q}) J_f(\mathbf{q}, \mathbf{k}_2) + 4 J_f(\mathbf{q}, \bar{\mathbf{q}}) W_f(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)\right] G^{(0)}_f(\mathbf{q}) , \tag{3.7}$$

where $\mathbf{q}$ sums over the UV states being integrated out, $\rho(\mathbf{q})$ is the density of states for the momentum states being integrated out and $G^{(0)}_f$ is the propagator for the excited intermediate state (for $J = 0$):

$$G^J_\alpha(\mathbf{q}) = \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha(\mathbf{q})/4 + W_\alpha(\mathbf{q})/2} . \tag{3.8}$$

An identical equations holds for ΔJ_d, with all $f-$quantities replaced with $d-$quantities. Switching on J modifies the excitation energy in the propagator, while either of the Ks do not enter the denominator because either the impurity or the conduction bath labels do not match the ones fluctuating in this process:

$$G_\alpha(\mathbf{q}) = \frac{1}{1/G^J_\alpha - J\vec{S}_f \cdot \vec{S}_d} . \tag{3.9}$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues, $-3/4$ in the singlet sector and $1/4$ in the triplet sector. Defining the projectors $\mathcal{P}_0$ and $\mathcal{P}_1$ for the two sectors, we have

$$\vec{S}_f \cdot \vec{S}_d = \frac{-3}{4}\mathcal{P}_0 + \frac{1}{4}\mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I} . \tag{3.10}$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G^J_f + 3J/4}\mathcal{P}_0 + \frac{1}{1/G^J_f - J/4}\mathcal{P}_1 \\ &= \left(\frac{1/4}{1/G^J_f + 3J/4} + \frac{3/4}{1/G^J_f - J/4}\right)\mathbb{I} + \left(\frac{1}{1/G^J_f + 3J/4} + \frac{1}{1/G^J_f - J/4}\right)\vec{S}_f \cdot \vec{S}_d . \end{aligned} \tag{3.11}$$

Only the first operator renormalises the Kondo interaction:

$$\begin{aligned} \frac{\Delta J_\alpha}{\Delta D} = -\int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \left[J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + 4 J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)\right] \left(\frac{1}{4}\frac{1}{1/G^J_\alpha(\mathbf{q}) + 3J/4}\right. \\ \left. + \frac{3}{4}\frac{1}{1/G^J_\alpha(\mathbf{q}) - J/4}\right) . \end{aligned} \tag{3.12}$$

We now simplify the second term, in conjunction with the two kinds of processes on either side: J^2 processes and JW processes. We first focus on the J^2 processes.

Effect of inter-layer term $\vec{S}_f \cdot \vec{S}_d$ on J^2 processes

Taking into account the other spin operators on either side of $G_f(\mathbf{q})$ in the full scattering process, we have

$$\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\beta}\sum_{\gamma,\delta} S_f^a\sigma^a_{\alpha,\beta}S_f^bS_d^bS_f^c\sigma^c_{\gamma,\delta}\left[\delta_{\beta,\gamma}c^\dagger_{k,\alpha,f}c_{q,\beta,f}c^\dagger_{q,\gamma,f}c_{k',\delta,f} + \delta_{\alpha,\delta}c^\dagger_{\bar{q},\alpha,f}c_{k',\beta,f}c^\dagger_{k,\gamma,f}c_{\bar{q},\delta,f}\right] . \tag{3.13}$$

Integrating out the UV states using $n_{\bar{q}\beta} = 1 - n_{q\beta} = 1$, we get

$$\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\beta,\gamma} S_f^aS_f^bS_d^bS_f^c\left[\sigma^a_{\alpha,\beta}\sigma^c_{\beta,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f} + \sigma^a_{\alpha,\beta}\sigma^c_{\gamma,\alpha}c_{k',\beta,f}c^\dagger_{k,\gamma,f}\right] . \tag{3.14}$$

Using $\sum_\beta \sigma^a_{\alpha,\beta}\sigma^c_{\beta,\gamma} = (\sigma^a\sigma^c)_{\alpha,\gamma} = \delta_{ac}\delta_{\alpha\gamma}+i\epsilon_{acm}\sigma^m_{\alpha,\gamma}$ in the first term and $\sum_\alpha \sigma^c_{\gamma,\alpha}\sigma^a_{\alpha,\beta} = (\sigma^c\sigma^a)_{\gamma,\beta} = \delta_{ca}\delta_{\gamma\beta} - i\epsilon_{acm}\sigma^m_{\gamma,\beta}$ in the second term, relabelling $\beta \to \gamma$ and $\gamma \to \alpha$ there and using $c_{k',\gamma}c^\dagger_{k,\alpha} = \delta_{k,k'}\delta_{\alpha,\gamma} - c^\dagger_{k,\alpha}c_{k',\gamma}$, we get

$$\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\gamma} S_f^aS_f^bS_d^bS_f^c\left[\delta_{a,c}\delta_{k,k'}\delta_{\alpha,\gamma} + 2i\epsilon_{acm}\sigma^m_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f}\right] . \tag{3.15}$$

We also use the spin-half identity

$$\begin{aligned}\sigma^a\sigma^b\sigma^c =&(\delta_{ab} + i\epsilon_{abp}\sigma^p)\sigma^c\\ &= \delta_{ab}\sigma^c + i\epsilon_{abp}(\delta_{pc} + i\epsilon_{pcr}\sigma^r)\\ &= \delta_{ab}\sigma^c - \delta_{ac}\sigma^b + \delta_{bc}\sigma^a + i\epsilon_{abc} ,\end{aligned} \tag{3.16}$$

which when contracted with ϵ_{acm} gives

$$\begin{aligned}\sum_{a,c}\epsilon_{acm}\sigma^a\sigma^b\sigma^c &= \sum_{a,c}\epsilon_{acm}(\delta_{ab}\sigma^c + \delta_{bc}\sigma^a + i\epsilon_{abc})\\ &= \sum_{a,c}(\epsilon_{acm}\delta_{ab}\sigma^c + \epsilon_{cam}\delta_{ba}\sigma^c - i\epsilon_{acm}\epsilon_{acb})\\ &= -2i\delta_{bm} .\end{aligned} \tag{3.17}$$

For the first term of eq. 3.15, we get

$$\begin{aligned}\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\gamma} S_f^aS_f^bS_d^bS_f^c\delta_{a,c}\delta_{k,k'}\delta_{\alpha,\gamma} &= \delta_{k,k'}\sum_{a,b} S_f^aS_f^bS_f^aS_d^b\\ &= \frac{1}{4}\delta_{k,k'}\sum_b(S_f^b - 3S_f^b + S_f^b)S_d^b\\ &= -\frac{1}{4}\delta_{k,k'}\sum_b S_f^bS_d^b .\end{aligned} \tag{3.18}$$

For the second term, we obtain

$$\begin{aligned}\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\gamma} S_f^a S_f^b S_d^b S_f^c 2i\epsilon_{acm}\sigma^m_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f} &= \frac{1}{4}\sum_b\sum_{\alpha,\gamma}\delta_{bm}S_d^b\sigma^m_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f} \\ &= \frac{1}{2}\sum_b S_d^b\frac{1}{2}\sum_{\alpha,\gamma}\sigma^b_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f}\end{aligned} \tag{3.19}$$

Substituting these in the expression for the renormalisation gives

$$\begin{aligned}&\frac{1}{2}\sum_{a,b,c}\sum_{\alpha,\gamma} S_f^a S_f^b S_d^b S_f^c\left[\delta_{a,c}\delta_{k,k'}\delta_{\alpha,\gamma} + 2i\epsilon_{acm}\sigma^m_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f}\right] \\ &= -\frac{1}{4}\mathbf{S}_f\cdot\mathbf{S}_d\delta_{k,k'} + \frac{1}{2}\sum_{\alpha,\gamma}\mathbf{S}_d\frac{1}{2}\sigma_{\alpha,\gamma}c^\dagger_{k,\alpha,f}c_{k',\gamma,f}\ .\end{aligned} \tag{3.20}$$

The first term leads to the renormalisation of the inter-layer hybridisation $J_\perp$:

$$\frac{\Delta J_\perp}{\Delta D} = -\frac{1}{4}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\int_{IR} d\mathbf{k}\rho(\mathbf{k})\sum_\alpha J_\alpha(\mathbf{k},\mathbf{q})^2\left(\frac{1}{1/G^J_\alpha + 3J/4} + \frac{1}{1/G^J_\alpha - J/4}\right)\ , \tag{3.21}$$

while the second term gives rise to the indirect impurity-bath Kondo couplings K_α mentioned earlier above eq. 3.2:

$$\frac{\Delta K_\alpha}{\Delta D} = \frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})J_{\bar\alpha}(\mathbf{k}_1,\mathbf{q})J_{\bar\alpha}(\mathbf{q},\mathbf{k}_2)\left(\frac{1}{1/G^J_{\bar\alpha} + 3J/4} + \frac{1}{1/G^J_{\bar\alpha} - J/4}\right)\ , \tag{3.22}$$

Effect of inter-layer term $\vec{S}_f\cdot\vec{S}_d$ on JW processes

We now turn to the effect of the $\vec{S}_f\cdot\vec{S}_d$ in eq. 3.11 on processes that involve a Kondo scattering J and a bath scattering W. The non-vanishing processes are of the form

$$\begin{aligned}P_1 &= (S_f^+\mathbf{S_f}\cdot\mathbf{S_d} + \mathbf{S_f}\cdot\mathbf{S_d}S_f^+)c^\dagger_{k,\downarrow}c_{k',\uparrow}, \\ P_2 &= P_1^\dagger, \\ P_z &= (S_f^z\mathbf{S_f}\cdot\mathbf{S_d} + \mathbf{S_f}\cdot\mathbf{S_d}S_f^z)\sum_\sigma \sigma c^\dagger_{k,\sigma}c_{k',\sigma}\ .\end{aligned} \tag{3.23}$$

In order to evaluate them, we note that for any $a\in x,y,z$, we have

$$S_1^a\mathbf{S_1}\cdot\mathbf{S_2} + \mathbf{S_1}\cdot\mathbf{S_2}S_1^a = \sum_b\{S_1^a,S_1^b\}S_2^b = \frac{1}{4}\sum_b 2\delta_{a,b}S_2^b = \frac{1}{2}S_2^a\ . \tag{3.24}$$

Using this, we obtain

$$\begin{aligned}P_1 &= \frac{1}{2}S_d^+c^\dagger_{k,\downarrow}c_{k',\uparrow}, \\ P_2 &= \frac{1}{2}S_d^-c^\dagger_{k,\uparrow}c_{k',\downarrow}, \\ P_z &= \frac{1}{2}S_d^z\sum_\sigma\sigma c^\dagger_{k,\sigma}c_{k',\sigma}\ .\end{aligned} \tag{3.25}$$

Since the bath sites are in the $f-$layer, these terms again lead to the renormalisation of the indirect couplings K_α. Combining with eq. 3.22, we get

$$\frac{\Delta K_\alpha}{\Delta D} = \frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\,[J_{\bar\alpha}(\mathbf{k}_1,\mathbf{q})J_{\bar\alpha}(\mathbf{q},\mathbf{k}_2) - 4J_{\bar\alpha}(\mathbf{q},\bar{\mathbf{q}})W_{\bar\alpha}(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)]\left(\frac{1}{1/G^J_{\bar\alpha}+3J/4} + \frac{1}{1/G^J_{\bar\alpha}-J/4}\right), \tag{3.26}$$

3.3.2 Particle/hole scattering processes: indirect with indirect

The processes that apply to this class are very similar to those participating in the direct-direct class; the only difference is that the bath degrees of freedom have to be transferred from $\alpha \to \bar\alpha$, due to the inter-layer nature of this coupling. As a result, we will have straightforward counterparts to the renormalisation equations obtained in the previous section, eqs. 3.12, 3.21 and 3.26. The first equation is for the renormalisation of the indirect coupling through itself,

$$\frac{\Delta K_\alpha}{\Delta D} = -\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\,[K_\alpha(\mathbf{k}_1,\mathbf{q})K_\alpha(\mathbf{q},\mathbf{k}_2) + 4K_\alpha(\mathbf{q},\bar{\mathbf{q}})W_{\bar\alpha}(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)]\left(\frac{1}{4}\frac{1}{1/G^K_{\bar\alpha}(\mathbf{q})+3J/4} + \frac{3}{4}\frac{1}{1/G^K_{\bar\alpha}(\mathbf{q})-J/4}\right). \tag{3.27}$$

where the indirect Greens function G^K_α is defined as

$$G^K_{\bar\alpha}(\mathbf{q}) = \frac{1}{\omega - |\epsilon_{\bar\alpha}(\mathbf{q})|/2 + K_\alpha(\mathbf{q})/4 + W_{\bar\alpha}(\mathbf{q})/2}, \tag{3.28}$$

and reflects the fact the excitations are happening on impurity and bath states in opposite layers. The second equation is for the effect of the inter-layer term through the denominator of such processes, leading to a renormalisation of the inter-layer itself:

$$\frac{\Delta J_\perp}{\Delta D} = -\frac{1}{4}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\int_{IR} d\mathbf{k}\rho(\mathbf{k})\sum_\alpha K_\alpha(\mathbf{k},\mathbf{q})^2\left(\frac{1}{1/G^K_{\bar\alpha}+3J/4} + \frac{1}{1/G^K_{\bar\alpha}-J/4}\right). \tag{3.29}$$

The final equation is for the renormalisation of the direct Kondo coupling arising from indirect processes. Similar to how the direct Kondo coupling renormalised the indirect coupling, this occurs through two routes. The complete equation is

$$\frac{\Delta J_\alpha}{\Delta D} = \frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\,[K_{\bar\alpha}(\mathbf{k}_1,\mathbf{q})K_{\bar\alpha}(\mathbf{q},\mathbf{k}_2) - 4K_{\bar\alpha}(\mathbf{q},\bar{\mathbf{q}})W_\alpha(\mathbf{k}_1,\mathbf{q},\mathbf{q},\mathbf{k}_2)]\left(\frac{1}{1/G^K_\alpha+3J/4} + \frac{1}{1/G^K_\alpha-J/4}\right). \tag{3.30}$$

3.3.3 Particle/hole scattering processes: direct with indirect

These processes are composed of a direct Kondo excitation process followed by an indirect Kondo de-excitation process, or vice versa. They can be of the particle excitation or the hole excitation kind. Within the particle or hole sector, we again have two sets of processes depending on whether the initial process is direct or indirect:

a. $P_{\text{dir}}(d)\ G_d\ H_{\text{ind}}(f) + \text{h.c.}$

b. $P_{\text{dir}}(f)\ G_f\ H_{\text{ind}}(d) + \text{h.c.}$

c. $H_{\text{ind}}(f)\ G_d\ P_{\text{dir}}(d) + \text{h.c.}$

d. $H_{\text{ind}}(d)\ G_f\ P_{\text{dir}}(f) + \text{h.c.}$

As mentioned above, $a(b)$ and $c(d)$ are time-reversed partners of each other. We will now show that the contributions from these processes lead to a renormalisation of the inter-layer Kondo term. We suppress the Greens function since it is the same for all the terms:

$$G^{JK}_\alpha = \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha/4 + K_{\bar\alpha}/4 + W_\alpha(\mathbf{q})/2 - J_\perp/4}\ , \tag{3.31}$$

containing antiferromagnetic contributions from J_α and K_α and ferromagnetic contribution in $J_\perp$. The contribution from the pair is of the form

$$\begin{aligned}
&\sum_\alpha \left[P_{\text{dir}}(\alpha)H_{\text{ind}}(\bar\alpha) + H_{\text{ind}}(\bar\alpha)P_{\text{dir}}(\alpha)\right] \\
&= \sum_{\alpha,a,b,\beta,\gamma,\delta} \frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)\left[\mathbf{S}^a_\alpha\sigma^a_{\beta,\gamma}c_{k',\gamma,\alpha}\mathbf{S}^b_{\bar\alpha}\sigma^b_{\delta,\beta}c^\dagger_{k,\delta,\alpha} + \mathbf{S}^b_{\bar\alpha}\sigma^b_{\delta,\beta}c^\dagger_{k,\delta,\alpha}\mathbf{S}^a_\alpha\sigma^a_{\beta,\gamma}c_{k',\gamma,\alpha}\right]
\end{aligned} \tag{3.32}$$

The sum over β can be carried out:

$$\begin{aligned}
&\sum_{\alpha,a,b,\beta,\gamma,\delta} \frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)\left[\mathbf{S}^a_\alpha\sigma^a_{\beta,\gamma}c_{k',\gamma,\alpha}\mathbf{S}^b_{\bar\alpha}\sigma^b_{\delta,\beta}c^\dagger_{k,\delta,\alpha} + \mathbf{S}^b_{\bar\alpha}\sigma^b_{\delta,\beta}c^\dagger_{k,\delta,\alpha}\mathbf{S}^a_\alpha\sigma^a_{\beta,\gamma}c_{k',\gamma,\alpha}\right] \\
&= \sum_{\alpha,a,b,\gamma,\delta} \frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)(\sigma^b\sigma^a)_{\delta,\gamma}\mathbf{S}^a_\alpha\mathbf{S}^b_{\bar\alpha}\left[c_{k',\gamma,\alpha}c^\dagger_{k,\delta,\alpha} + c^\dagger_{k,\delta,\alpha}c_{k',\gamma,\alpha}\right] \\
&= \sum_{\alpha,a,b,\gamma,\delta} \frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)(\sigma^b\sigma^a)_{\delta,\gamma}\mathbf{S}^a_\alpha\mathbf{S}^b_{\bar\alpha}\delta_{k,k'}\delta_{\delta,\gamma} \\
&= \sum_{\alpha,a,b} \frac{1}{4}J_\alpha(q,k')K_{\bar\alpha}(k,q)\text{Trace}(\sigma^b\sigma^a)\mathbf{S}^a_\alpha\mathbf{S}^b_{\bar\alpha}\delta_{k,k'}\ .
\end{aligned} \tag{3.33}$$

Adding the contribution from the hermitian conjugate and restoring the Greens function, we get the total contribution as

$$\sum_\alpha \left[P_{\text{dir}}(\alpha)H_{\text{ind}}(\bar\alpha) + H_{\text{ind}}(\bar\alpha)P_{\text{dir}}(\alpha)\right] + \text{h.c.} = \frac{1}{2}\mathbf{S}_d\cdot\mathbf{S}_f\sum_{\alpha,q,k}\frac{J_\alpha(q,k)K_{\bar\alpha}(k,q)}{1/G^{JK}_\alpha - J_\perp/4}\ . \tag{3.34}$$

The renormalisation of the inter-layer coupling arising from this process is

$$\frac{\Delta J_\perp}{\Delta D} = \int_{UV} d\mathbf{q}\,\rho(\mathbf{q}) \int_{IR} d\mathbf{k}\rho(\mathbf{k}) \sum_{\alpha,q,k} \frac{J_\alpha(q,k)K_{\bar{\alpha}}(k,q)}{1/G^{JK}_\alpha - J_\perp/4}\ . \tag{3.35}$$

3.3.4 Mixed processes

We now turn to the renormalisation arising from particle-hole mixed processes (designated as M_α above). There are four types of process:

$$\begin{aligned}
P_1 &: \sum_\alpha M_{\text{dir}}(\alpha)GM_{\text{dir}}(\alpha)\\
P_2 &: \sum_\alpha M_{\text{ind}}(\alpha)GM_{\text{ind}}(\alpha)\\
P_3 &: \sum_\alpha [M_{\text{dir}}(\alpha)GM_{\text{ind}}(\bar{\alpha}) + \text{h.c.}]\\
P_4 &: \sum_\alpha [M_{\text{dir}}(\alpha)GM_{\text{ind}}(\alpha) + \text{h.c.}]\\
P_5 &: \sum_\alpha [M_{\text{dir}}(\alpha)GM_{\text{dir}}(\bar{\alpha}) + M_{\text{ind}}(\alpha)GM_{\text{ind}}(\bar{\alpha})]\ .
\end{aligned} \tag{3.36}$$

We first note that P_1 and P_2 are equivalent to the ones considered in the direct-direct and indirect-indirect subsections, specifically the renormalisation of $J_\perp$ in eq. 3.21 and its indirect counterpart, the only difference being that k will now range over the UV subspace instead of IR:

$$\begin{aligned}
\frac{\Delta J_\perp}{\Delta D} = -\frac{1}{4}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q}') \int_{UV} d\mathbf{q}'\rho(\mathbf{q}') \sum_\alpha \Bigg[J_\alpha(\mathbf{q}',\mathbf{q})^2 \left(\frac{1}{1/G^J_\alpha + 3J/4} + \frac{1}{1/G^J_\alpha - J/4}\right)\\
+ K_\alpha(\mathbf{k},\mathbf{q})^2 \left(\frac{1}{1/G^K_{\bar{\alpha}} + 3J/4} + \frac{1}{1/G^K_{\bar{\alpha}} - J/4}\right)\Bigg]\ .
\end{aligned} \tag{3.37}$$

The other equations in the same subsection – the ones pertaining to the renormalisation of J_α or K_α are not relevant here because they require the participation of IR bath operators. Similarly, P_3 and P_4 are equivalent to the ones considered in the previous subsection (eq. 3.35), the only difference being that the dummy variable k in the previous subsection would now have to range only over the UV subspace:

$$\frac{\Delta J_\perp}{\Delta D} = \int_{UV} d\mathbf{q}\,\rho(\mathbf{q}) \int_{UV} d\mathbf{q}'\rho(\mathbf{q}') \sum_{\alpha} \frac{J_\alpha(q,q')K_{\bar{\alpha}}(q',q)}{1/G^{JK}_\alpha - J_\perp/4}\ . \tag{3.38}$$

We finally evaluate the contribution from processes P_4 and P_5. In order to capture coherent scattering processes between the two layers arising from P_4, we project onto the following rotated basis of excited states:

$$\begin{aligned}
|H\rangle_\pm &= \frac{1}{\sqrt{2}}\left(|H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d\right)\ ,\\
\mathcal{P}(\mathbf{q})_H &= |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_-\ ,
\end{aligned} \tag{3.39}$$

where $|H(\mathbf{q})\rangle_\alpha$ is the state obtained upon exciting both states in the layer α: $|H(q)\rangle_\alpha = c^\dagger_{\mathbf{q},\alpha} c_{\bar{\mathbf{q}},\alpha} |L\rangle_f |L\rangle_d$, and $\mathcal{P}(\mathbf{q})_H$ projects onto this rotated excited basis. We focus on the spin-flip component $JS^+_f S^-_d$ of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the $d-$layer followed by a spin-flip of the $f-$layer:

$$\Delta H = \frac{1}{N^2}\sum_{\mathbf{q}\in \text{UV}} \langle L|\frac{1}{2}J_f(\bar{\mathbf{q}},\mathbf{q}) S^+_f c^\dagger_{\mathbf{q}-\downarrow,f} c_{\mathbf{q}+\uparrow,f}\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H \frac{1}{2}J_d(\mathbf{q},\bar{\mathbf{q}}) S^-_d c^\dagger_{\mathbf{q}\uparrow,d} c_{\bar{\mathbf{q}}\downarrow,d}|L\rangle \ , \quad (3.40)$$

where $|L\rangle = |L\rangle_f |L\rangle_d$ is the initial configuration for both layers. G is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D} \ , \quad (3.41)$$

where H_D is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q},\sigma\in\mathbf{q}_\pm,\alpha} \epsilon_\alpha(\mathbf{q})\tau_{\mathbf{q},\sigma} + \frac{1}{2}\sum_\alpha J_\alpha S^z_\alpha \sum_{\mathbf{q},\sigma} \sigma n_{\mathbf{q},\sigma,\alpha} + J_\perp \vec{S}_f\cdot\vec{S}_d \ . \quad (3.42)$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\begin{aligned}\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H =& \frac{1}{2}\left(\frac{1}{1/G^J_{ff} - J_\perp/4} + \frac{1}{1/G^J_{dd} - J_\perp/4}\right)\left(|H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_-\right)\\ &+ \frac{1}{2}\left(\frac{1}{1/G^J_{ff} - J_\perp/4} - \frac{1}{1/G^J_{dd} - J_\perp/4}\right)\left(|H\rangle_+ \langle H|_- + |H\rangle_- \langle H|_+\right) \ ,\end{aligned} \quad (3.43)$$

where

$$G^J_{\alpha\alpha}(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2}J_\alpha(\mathbf{q}) + W_\alpha(\mathbf{q})} \ . \quad (3.44)$$

where $\bar{\epsilon}_\alpha(\mathbf{q}) = \epsilon_f(\mathbf{q}) - \epsilon_f(\bar{\mathbf{q}})$ is the total excitation energy for the pair of states $\mathbf{q},\bar{\mathbf{q}}$.

In eq. 3.40, since the first process is an excitation into a $d-$state and the second process is a de-excitation from an $f-$state, the expression can be simplified into

$$\begin{aligned}\langle L|S^+_f c^\dagger_{\mathbf{q}-\downarrow,f} c_{\mathbf{q}+\uparrow,f}\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H S^-_d c^\dagger_{\mathbf{q}+\uparrow,d} c_{\mathbf{q}-\downarrow,d}|L\rangle &= S^+_f S^-_d \langle H_f|\mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H|H_d\rangle\\ &= \frac{1}{2}\left(\frac{1}{1/G^J_{ff} - J_\perp/4} + \frac{1}{1/G^J_{dd} - J_\perp/4}\right) S^+_f S^-_d \ .\end{aligned} \quad (3.45)$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2}\sum_{\mathbf{q}\in\text{UV}} J_f(\bar{\mathbf{q}},\mathbf{q}) J_d(\mathbf{q},\bar{\mathbf{q}})\left(\frac{1}{1/G^J_{ff} - J_\perp/4} + \frac{1}{1/G^J_{dd} - J_\perp/4}\right) S^+_f S^-_d \ . \quad (3.46)$$

Converting the sums to integrals (assuming a density of states $\rho(\mathbf{q})$) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N} S_f^+ S_d^- \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \left(\frac{1}{1/G^J_{ff}(q) - J_\perp/4} + \frac{1}{1/G^J_{dd}(q) - J_\perp/4} \right) . \quad (3.47)$$

The time-reversed partner of this process can be obtained simply by exchanging the $d-$interaction with the $f-$interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term $S_f^+ S_d^-$ is therefore

$$\frac{\Delta J_\perp}{\Delta D} = \frac{1}{2} \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \left(\frac{1}{1/G^J_{ff}(q) - J_\perp/4} + \frac{1}{1/G^J_{dd}(q) - J_\perp/4} \right) . \quad (3.48)$$

Taking into account P_5 by replacing couplings J_α with K_α and the Greens function $G^J_{\alpha\alpha}$ with

$$G^K_{\alpha\alpha}(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2} K_{\bar{\alpha}}(\mathbf{q}) + W_\alpha(\mathbf{q})} , \quad (3.49)$$

we have the total renormalisation

$$\begin{aligned} \frac{\Delta J_\perp}{\Delta D} = \frac{1}{2} \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \Bigg[& J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \sum_\alpha \frac{1}{1/G^J_{\alpha\alpha}(\mathbf{q}) - J_\perp/4} \\ & + K_f(\bar{\mathbf{q}}, \mathbf{q}) K_d(\mathbf{q}, \bar{\mathbf{q}}) \sum_\alpha \frac{1}{1/G^K_{\alpha\alpha}(\mathbf{q}) - J_\perp/4} \Bigg] . \end{aligned} \quad (3.50)$$

3.3.5 Complete RG Equations

Inter-layer Kondo coupling $J_\perp$

$$\begin{aligned} \frac{\Delta J_\perp}{\Delta D} = & -\frac{1}{4} \int_{\text{UV}} d\mathbf{q}\, \rho(q) \int_{\text{UV + IR}} d\mathbf{k}\, \rho(k) \sum_\alpha \Bigg[J^2_\alpha(q,k) \left(\frac{1}{1/G^J_\alpha + 3J_\perp/4} + \frac{1}{1/G^J_\alpha - J_\perp/4} \right) \\ & K^2_\alpha(q,k) \left(\frac{1}{1/G^K_{\bar{\alpha}} + 3J_\perp/4} + \frac{1}{1/G^K_{\bar{\alpha}} - J_\perp/4} \right) \Bigg] \\ & + \int_{\text{UV}} d\mathbf{q}\, \rho(q) \int_{\text{UV + IR}} d\mathbf{k}\, \rho(k) \sum_\alpha \frac{J_\alpha(q,k) K_{\bar{\alpha}}(k,q)}{1/G^{JK}_\alpha - J_\perp/4} \\ & + \frac{1}{2} \int_{UV} d\mathbf{q}\, \rho(\mathbf{q}) \sum_\alpha \left[\frac{J_\alpha(\bar{\mathbf{q}}, \mathbf{q}) J_{\bar{\alpha}}(\mathbf{q}, \bar{\mathbf{q}})}{1/G^J_{\alpha\alpha}(\mathbf{q}) - J_\perp/4} + \sum_\alpha \frac{K_\alpha(\bar{\mathbf{q}}, \mathbf{q}) K_{\bar{\alpha}}(\mathbf{q}, \bar{\mathbf{q}})}{1/G^K_{\alpha\alpha}(\mathbf{q}) - J_\perp/4} \right] \end{aligned} \quad (3.51)$$

Direct Kondo coupling J_α

$$\begin{aligned}\frac{\Delta J_\alpha(k_1,k_2)}{\Delta D} = &-\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\,[J_\alpha(\mathbf{k}_1,\mathbf{q})J_\alpha(\mathbf{q},\mathbf{k}_2) + 4J_\alpha(\mathbf{q},\bar{\mathbf{q}})W_\alpha(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)]\\ &\left(\frac{1/4}{1/G^J_\alpha(\mathbf{q}) + 3J/4} + \frac{3/4}{1/G^J_\alpha(\mathbf{q}) - J/4}\right)\\ &+\frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\,[K_{\bar\alpha}(\mathbf{k}_1,\mathbf{q})K_{\bar\alpha}(\mathbf{q},\mathbf{k}_2) - 4K_{\bar\alpha}(\mathbf{q},\bar{\mathbf{q}})W_\alpha(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)]\\ &\left(\frac{1}{1/G^K_\alpha + 3J/4} + \frac{1}{1/G^K_\alpha - J/4}\right)\ .\end{aligned} \quad (3.52)$$

Indirect Kondo coupling K_α

$$\begin{aligned}\frac{\Delta K_\alpha(k_1,k_2)}{\Delta D} = &-\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\,[K_\alpha(\mathbf{k}_1,\mathbf{q})K_\alpha(\mathbf{q},\mathbf{k}_2) + 4K_\alpha(\mathbf{q},\bar{\mathbf{q}})W_{\bar\alpha}(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)]\\ &\left(\frac{1}{4}\frac{1}{1/G^K_{\bar\alpha}(\mathbf{q}) + 3J/4} + \frac{3}{4}\frac{1}{1/G^K_{\bar\alpha}(\mathbf{q}) - J/4}\right)\\ &+\frac{1}{2}\int_{UV} d\mathbf{q}\,\rho(\mathbf{q})\,[J_{\bar\alpha}(\mathbf{k}_1,\mathbf{q})J_{\bar\alpha}(\mathbf{q},\mathbf{k}_2) - 4J_{\bar\alpha}(\mathbf{q},\bar{\mathbf{q}})W_{\bar\alpha}(\mathbf{k}_1,\mathbf{q},\bar{\mathbf{q}},\mathbf{k}_2)]\\ &\left(\frac{1}{1/G^J_{\bar\alpha} + 3J/4} + \frac{1}{1/G^J_{\bar\alpha} - J/4}\right)\ .\end{aligned} \quad (3.53)$$

Propagators

$$\begin{aligned}\text{Intra-layer spin-flip excitation: } G^J_\alpha(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha(\mathbf{q})/4 + W_\alpha(\mathbf{q})/2}\ ,\\ \text{Inter-layer spin-flip excitation: } G^K_\alpha(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + K_{\bar\alpha}(\mathbf{q})/4 + W_\alpha(\mathbf{q})/2}\ ,\\ \text{Both spins, one of the baths: } G^{JK}(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha(\mathbf{q})/4 + K_{\bar\alpha}(\mathbf{q})/4 + W_\alpha(\mathbf{q})/2}\ ,\\ \text{Intra-layer spin-flip, two UV excitation: } G^J_{\alpha\alpha}(\mathbf{q}) &= \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2}J_\alpha(\mathbf{q}) + W_\alpha(\mathbf{q})}\ ,\\ \text{Inter-layer spin-flip, two UV excitation: } G^K_{\alpha\alpha}(\mathbf{q}) &= \frac{1}{\omega - \frac{1}{2}\left[\epsilon_\alpha(\mathbf{q}) - \epsilon_\alpha(\bar{\mathbf{q}})\right] + \frac{1}{2}K_{\bar\alpha}(\mathbf{q}) + W_\alpha(\mathbf{q})}\ .\end{aligned} \quad (3.54)$$

3.4 Analysis for the simplified case of infinite dimensions

We first investigate the properties of the impurity model in the limit of infinite dimensions, on a Bethe lattice. This limit suppresses any momentum-space differentiation arising from low-dimensional lattices, and allows the study of real-space local properties.

We simulate the RG equations by adding small *negative* seed values of K_f and K_d. The other couplings are set as follows, in units of the conduction electron bandwidth: $J_f = J_d = 0.1, W_d = 0.0$. The results are shown in Fig. 3.1, in the form of the renormalised low-energy fixed point values of the Kondo couplings J_f and J_d and the inter-layer coupling $J_\perp$, all as functions of W_f and $J_\perp$. We find that the $d-$layer Kondo coupling J_d (central panel) undergoes a suppression at larger values of the inter-layer coupling $J_\perp$. This coincides with an increase in the renormalised value of $J_\perp$ (right panel) itself, indicating a transfer of correlations from within the $d-$layer to between the $d-$ and $f-$layers. The couplings J_d is, however, largely unaffected by the bath correlation of the $f-$layer, as seen from the lack of any dependence of the boundary on W_f.

The $f-$layer coupling J_f (left panel) is suppressed with increasing $|W_f|$ as well as increasing $J_\perp$. The phase space for J_f therefore splits, roughly, into four regions. The lower left region involves a dominant J_f coupling and displays local Fermi liquid excitations within the $f-$layer. The upper left region has the in-plane correlation W_f driving J_f to irrelevance and leads to a Mott insulator within the layer, which can order magnetically if spontaneous symmetry breaking is allowed. The lower right region has the influence of both J_f and $J_\perp$ and therefore involves the hybridisation of the local Fermi liquids in the two layers. Finally, the upper right region has the Mott state of the $f-$layer hybridising with the $d-$layer through $J_\perp$

3.5 Infinite dimensions: Real-space correlation and phase diagram

In order to characterise the various phases, we compute a set of real-space correlations involving the impurity sites and the conduction bath sites hybridising with them. For small $|W_f|$, the $f-$layer is highly correlated but still itinerant due to the presence of local Fermi liquid excitations. In order to capture the $J_\perp$-driven transition in this regime, we track the inter-impurity compensation factor $\langle \mathbf{S}_f \cdot \mathbf{S}_d \rangle$ that quantifies the strength of the singlet state between these two sites. As seen from the left panel in Fig. 3.2, the inter-impurity correlation (red curve) grows to large values as $J_\perp$ is increased and eventually saturates close to the maximum value. This marks a transition from a phase in which the two metallic layers are decoupled and participate in transport independently, to one in which the local Fermi liquids hybridise and form local bound states.

A complimentary picture is obtained in the same regime by studying the one-particle hybridisa-

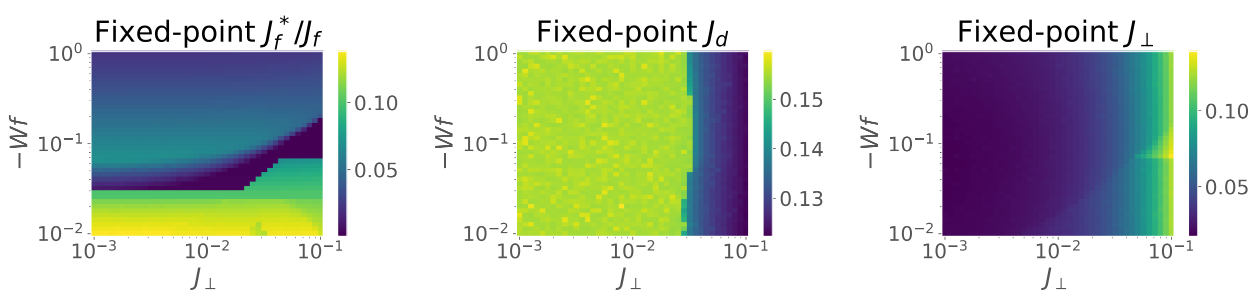

Figure 3.1: *Fixed-point values for the direct Kondo couplings J_α and $J_\perp$.*

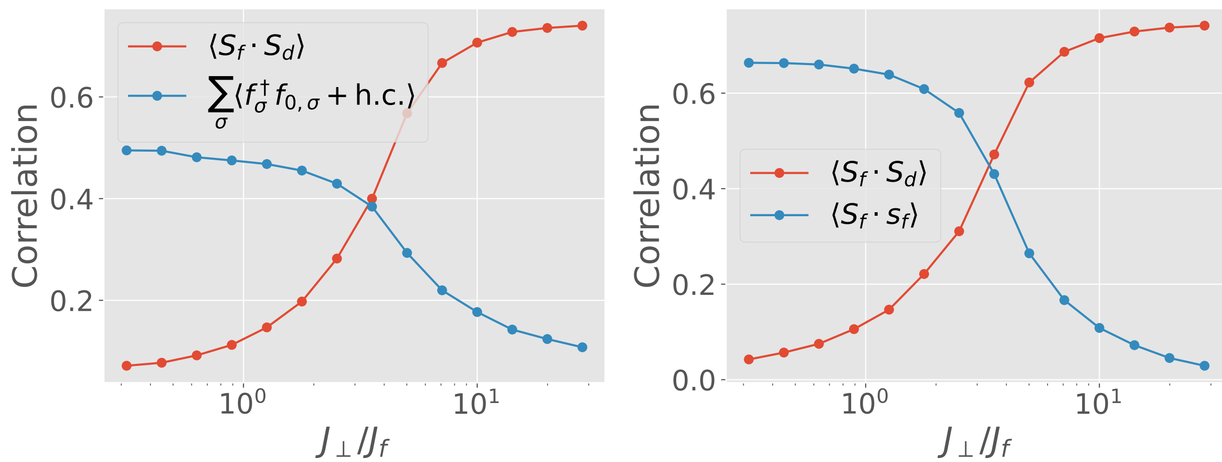

Figure 3.2

tion between the $f-$impurity and its nearest-neighbour site f_0. While this measure is large for small $J_\perp$, indicating the presence of in-plane Kondo screening, and it gets suppressed as $J_\perp$ is increased, revealing that the $f-$impurity is no longer hybridising with its own bath.

For large $|W_f|$, the $f-$layer is sufficiently strongly correlated to form a Mott insulator where the charge excitations are gapped, and the low-energy theory is described by a Heisenberg model. Here, we retain the inter-layer compensation factor as a measure but we replace the one-particle hybridisation with the spin-spin correlation $\langle \mathbf{S}_f \cdot \mathbf{s}_{f0} \rangle$ between the $f-$spin and it's nearest spin; this change is necessitated by the charge gap in the spectrum. We again find that the inter-layer correlation grows as $J_\perp$ is increased, while the intra-layer spin correlation shrinks. This marks a transition between a phase with no Kondo screening between the layers (and where the $f-$layer is potentially magnetically ordered) to one where the $f-$spin is Kondo screened by the other layer. In the context of heavy fermions, this marks a Kondo breakdown QCP between a magnetically-ordered phase with no (static) Kondo screening versus a heavy Fermi liquid phase where the $f-$spin participates in transport through its hybridisation with the other layer.

These conclusions are crystallised in the phase diagram in Fig. 3.3, computed using the three measures described above. For small $J_\perp$ and $|W_f|$ (lower left region), the two layers are metallic and mostly decoupled. Increasing $J_\perp$ along the $y-$axis leads to a heavy Fermi liquid phase, while increasing $|W_f|$ along the $x-$axis leads to the $f-$layer becoming insulating and decoupled from the other layer.

We have also characterised the transition using entanglement measures. Specifically, the mutual information (Fig. 3.4 between the $f-$ and $d-$spins grows during the emergence of the heavy Fermi liquid, as $J_\perp$ is increased. While this holds for both large and small $|W_f|$, there's one difference. Upon lowering $J_\perp$, the I_2 for the case of small $|W_f|$ vanishes much more rapidly compared to the case of large $|W_f|$. We believe the former is due to the gapless excitations present within the $f-$layer at small $|W_f|$ that promote strong entanglement within the layer and hence compete with the inter-layer entanglement.

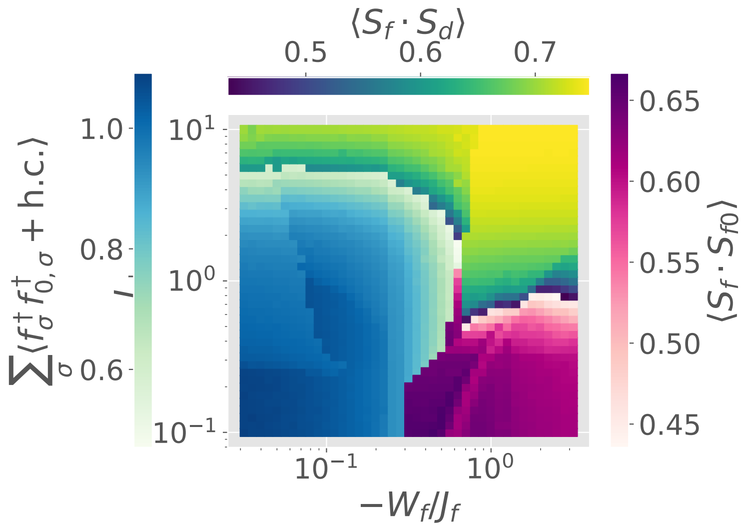

Figure 3.3

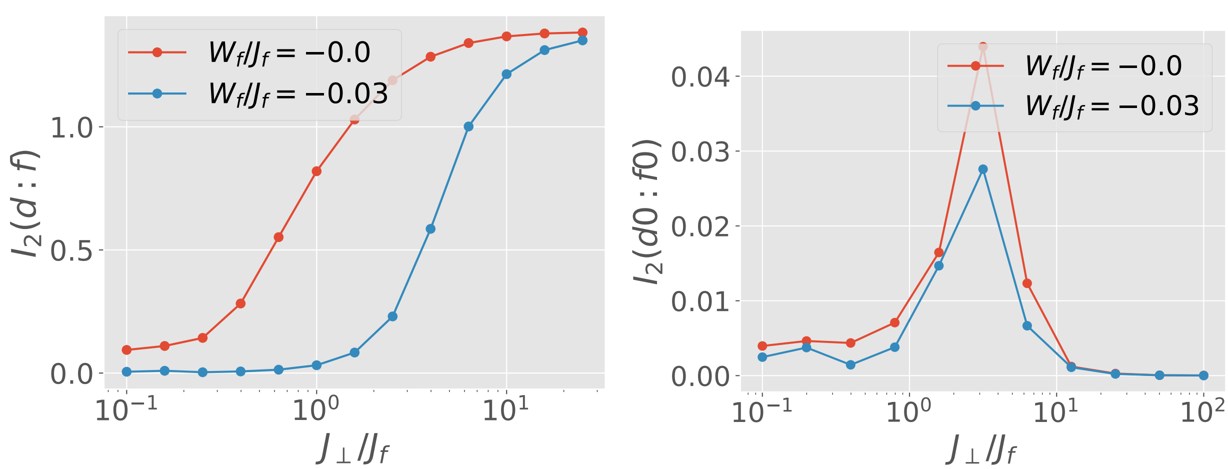

Figure 3.4

3.6 Infinite dimensions: Local spectral function

To further investigate the phases, we compute the local spectral functions of the $f-$ and $d-$impurities. For weak $J_\perp$, the $d-$impurity always exhibits a central Kondo resonance, while the $f-$layer supports it only for weak W_f. For larger $J_\perp$, this resonance splits into two bands, in both the layers. These bands correspond to the two hybridised levels formed locally due to the mixing of the $f-$ and $d-$impurities. For weak $|W_f|$, this is essentially the mixing of the two local Fermi liquids, while for large $|W_f|$, it arises purely from the LFL of the $d-$layer. At half-filling, the splitting of the central resonance into bands leads to a gap at zero frequency, and we get a Kondo insulator.

Away from half-filling, if the chemical potential lies in one of the bands, the systems becomes a heavy Fermi liquid. The value of W_f determines the amount of correlation in the heavy Fermi liquid. Larger values of $|W_f|$ lead to narrow bands for the metal.

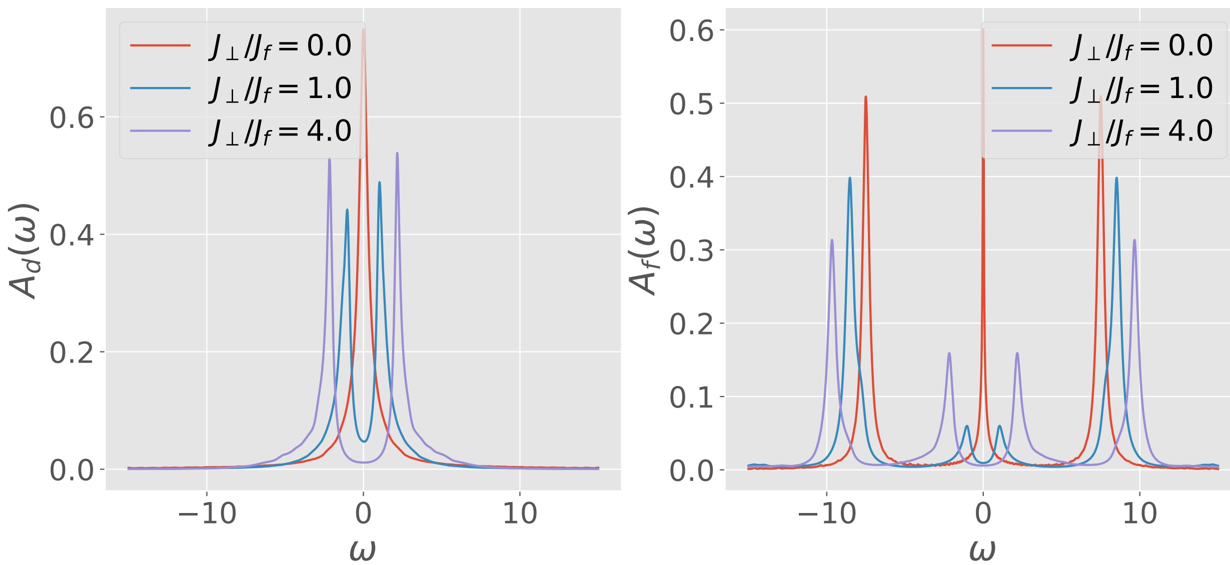

Figure 3.5

3.7 Renormalisation Group Flows of Kondo Couplings: RG Phase Diagram

We first analyse the model by obtaining unitary RG flows of three kinds of Kondo couplings. Fig. 3.7 shows the fixed point values of the couplings under renormalisation to a low-energy theory. We view these values as a function of the $f-$layer bath interaction W_f and inter-layer Kondo coupling $J_\perp$. In each panel, the right (left) colorbar quantifies the background (marker) color of the plot. The background displays the fixed point value of each coupling whereas the marker color tracks whether the couplings is relevant (1), intermediate coupling (0.5) or irrelevant (0).

The Kondo coupling J_d (middle panel) of the $d-$layer remains relevant for the entirety of the chosen parameter regime. The Kondo coupling J_f (left panel) of the $f-$layer becomes RG-irrelevant

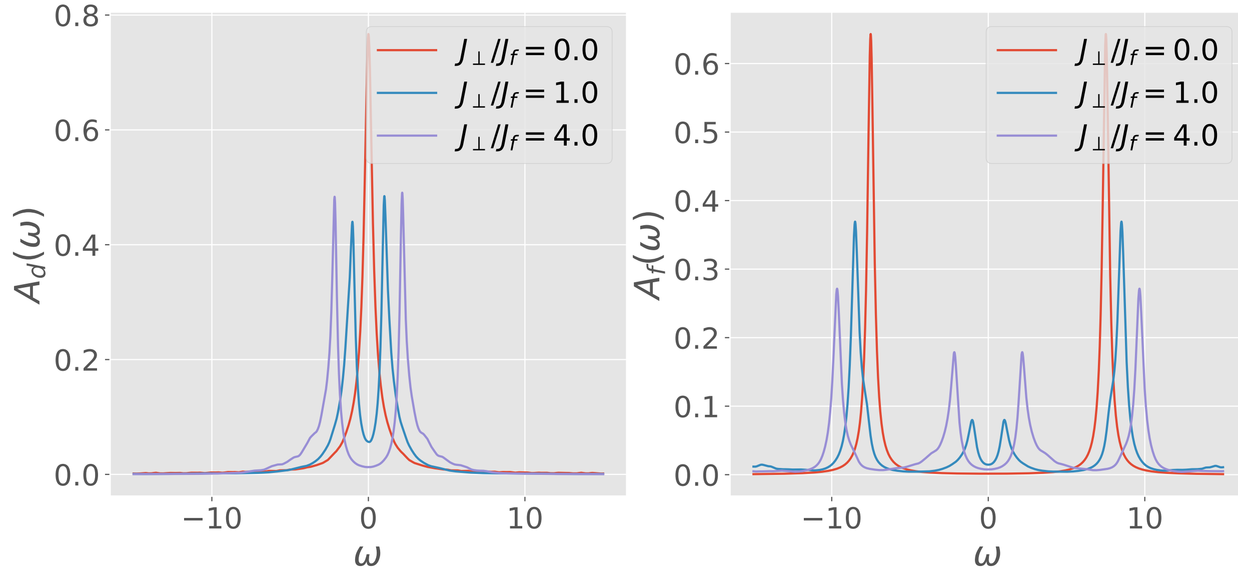

Figure 3.6

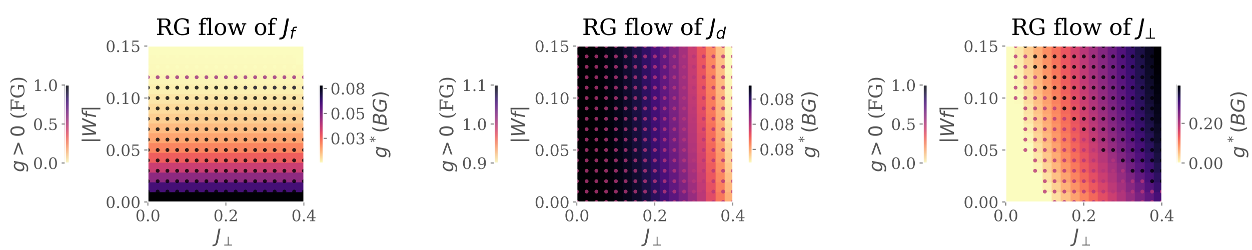

Figure 3.7: *RG phase diagrams for various Kondo couplings: $f-$layer (left), $d-$layer (middle) and inter-layer (right). For each plot, the background color (quantified by the right colorbar) tracks the fixed point value of the coupling as a function of the inter-layer couplings $J_\perp$ and the $f-$layer bath interaction $|W_f|$. The marker color complements this by tracking whether the coupling is relevant, intermediate-coupling or irrelevant in the RG sense, by taking the values 1, $1/2$ or 0 respectively.*

for larger values of $|W_f|$; this leads to an in-plane Mott transition of the kind observed in our earlier work, where the local Fermi liquid excitations of the $f-$impurity give way to a pseudogapped non-Fermi liquid Mott metal phase which ultimately destabilises into a local moment phase. The RG flow of J_f does not appear to be significantly affected by the variation of $J_\perp$. Concomitant with the irrelevance of J_f under increasing $|W_f|$, the inter-plane coupling $J_\perp$ (right panel) becomes RG-relevant, as seen from the darkening markers as we go to higher $|W_f|$.

These simulations motivate an RG-based phase diagram with the following regimes: (i) weak $J_\perp$, weak $|W_f|$: decoupled metallic layers, each layer participating in transport independently, (ii) strong $J_\perp$, strong $|W_f|$: Kondo insulator regime, where the local moments of the $f-$layer hybridise with the itinerant electrons of the $d-$layer to form a band insulator with the chemical potential situated in the hybridisation gap, and (iii) weak $J_\perp$, strong $|W_f|$: the so-called RKKY regime, where the $f-$layer forms local moments which decouple from the $d-$layer and are then free to order magnetically, and (iv) strong $J_\perp$, weak $|W_f|$: this is similar to the Kondo insulator regime, except here the $f-$electrons remain itinerant and it is they who hybridise with the $d-$electrons to again form two bands separated by a hybridisation gap.

3.8 Results at half-filling

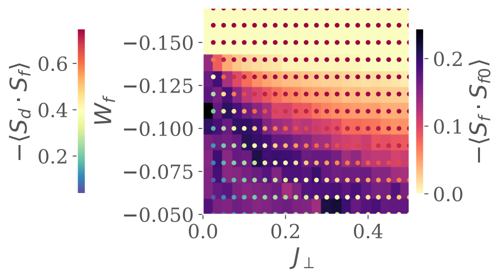

Figure 3.8: *Left: Real-space spin correlations for the auxiliary model. Background colour (right colourbar) represents that between the $f-$impurity and its nearest-neighbour sites, while the marker colour (left colourbar) represents that between the $f-$impurity and the $d-$impurity. While the former shrinks as $J_\perp$ or $|W_f|$ is increased, the latter grows. This marks a transfer of entanglement from intra-plane channel into inter-plane channel. Right:*

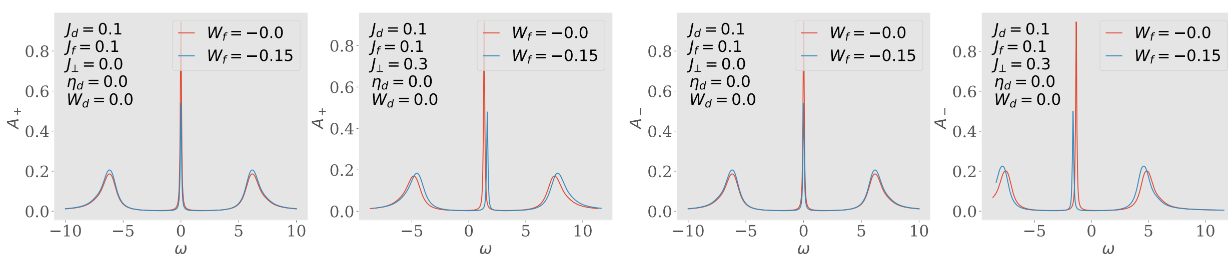

Figure 3.9: *Spectral functions of the tiled model, for excitations in the bonding (two on the left) and antibonding (two on the right) basis, defined by the Greens function $G_{k,\pm} = G(k_d \pm k_f; \omega)$. There is no hybridisation for $J_\perp = 0$ (first and third plots), and the only effect of W_f is to shrink the height of the central resonance. Introducing a non-zero $J_\perp$ splits the central resonance into an upper band (second from left) and a lower band (first from right). This leads to a Kondo insulator, where $\omega = 0$ resides inside the hybridisation gap.*

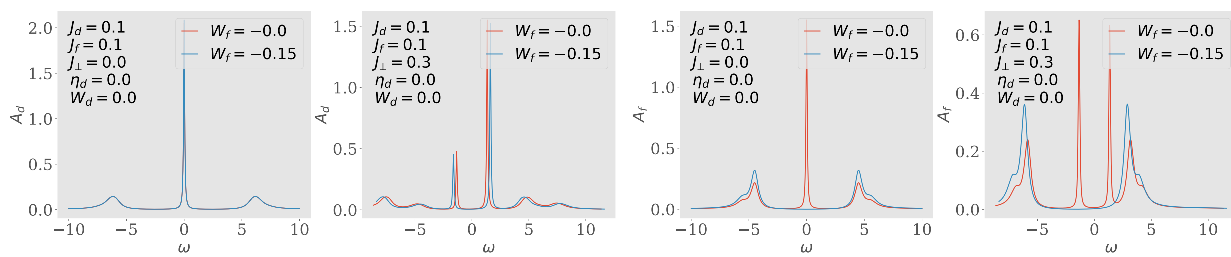

Figure 3.10: *Spectral functions A_d and A_f of the auxiliary model, for excitations in the $d-$ and $f-$layers respectively. Both show a central Kondo resonance for $J_\perp = 0$ which splits into two bands upon adding a non-zero $J_\perp$. Increasing $|W_f|$ to large values gaps out the local Fermi liquid excitations of A_f.*

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